



Chapter # 05

Nervous System of Man



Student Learning Outcomes

At the end of this chapter, students will be able to:

- [B-12-G-01] Recognize receptors as transducers sensitive to various stimuli.
- [B-12-G-02] Trace the path of a message transmitted to the CNS (central nervous system) for processing.
- [B-12-G-03] Identify the three neurons (sensory, intermediate, motor) involved in nervous transmission.
- [B-12-G-04] Identify muscles and glands as the effectors.
- [B-12-G-05] Annotate the detailed structure of a sensory neuron, associative and motor neuron.
- [B-12-G-06] Relate the structure of neurons with functions.
- [B-12-G-07] Differentiate between myelinated and non-myelinated neurons.
- [B-12-G-08] Explain the function of the three types of neurons with the help of a reflex arc.
- [B-12-G-09] Define nerve impulse.
- [B-12-G-10] Describe the generation and transmission of nerve impulse.
- [B-12-G-11] Name the factors responsible for the resting membrane potential of neuron.
- [B-12-G-12] Evaluate from a graph the phenomena of polarization, depolarization and hyperpolarization of membrane.
- [B-12-G-13] Compare the velocities of nerve impulse in the axon membrane and in the synaptic cleft.
- [B-12-G-14] Describe the role of local circuits in saltatory conduction of nerve impulse.
- [B-12-G-15] Outline the structure of synapse.
- [B-12-G-16] Explain synaptic transmission of nerve impulse.
- [B-12-G-17] Classify neurotransmitters as inhibitory and excitatory and list some common examples.
- [B-12-G-18] Identify the main components of the nervous system.
- [B-12-G-19] Explain briefly the major parts, functions of major divisions of the brain and its functions of brain.
- [B-12-G-20] Describe the architecture of human brain.
- [B-12-G-21] Describe cranial and spinal nerves in man.
- [B-12-G-22] Explain the structure types and functions of the autonomic nervous system.
- [B-12-G-29] Explain the structure and functioning of the receptors for smell, taste, and touch/pain.
- [B-12-G-30] Define narcotic drugs as agents that interact with normal nervous activity.
- [B-12-G-31] Compare the use and abuse of drugs with respect to heroine, Cannabis, nicotine, alcohol and inhalants like nail polish remover and glue.
- [B-12-G-32] Explain the terms drug addiction and drug tolerance with reference to caffeine and nicotine and their adverse effects.
- [B-12-G-33] Associate the effects of drug addiction and tolerance with the functioning of the nervous system.
- [B-12-G-34] Describe the way how pain medicines can reduce or numb pain in the human body.
- [B-12-G-35] Discuss that certain pain medications are addictive.
- [B-12-G-36] Classify nervous disorders into vascular, infectious, structural, functional and degenerative disorders.
- [B-12-G-37] Describe the causes, symptoms and treatment of one type of each category of disorders outlined above (e.g., stroke as vascular, meningitis as infectious, brain tumor as structural, headache as functional, and Alzheimer disease as degenerative disorder).
- [B-12-G-38] Explain the principles of important diagnostic tests for nervous disorders i.e., EEG, CT scan and MRI.



The nervous system of man is a highly specialized and complex network that controls and coordinates all activities of the body. It enables the body to sense changes in the internal and external environment and respond to them appropriately, thereby maintaining balance and ensuring survival. The nervous system carries out three essential roles: it detects stimuli through sense organs, processes and interprets the information in the brain and spinal cord, and initiates suitable responses by stimulating muscles or glands. Structurally, it is divided into the central nervous system, which includes the brain and spinal cord and is responsible for integration and control, and the peripheral nervous system, which consists of nerves and ganglia and connects the central nervous system with the rest of the body. In addition, the autonomic nervous system regulates involuntary activities such as heartbeat, breathing, and digestion, thus ensuring the smooth functioning of vital organs. Altogether, the nervous system serves as the body's main control and communication center, making it possible for humans to think, learn, remember, and adapt to their environment.

SLO [B-12-G-01]

Recognize receptors as transducers sensitive to various stimuli.

RECEPTORS AS BIOLOGICAL TRANSDUCERS

What is a Transducer?

A transducer is any device that can convert energy from one form to another. In engineering, microphones convert sound into electrical signals and solar panels convert light into electricity. Similarly, in biology, **receptors** act as natural transducers that convert environmental energies — such as light, sound, temperature, pressure, or chemicals — into electrochemical impulses that the nervous system can understand.

► Definition of Receptor

A receptor is a **cell, tissue, or organ capable of detecting a stimulus** in the environment and converting it into an electrical message (nerve impulse). These impulses are then carried by sensory neurons to the central nervous system (CNS), where they are interpreted and transformed into appropriate responses.

► Types of Stimuli

Receptors can detect both **internal stimuli** (arising inside the body) and **external stimuli** (coming from the surroundings).

- **Internal Stimuli:** Any change occurring inside the organism is called an internal stimulus. Examples include hunger, thirst, fluctuations in blood pH, blood pressure, or the presence of pathogens such as during a viral infection. Detecting such stimuli helps the body maintain homeostasis.
- **External Stimuli:** Any change in the external environment is called an external stimulus. Examples include light, sound, heat, touch, and mechanical pressure. Detecting such stimuli allows the organism to interact effectively with its environment and ensures survival.

► Structural Organization of Receptors

Receptors exist in two main forms:

1. **Specialized Receptors in Sense Organs:** These are highly organized and sensitive to specific stimuli. For example, **photoreceptors** in the retina detect light, **mechanoreceptors** in the inner ear detect sound and balance, and **chemoreceptors** in the nose and tongue detect smell and taste.
2. **Receptors Dispersed in Body Tissues:** These are scattered throughout the body, such as mechanoreceptors and thermoreceptors in the skin (for touch, pain, and temperature) or baroreceptors in blood vessels (for blood pressure regulation). They provide continuous feedback to maintain internal balance.

► Examples of Major Receptor Types

- **Photoreceptors:** Detect light (e.g., rods and cones in the retina).
- **Mechanoreceptors:** Detect pressure, vibration, touch, and sound (e.g., skin receptors, ear hair cells).
- **Thermoreceptors:** Detect changes in temperature (e.g., in skin and hypothalamus).
- **Chemoreceptors:** Detect chemicals (e.g., taste buds, olfactory cells, oxygen receptors in blood vessels).
- **Nociceptors:** Detect pain caused by harmful stimuli (e.g., tissue damage).



► Real-Life Applications and Impact

Receptors are not just biological structures — they are the reason humans and other organisms can survive, adapt, and thrive.

- **Driving a Car:** Photoreceptors detect light signals, mechanoreceptors in muscles and joints sense movement, and auditory receptors detect horns or sirens. Together, they allow safe and quick reactions.
- **Medical Diagnostics:** Understanding chemoreceptors has led to glucose sensors for diabetic patients and pH monitoring devices for patients with acid–base imbalances.
- **Survival Instincts:** Nociceptors alert the body to potential harm. For example, when touching a hot object, thermoreceptors and nociceptors instantly send signals that trigger a withdrawal reflex before conscious thought occurs.
- **Homeostasis:** Internal receptors such as baroreceptors and osmoreceptors continuously monitor blood pressure and osmotic balance, ensuring stable internal conditions essential for life.

Receptors act as the **gateways of perception** by functioning as transducers that convert different forms of environmental energy into electrical impulses. They enable humans to sense, interpret, and respond to both internal and external changes. By doing so, receptors play a vital role in survival, adaptation, and the maintenance of homeostasis.

SLO [B-12-G-02]

Trace the path of a message transmitted to the CNS (central nervous system) for processing.

PATH OF A MESSAGE TO THE CENTRAL NERVOUS SYSTEM

The central nervous system (CNS) cannot respond to the environment unless it receives information from receptors. The journey of this information — called sensory transmission — begins at the receptor, travels along sensory pathways, and ends in the brain, where it is interpreted. This organized flow ensures that the body can detect, analyze, and respond effectively to both internal and external changes.

Transmission of Sensory Information: Neural Pathway to the Brain

Sensory perception begins with the **detection of stimuli** and progresses through a highly organized neural pathway to reach the brain, where the information is interpreted. This pathway usually follows a **three-neuron relay system**:

1. First-Order Neuron (Sensory Neuron)

The first-order neuron is the initial component of the sensory pathway. It originates in peripheral tissues such as the skin, muscles, joints, or internal organs and transmits impulses to the CNS. These neurons detect various stimuli including touch, pain, pressure, temperature, and body position (**proprioception**) through specialized receptor cells.

The cell bodies of first-order neurons are found in **dorsal root ganglia** (for the body) or **cranial nerve ganglia** (for the head region). Their axons synapse with second-order neurons in the spinal cord or brainstem.

Example: When you touch a sharp pin, receptors in the skin activate first-order sensory neurons, which carry the impulse toward the spinal cord.

2. Second-Order Neuron (Interneuron)

The second-order neuron receives input from the first-order neuron. It is usually located in the **dorsal horn of the spinal cord** or in the **brainstem**, depending on the pathway. Its main function is to relay and sometimes integrate sensory input before forwarding it to higher brain centers.

In most pathways, the axons of second-order neurons **decussate (cross over)** to the opposite side of the CNS and then ascend to the thalamus, which acts as the main relay station for sensory information.

Example: In pain perception, the second-order neuron in the spinal cord transmits the impulse to the thalamus after crossing to the opposite side. This explains why the left side of the brain processes pain from the right side of the body.

3. Third-Order Neuron

The third-order neuron resides in the **thalamus** and serves as the final relay in the ascending sensory pathway. It projects axons to the **primary somatosensory cortex** in the parietal lobe of the cerebrum.



(postcentral gyrus). This is the region responsible for the **conscious perception** of sensations such as pain, pressure, touch, temperature, and body position.

Example: When you feel the heat of a cup of tea, the third-order neuron in the thalamus projects the signal to your cerebral cortex, allowing you to consciously recognize the warmth.

► Applications and Impact

- **Driving a Vehicle:** Visual information travels via photoreceptors → first-order neurons (optic nerve) → second-order neurons (optic chiasma, thalamus) → third-order neurons projecting to the visual cortex. This enables drivers to process traffic lights and road signs instantly.
- **Reflex vs. Conscious Action:** Reflexes are processed at the spinal cord level (bypassing higher-order neurons) for speed, while conscious sensations like recognizing pain or texture require the full three-neuron relay.
- **Medical Relevance:** Damage to first-order neurons (e.g., peripheral neuropathy in diabetes) causes numbness, while injury to second-order pathways (e.g., spinal cord trauma) can block sensations from large parts of the body. Stroke affecting third-order pathways (thalamus or cortex) may lead to loss of conscious perception despite intact receptors.

► Summary of the Pathway

The transmission of sensory information to the brain follows this sequence:

Stimulus → Receptor → First-Order Neuron (to spinal cord/brainstem) → Second-Order Neuron (crosses & ascends to thalamus) → Third-Order Neuron (projects to cerebral cortex) → Conscious Perception

This well-organized system ensures that the body not only reacts quickly but also consciously understands and interprets its environment.

Summary of the Sensory Pathway

Neuron	Location	Function
First-order neuron	Peripheral nervous system to spinal cord or brainstem	Detects sensory stimuli and transmits impulses to CNS
Second-order neuron	Spinal cord or brainstem	Relays and integrates information; transmits to thalamus (often after decussation)
Third-order neuron	Thalamus	Sends processed information to the somatosensory cortex for perception

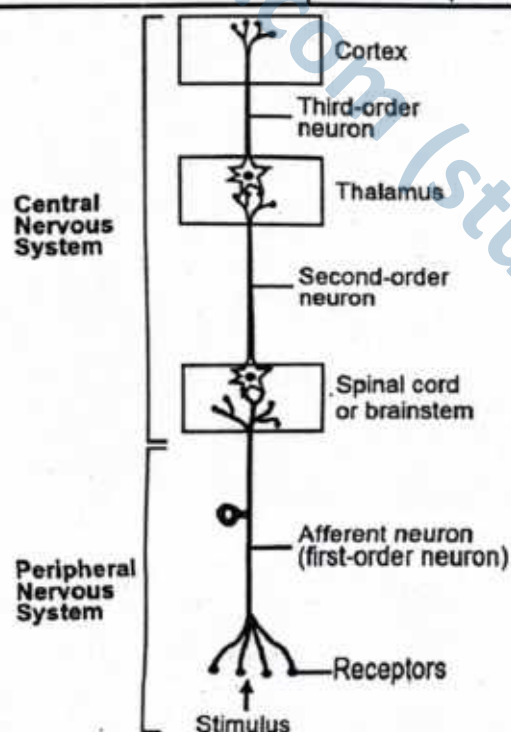


Figure: Pathway of message transmitted to the CNS

RECEPTORS AS BIOLOGICAL TRANSDUCERS, PATHWAY OF MESSAGE TRANSMITTED TO THE CNS AND STRUCTURE OF NEURON

1. Which of the following best describes the role of a receptor in biological systems?
 - A. It stores sensory information for later use.
 - B. It converts energy from one form to another.
 - C. It transports neurotransmitters to the brain.
 - D. It amplifies action potentials in neurons.
2. What is the primary function of the axon in a neuron?
 - A. Receiving signals from adjacent neurons
 - B. Synthesizing neurotransmitter proteins
 - C. Conducting impulses away from the cell body
 - D. Supporting metabolic activity of the cell
3. Where are the cell bodies of first-order sensory neurons located?
 - A. Thalamus
 - B. Cerebral cortex
 - C. Dorsal root ganglia or cranial nerve ganglia
 - D. Anterior horn of spinal cord
4. In most sensory pathways, the axon of the second-order neuron:
 - A. Projects directly to the cerebral cortex
 - B. Terminates in the spinal ganglion
 - C. Decussates and travels to the thalamus
 - D. Forms synapses with motor neurons
5. Which structural feature of a neuron contains Nissl bodies?
 - A. Dendrite
 - B. Axon hillock
 - C. Cell body
 - D. Axon terminal
6. Receptors present in the skin that respond to pressure or pain are best described as:
 - A. Photoreceptors
 - B. Chemoreceptors
 - C. Mechanoreceptors
 - D. Thermoreceptors
7. Which region of the brain receives projections from third-order sensory neurons?
 - A. Hypothalamus
 - B. Midbrain
 - C. Somatosensory cortex
 - D. Corpus callosum

Answer Key

1. B 2. C 3. C 4. C 5. C 6. C 7. C

Short Questions

Q.1 What is a biological transducer?

Ans. A biological transducer is a receptor cell that converts physical or chemical environmental stimuli into electrical signals that the nervous system can interpret. For example, photoreceptors in the retina transform light energy into nerve impulses, mechanoreceptors in the skin detect pressure or vibration, and chemoreceptors in the nose or tongue respond to chemical molecules. These cells function as "biological converters," bridging the gap between external changes and internal responses. The electrochemical signals produced by these receptors are then transmitted via sensory neurons to the brain or spinal cord for interpretation. Without biological transducers, organisms would not be able to perceive their surroundings, respond to changes, or maintain survival-related behaviors such as feeding, defense, and navigation.

MARKING SCHEME (3 Marks)

- Definition (2), examples (1).

Q.2 Describe the role of a first-order neuron.

Ans. A first-order neuron is the starting point of a sensory pathway. It directly receives input from receptor cells located in peripheral tissues, such as the skin, muscles, or sensory organs. Its dendrites detect specific stimuli (e.g., touch, temperature, pain), while its cell body is located in the dorsal root ganglia of spinal nerves or in cranial nerve ganglia. The axon of the first-order neuron carries this information into the spinal cord or brainstem, where it synapses with a second-order neuron. By transmitting raw sensory data from the environment to the central nervous system, the first-order neuron ensures that higher centers receive accurate information to process. This forms the basis for perception, reflexes, and voluntary responses.

MARKING SCHEME (3 Marks)

- Definition (2), function (1).



Q.3 What is the function of the second-order neuron in sensory pathways?

Ans. The second-order neuron acts as the next relay station in the sensory pathway. Located in the dorsal horn of the spinal cord or in nuclei of the brainstem, it receives signals from the first-order neuron and processes them further. A key characteristic of second-order neurons is that most of their axons cross to the opposite side (decussation), ensuring that sensations from one side of the body are interpreted by the opposite hemisphere of the brain. The axons then ascend toward the thalamus, the brain's main sensory relay center. By filtering and relaying impulses to higher structures, second-order neurons ensure proper transmission of sensory information needed for awareness and appropriate response.

MARKING SCHEME (3 Marks)

- Definition (2), function (1).

Q.4 What are Nissl bodies and where are they found?

Ans. Nissl bodies are specialized structures found in the cytoplasm of neurons, specifically within the cell body and dendrites, but absent from the axon and axon hillock. They consist of rough endoplasmic reticulum with ribosomes attached, along with free ribosomes. Their main function is protein synthesis, particularly enzymes and neurotransmitters essential for neuronal activity. Because neurons require continuous protein production to maintain signaling, Nissl bodies are highly prominent under a microscope when stained. They also play a vital role in neuronal repair: following injury, they increase protein synthesis to help restore damaged regions. Thus, Nissl bodies are critical for both routine neuronal function and recovery after damage.

MARKING SCHEME (3 Marks)

- Definition (2), location (1).

Q.5 How do dendrites contribute to neural communication?

Ans. Dendrites are branched extensions of a neuron that serve as the main receptive surface for incoming signals. They receive impulses from surrounding neurons through synapses, often collecting input from multiple sources at once. These signals are then conducted toward the neuron's cell body, where they are integrated and processed. By allowing a single neuron to receive input from thousands of other neurons, dendrites make complex communication possible. This integration determines whether the neuron will generate its own action potential and pass the message forward through its axon. Thus, dendrites play a vital role in synaptic transmission, coordination of signals, and the overall connectivity of neural networks.

MARKING SCHEME (3 Marks)

- Definition (2), function (1).

Q.6 What is the difference between receptors located in sense organs and those dispersed in tissues?

Ans. Receptors in sense organs are highly specialized, concentrated, and adapted to detect specific environmental stimuli with great precision. For example, photoreceptors in the retina detect light, hair cells in the cochlea detect sound vibrations, and olfactory receptors in the nasal epithelium sense odors. These receptors allow organisms to interact effectively with the external environment. In contrast, receptors dispersed in tissues are distributed throughout the body and play a more general role in maintaining homeostasis. Examples include baroreceptors in blood vessels, which monitor blood pressure, and chemoreceptors, which sense pH and oxygen levels. Unlike sense organ receptors, they are not localized but scattered, ensuring continuous monitoring of the body's internal environment. Both types together allow survival by integrating external perception with internal regulation.

MARKING SCHEME (3 Marks)

- Sense organ receptors (1.5), dispersed tissue receptors (1.5).

Q.7 What is the function of the third-order neuron?

Ans. The third-order neuron represents the final relay point in most ascending sensory pathways. It is located in the thalamus, which serves as the brain's central relay station for sensory signals. These neurons receive input from second-order neurons that have crossed over from the opposite side of the body and then transmit this information to the primary somatosensory cortex of the cerebral hemisphere. This transmission allows for the conscious perception of sensations such as touch, pressure, vibration, pain, temperature, and proprioception (awareness of body position). The third-order neuron is therefore essential for transforming raw sensory input into conscious awareness and enabling appropriate voluntary or reflexive responses. Without its role, perception and higher processing of stimuli would not occur.

MARKING SCHEME (3 Marks)

- Definition (2), function (1).



NEURON: THE BASIC UNIT OF NERVOUS COORDINATION

A neuron is the fundamental structural and functional unit of the nervous system. It is specialized to **receive, process, and transmit information** using electrical and chemical signals, enabling the body to respond to internal and external stimuli.

Functional Types of Neurons

Neurons are highly diverse in size, shape, and structure, reflecting their specific functional roles. Based on function, neurons are classified into **three major types** involved in nervous transmission:

1. **Sensory Neurons (Afferent Neurons)**
2. **Motor Neurons (Efferent Neurons)**
3. **Interneurons (Association or Relay Neurons)**

The interaction of these three neurons forms the **basic pathway for nervous transmission**, allowing signals to travel from a sensory receptor to a motor effector via the central nervous system (CNS).

1. Sensory Neurons (Afferent Neurons)

Role in Nervous Transmission:

Sensory neurons detect stimuli from the **external environment** (e.g., light, sound, temperature, pressure) and **internal environment** (e.g., blood pressure, muscle stretch) and transmit this information to the CNS for processing.

Location:

- Cell bodies are located in **dorsal root ganglia** (spinal nerves) or **cranial nerve ganglia**.
- Peripheral processes connect to sensory receptors; central processes enter the CNS.

Structure:

- Most sensory neurons are **pseudounipolar**, meaning a single process emerges from the cell body and **branches into two**:
 1. One branch extends toward the sensory receptor.
 2. The other branch enters the CNS.
- Some sensory neurons (in special senses like vision or olfaction) are **bipolar**, with one dendrite and one axon.
- Many axons are **myelinated**, which accelerates signal transmission.

Function in Transmission:

- Convert sensory stimuli into **electrical impulses**.
- Transmit impulses from the receptor to the CNS.
- Serve as the **first neuron** in the reflex arc or sensory pathway.

Feature	Sensory Neuron
Structure	Pseudounipolar (mostly), some bipolar
Function	Transmit sensory information to CNS
Location	Dorsal root ganglia, cranial nerve ganglia
Role	First neuron in nervous transmission

2. Motor Neurons (Efferent Neurons)

Role in Nervous Transmission:

Motor neurons carry commands **from the CNS to peripheral effectors**, such as muscles or glands, enabling responses to stimuli detected by sensory neurons.

Location:

- **Cell bodies:** CNS (spinal cord or brain).
- **Axons:** Extend to muscles or glands.
- **Upper motor neurons:** originate in the cerebral cortex.
- **Lower motor neurons:** reside in spinal cord or brainstem.

Structure:

- Typically multipolar neurons, with one long axon and multiple dendrites.
- Axons are often myelinated for rapid impulse conduction.
- Dendrites integrate inputs from interneurons and sensory neurons.

Function in Transmission:

- Transmit impulses from CNS to effectors.
- Control voluntary movements (skeletal muscles) and involuntary responses (autonomic motor neurons).
- Serve as the third neuron in a reflex arc.

Feature	Motor Neuron
Structure	Multipolar
Function	Transmit commands from CNS to effectors
Location	CNS → muscles or glands
Role	Last neuron in nervous transmission

3. Interneurons (Relay or Association Neurons)**Role in Nervous Transmission:**

Interneurons act as **connectors** between sensory and motor neurons, integrating and processing information within the CNS. They are essential for **reflex arcs, learning, memory, and decision-making**.

Location:

- Entirely within the **CNS** (brain and spinal cord).

Structure:

- Multipolar neurons with highly branched dendrites.
- Short or moderately long axons, often unmyelinated.
- Form extensive synaptic connections with multiple neurons.

Function in Transmission:

- Relay impulses from sensory neurons to motor neurons.
- Integrate information from multiple sources.
- Participate in complex processing such as reflex coordination, learning, and memory.

Additional Classification:

- **Local interneurons:** Short axons; process information locally.
- **Relay interneurons:** Long axons; connect distant neural circuits.

Feature	Interneuron
Structure	Multipolar, highly branched dendrites
Function	Integrate and relay signals between neurons
Location	Brain and spinal cord
Role	Middle neuron in nervous transmission

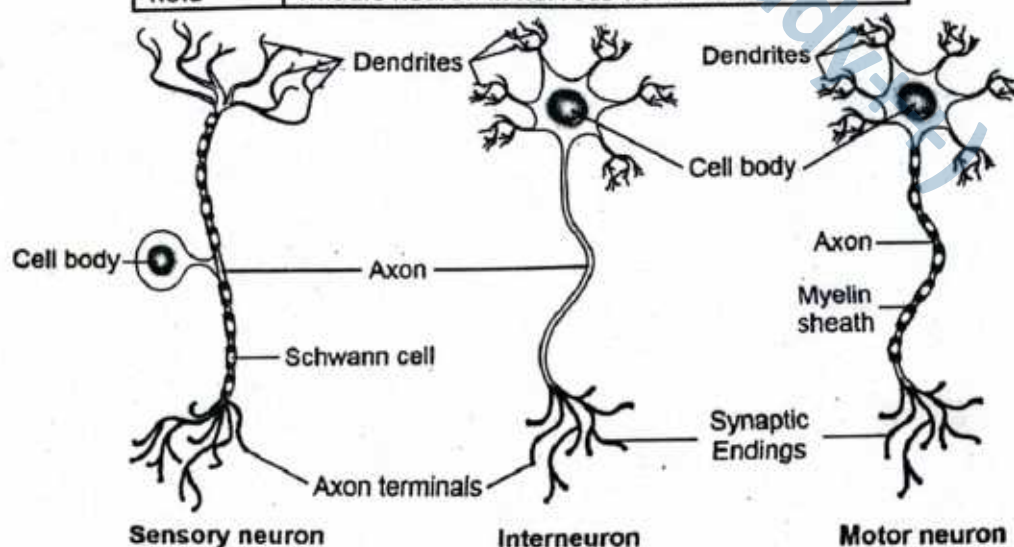


Figure: Types of neuron based on function

► Relationship Between the Three Neurons in Nervous Transmission

The **typical reflex pathway** illustrates the collaboration of these three neurons:

1. **Sensory neuron** detects stimulus → transmits impulse to CNS.
2. **Interneuron** receives sensory input → integrates and relays signal.
3. **Motor neuron** receives processed signal → activates effector to produce response.

This **three-neuron model** ensures rapid and coordinated reactions to stimuli.

Step	Neuron Type	Function
1	Sensory	Detect stimulus, transmit to CNS
2	Interneuron	Integrate signal, relay to motor neuron
3	Motor	Transmit command to effector organ

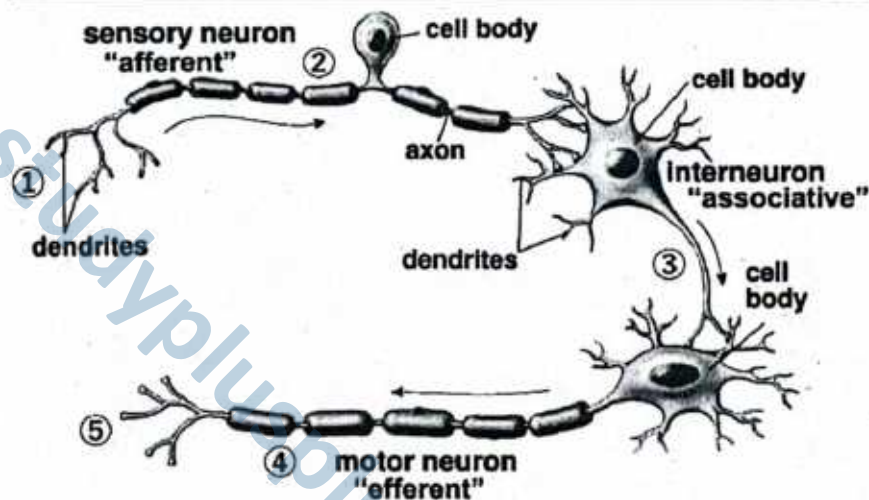


Figure: Relationship between three types of neuron

SLO [B-12-G-04]

Identify muscles and glands as the effectors.

MUSCLES AND GLANDS AS EFFECTORS

The nervous system not only detects and processes information but also produces a **response** to maintain balance and ensure survival. This response is carried out by **effectors** — specialized structures that bring about an action when stimulated by nerve impulses. In humans, the two main effectors are **muscles** and **glands**.

What are Effectors?

Effectors are organs or tissues that receive commands from the central nervous system through motor neurons and then carry out the required response. They serve as the final link in the pathway of information flow, converting nerve impulses into visible actions such as movement, secretion, or regulation of body functions.

Muscles as Effectors

Muscles serve as the **mechanical effectors** of the nervous system. When stimulated by motor neurons, they convert electrical impulses into mechanical action by contracting and generating force. This allows the body to move, maintain posture, circulate blood, and regulate essential internal processes. Muscles are essential not only for voluntary actions like walking or speaking but also for involuntary functions such as breathing, digestion, and heartbeat.

► Types of Muscles as Effectors

1. Skeletal Muscles

- **Structure and Control:** Skeletal muscles are attached to bones via tendons and are under voluntary control through the **somatic nervous system**.



- **Function:** They are responsible for conscious actions such as walking, writing, speaking, and maintaining body posture. They respond quickly but can fatigue easily.
- **Example as Effector:** When you accidentally touch a hot surface, the sensory message is relayed to the spinal cord, and motor neurons activate skeletal muscles in your arm to withdraw the hand immediately (reflex action).
- **Application/Impact:** Skeletal muscles allow humans to interact with their environment. Disorders such as muscular dystrophy weaken skeletal muscles, showing their vital role in everyday life.

2. Smooth Muscles

- **Structure and Control:** Smooth muscles are found in the walls of internal organs such as the stomach, intestines, urinary bladder, and blood vessels. They are **involuntary muscles** controlled by the **autonomic nervous system**.
- **Function:** They carry out slow, sustained actions like peristalsis in the digestive tract, constriction and dilation of blood vessels, and regulation of airways in the lungs.
- **Example as Effector:** When blood pressure falls, smooth muscles in the blood vessels constrict to restore normal circulation.
- **Application/Impact:** Smooth muscle control is essential for homeostasis. Overconstriction of airway smooth muscles, as in asthma, shows how malfunction can endanger life.

3. Cardiac Muscle

- **Structure and Control:** Found only in the heart, cardiac muscle fibers are striated but involuntary. They have **intrinsic pacemaker activity** and are also regulated by the **autonomic nervous system**.
- **Function:** Cardiac muscle contracts rhythmically and continuously throughout life, pumping blood to every part of the body. It is fatigue-resistant, ensuring uninterrupted circulation.
- **Example as Effector:** During exercise, sympathetic nerves stimulate cardiac muscle to increase heart rate and stroke volume, delivering more oxygen and nutrients to muscles.
- **Application/Impact:** Cardiac muscle is critical for survival. Conditions like arrhythmias or heart failure demonstrate the importance of precise nervous and hormonal regulation of this effector.

► Integration of Muscles as Effectors

Muscles act as the **final executors of motor commands** from the nervous system. For instance, in the simple act of running:

- **Skeletal muscles** move the limbs,
- **Smooth muscles** in blood vessels adjust blood flow, and
- **Cardiac muscle** increases output to meet oxygen demand.

This shows how all three muscle types coordinate as effectors to carry out complex body responses.

Glands as Effectors

Glands are specialized structures in the body that produce and secrete substances essential for regulation, coordination, and survival. Unlike muscles, which bring about movement, glands act as **chemical effectors**. They release their secretions either into ducts or directly into the bloodstream, influencing processes such as digestion, temperature regulation, stress response, metabolism, and reproduction. Glands may be stimulated by **nerve impulses** (neurogenic control) or by **chemical messengers** (hormonal feedback).

1. Exocrine Glands

Exocrine glands discharge their secretions into ducts that open either onto the body surface or into body cavities. These secretions perform **local and immediate actions**.

- **Examples and Roles:**
 - **Salivary glands:** Secrete saliva into the oral cavity, moistening food and initiating starch digestion through the enzyme amylase.
 - **Sweat glands:** Release sweat onto the skin surface, playing a vital role in cooling the body during exercise or heat stress.
 - **Gastric glands:** Secrete digestive juices containing hydrochloric acid and pepsin, aiding in protein digestion.
 - **Sebaceous glands:** Secrete sebum, an oily substance that lubricates and protects the skin and hair.
- **Application and Impact:** Exocrine gland function is crucial for everyday life. For example, **reduced saliva secretion** (xerostomia) can lead to difficulty in chewing and swallowing, while **blocked sweat glands** can cause overheating of the body.



2. Endocrine Glands

Endocrine glands are **ductless glands** that secrete hormones directly into the bloodstream. Unlike exocrine secretions, hormones act on **distant target organs** and regulate long-term and wide-ranging processes.

• Examples and Roles:

- **Adrenal glands:** Secrete adrenaline (epinephrine) during stress, preparing the body for “fight or flight” by increasing heart rate, blood pressure, and glucose availability.
- **Pancreas (Islets of Langerhans):** Secretes insulin and glucagon, hormones essential for regulating blood glucose levels.
- **Thyroid gland:** Produces thyroxine, which regulates metabolism, growth, and energy balance.
- **Pituitary gland:** Known as the “master gland,” it secretes hormones that control other endocrine glands and regulate growth, reproduction, and water balance.

- **Application and Impact:** Endocrine control is vital for maintaining **homeostasis**. For instance, a deficiency of insulin leads to **diabetes mellitus**, while overproduction of thyroxine causes **hyperthyroidism**. Medical treatment often involves replacing or regulating these hormones to restore balance.

Together, exocrine and endocrine glands ensure the body's adaptation to both internal and external demands. For example, when running on a hot day, **sweat glands** (exocrine) release sweat to cool the body, while **adrenal glands** (endocrine) secrete adrenaline to increase stamina and prepare muscles for exertion. This shows how glands function as **chemical executors of the nervous system's commands**.

► Real-Life Applications and Impact

- **Muscle Effectors in Reflexes:** When stepping on a sharp object, receptors in the skin detect pain, sensory neurons carry the message to the spinal cord, and motor neurons activate leg muscles to withdraw the foot instantly.
- **Gland Effectors in Stress:** When facing danger, the hypothalamus signals the adrenal glands to release adrenaline. This causes increased heart rate, faster breathing, and energy mobilization — preparing the body for “fight or flight.”
- **Medical Relevance:** Diseases such as myasthenia gravis (affecting skeletal muscle response) or diabetes mellitus (due to insulin deficiency from pancreatic glands) show the critical role of effectors in maintaining normal physiology.
- **Everyday Example:** Sweating while running on a hot day involves both muscles (increased heart and breathing rates) and glands (sweat secretion) working together to protect the body.

Effectors are the organs that **execute the commands of the nervous system**. Muscles respond by contracting and causing movement, while glands respond by secreting essential substances. Together, they ensure that the body can interact with the environment, maintain internal balance, and survive under varying conditions.

SLO [B-12-G-05]

Annotate the detailed structure of a sensory neuron, associative and motor neuron.

DETAILED STRUCTURE OF SENSORY, ASSOCIATIVE AND MOTOR NEURONS

Although neurons vary in their size and shape, they all share the same fundamental plan: a cell body (soma) with a nucleus and Nissl bodies, dendrites that receive information, and an axon that conducts impulses to another cell. Within this general plan, sensory neurons, associative (interneurons), and motor neurons each show distinct structural features that enable them to perform their specialized functions. The following sections describe their detailed structures, as required for annotation and examination.

Sensory Neurons (Afferent Neurons)

Sensory neurons are responsible for transmitting information from receptors in the skin, muscles, joints, and internal organs to the central nervous system (CNS). Most vertebrate sensory neurons are **pseudounipolar**, a design well suited for rapid conduction.

The **cell body** of a sensory neuron is located in the **dorsal root ganglion** of spinal nerves or in the sensory ganglia of cranial nerves. This soma is typically round, with a large nucleus and a prominent nucleolus, and is encircled by **satellite glial cells**. A key feature of these ganglion cells is that they do not receive synapses directly on their soma.



- From the cell body arises a single process, which quickly bifurcates in a T-shaped fashion into two branches:
- the **peripheral process**, which extends to a sensory receptor such as a free nerve ending, a pressure corpuscle, or a muscle spindle,
 - and the **central process**, which enters the spinal cord through the dorsal root and terminates in the dorsal horn.

Both branches are structurally axons and are usually **myelinated**, with **nodes of Ranvier** at regular intervals to allow saltatory conduction. This arrangement permits impulses to travel rapidly and with minimal delay from the receptor to the CNS.

Thus, sensory neurons are highly specialized for **uninterrupted transmission of impulses** from the body's periphery to the spinal cord and brain.

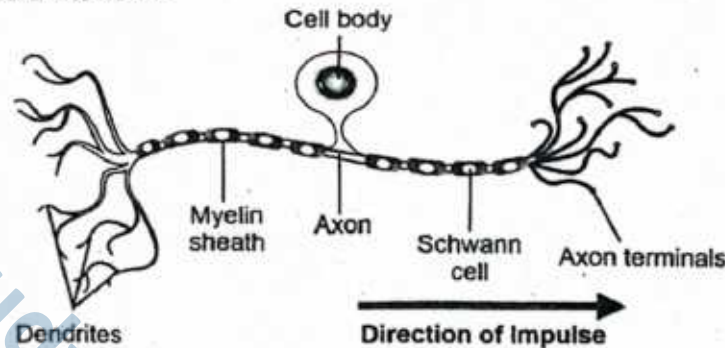


Figure: Sensory neuron

Associative Neurons (Interneurons)

Interneurons, also called **relay neurons** or **association neurons**, are located entirely within the CNS. They form the largest class of neurons and are essential for the integration, processing, and relay of information between sensory and motor pathways.

Structurally, interneurons are usually **multipolar**, with a relatively small cell body containing abundant Nissl substance. From the soma, **multiple dendrites** arise, often bearing dendritic spines that increase synaptic contact. These dendrites receive signals from sensory neurons or other interneurons. The **axon** of an interneuron is typically short and may be **unmyelinated** or only thinly myelinated. It gives off numerous **collaterals** that form synapses locally within the grey matter of the brain or spinal cord.

This dense network of dendrites and short axons enables interneurons to establish thousands of synaptic connections. Their structural plan perfectly matches their role in **reflex arcs**, **local circuit activity**, and **higher brain functions** such as memory, learning, and decision-making.

In summary, interneurons are designed for **integration and coordination** rather than long-distance transmission.

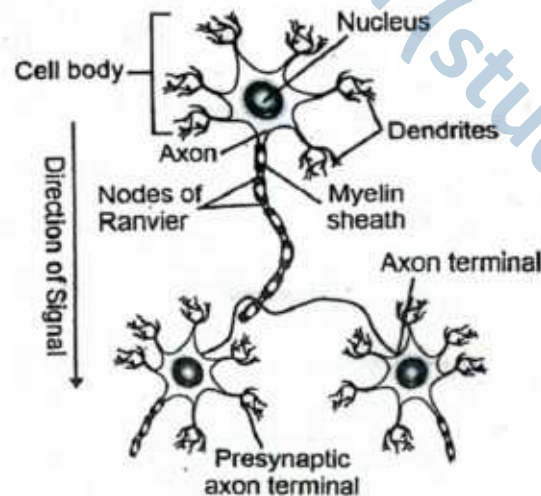


Figure: Interneuron

Motor Neurons (Efferent Neurons)

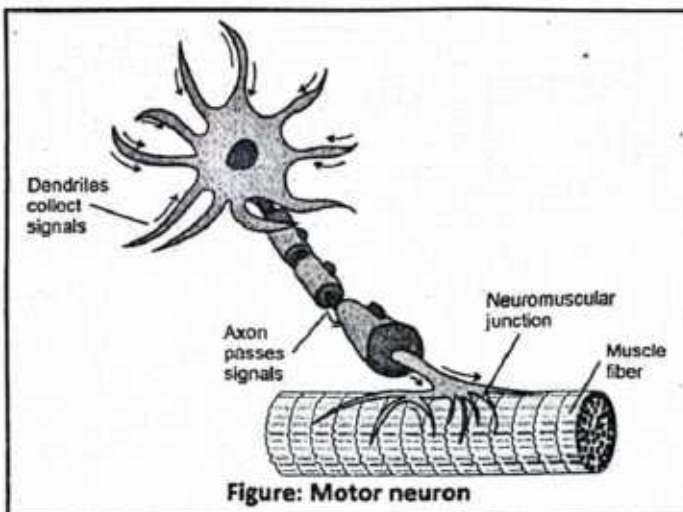
Motor neurons carry impulses from the CNS to effector organs, such as muscles and glands, thereby converting neural activity into action. The best-studied example is the **lower motor neuron** whose cell body lies in the **anterior (ventral) horn of the spinal cord**.

Motor neurons are characteristically **multipolar**, with a large polygonal cell body, a prominent central nucleus, and coarse **Nissl bodies** in the cytoplasm. From the soma emerge numerous dendrites, which form an extensive dendritic tree capable of receiving input from thousands of synapses.

At one side of the soma lies the **axon hillock**, a Nissl-free zone that gives rise to the **axon initial segment**, the site where action potentials are generated. The axon is typically very long and **thickly myelinated**, ensuring rapid conduction over considerable distances. Along its course, the axon exhibits **nodes of Ranvier**, where ionic exchange occurs during saltatory conduction.

The axon eventually branches into **telodendria**, each ending in a **synaptic bouton** at a **neuromuscular junction (NMJ)**. Here, the motor neuron releases acetylcholine into the synaptic cleft, initiating contraction of the skeletal muscle fiber.

This structural plan reflects the motor neuron's essential role: to **transmit fast, precise commands from the CNS to muscles or glands**, thereby enabling both voluntary and involuntary responses.



Structural Types of Neurons

Neuron Type	Processes	Function	Example/Location
Unipolar	One process	Sensory (in invertebrates)	Invertebrate nervous systems
Pseudounipolar	One bifurcated process	General sensory input	Dorsal root ganglia, cranial nerve ganglia
Bipolar	One axon, one dendrite	Special sensory input	Retina, olfactory mucosa, inner ear
Multipolar	One axon, multiple dendrites	Motor and integrative functions	Brain, spinal cord, motor and association pathways

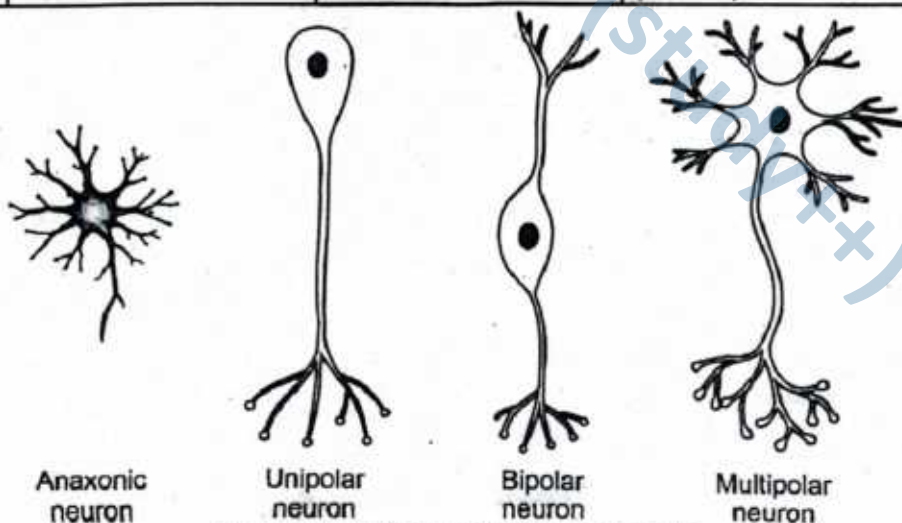


Figure: Types of neuron based on structure

► Summary

- **Sensory neurons** are pseudounipolar, with their soma in dorsal root ganglia and processes specialized for rapid transmission of receptor input to the CNS.

- **Interneurons** are multipolar, located entirely within the CNS, and structurally adapted for integration through their short axons and highly branched dendrites.
 - **Motor neurons** are large multipolar neurons in the anterior horn of the spinal cord, with long myelinated axons that terminate in neuromuscular junctions, enabling effectors to respond to CNS commands.
- These structural differences clearly illustrate how the **form of each neuron is adapted to its function** within the nervous system.

Comparative Structure of Sensory, Associative, and Motor Neurons

Feature	Sensory Neuron (Afferent, pseudounipolar)	Associative Neuron (Interneuron, multipolar)	Motor Neuron (Efferent, multipolar)
Location of Cell Body	Dorsal root ganglia of spinal nerves or cranial sensory ganglia	Entirely within CNS (grey matter of brain and spinal cord)	Anterior (ventral) horn of spinal cord or cranial motor nuclei
Shape / Type	Pseudounipolar – single process bifurcating into peripheral and central branches	Multipolar – many dendrites and one short axon	Multipolar – many dendrites and one long axon
Dendrites	Absent on soma (peripheral branch functions as input)	Numerous, branched, often with spines	Numerous, extensively branched (for wide synaptic input)
Axon	Central branch enters CNS; usually myelinated with nodes of Ranvier	Short, often unmyelinated or thinly myelinated; local collaterals	Long, thick, myelinated; may give collaterals
Myelination	Present (Schwann cells in PNS); rapid conduction	Often absent or thin; suited for local processing	Prominent (Schwann cells in PNS); ensures fast saltatory conduction
Specialized Termination	Peripheral branch ends in sensory receptors; central branch synapses in dorsal horn of spinal cord/brainstem	Axon terminals synapse with other interneurons or motor neurons within CNS	Axon terminals form neuromuscular junctions with skeletal muscle fibers (or synapse on glands)
Primary Function	Conduct impulses from receptors to CNS	Integrate and relay information between sensory and motor neurons	Transmit impulses from CNS to effectors (muscles, glands)
Structural Adaptation	T-shaped bifurcation minimizes synaptic delay; soma insulated by satellite cells	Dense dendritic arborization allows extensive synaptic integration	Large soma with abundant Nissl substance for high protein demand; long axon for distant muscle innervation

SLO [B-12-G-06]

Relate the structure of neurons with functions.

RELATING THE STRUCTURE OF NEURONS WITH THEIR FUNCTIONS

Neurons are specialized cells whose structures are perfectly adapted to their functions in the nervous system. Each part of a neuron—cell body, dendrites, axon, and synaptic terminals—plays a *distinct role* in receiving, processing, and transmitting information. The efficiency of the nervous system depends on this precise relationship between structure and function.



Cell Body (Soma or Perikaryon)

The **cell body** serves as the **metabolic and synthetic center** of the neuron. It contains a large **nucleus** with a prominent **nucleolus**, indicating high levels of gene activity. The cytoplasm, also called **neuropasm**, is rich in **Nissl bodies** (aggregations of rough endoplasmic reticulum and ribosomes), which produce structural proteins and **neurotransmitters** required for signal transmission.

Other organelles, such as numerous **mitochondria**, provide the ATP needed for ion pumping, axonal transport, and synaptic activity. The **Golgi apparatus** packages proteins into vesicles for delivery to axon terminals. A robust **cytoskeleton** (microtubules, actin filaments, and neurofilaments) maintains the shape of the cell and forms tracks for **axonal transport**. Together, these features ensure that the soma supports the enormous metabolic demands of the neuron and sustains long-distance communication.

Dendrites: The Receiving Structures

Dendrites are short, tapering, and highly branched extensions that provide a **large surface area** for synaptic input. Many dendrites bear **dendritic spines**, which are small projections packed with **postsynaptic receptors** and signaling molecules. These spines are sites of **synaptic plasticity**, meaning their structure changes during processes like learning and memory.

Dendrites act as the **receptive region** of the neuron. They receive signals from thousands of presynaptic terminals, and their branching pattern allows **spatial and temporal summation** of incoming impulses. In this way, dendrites filter and integrate information before it reaches the cell body and axon hillock.

Axon: The Conducting Structure

Each neuron has a single **axon**, which conducts impulses **away from the cell body**. The axon arises from a specialized region called the **axon hillock**, and its adjacent **initial segment** is rich in **voltage-gated sodium channels**, making it the **trigger zone** for action potential initiation.

The **diameter of the axon** strongly influences conduction speed; larger diameters reduce resistance and conduct impulses faster. Within the axon, bundles of **microtubules** and **neurofilaments** serve as tracks for **anterograde transport** (moving vesicles and organelles from the soma to terminals via kinesin) and **retrograde transport** (carrying signals and waste back to the soma via dynein).

Myelination and Saltatory Conduction

Many axons are covered by a myelin sheath, formed by Schwann cells in the **peripheral nervous system (PNS)** or by oligodendrocytes in the **central nervous system (CNS)**. Myelin acts as an insulating layer, interrupted at intervals by **Nodes of Ranvier**, where the axonal membrane is exposed.

This arrangement allows **saltatory conduction**, in which action potentials “jump” from one node to the next, greatly increasing the speed and energy efficiency of impulse transmission. Myelinated axons make up the **white matter** of the nervous system, while non-myelinated axons conduct impulses more slowly by continuous conduction and are common in the autonomic system.

Axon Terminals (Synaptic Boutons)

At its end, the axon branches into numerous **axon terminals** (or synaptic boutons). Each terminal contains **synaptic vesicles**, **mitochondria**, and **specialized proteins** for neurotransmitter release. When an **action potential** arrives, voltage-gated **calcium channels** open, triggering the fusion of vesicles with the presynaptic membrane. Neurotransmitters are then released into the **synaptic cleft**, allowing the conversion of an **electrical signal** into a chemical signal.

Synapses: The Communication Points

Neurons communicate at specialized junctions called **synapses**. Most are **chemical synapses**, which consist of a presynaptic terminal, a **synaptic cleft**, and a postsynaptic membrane rich in receptors. These synapses allow **unidirectional, modifiable, and amplifiable** transmission, essential for processes such as learning and memory.

In contrast, **electrical synapses** (formed by **gap junctions**) allow **direct cytoplasmic continuity** between neurons. They enable **fast and often bidirectional** transmission, useful for synchronization of neuronal activity.



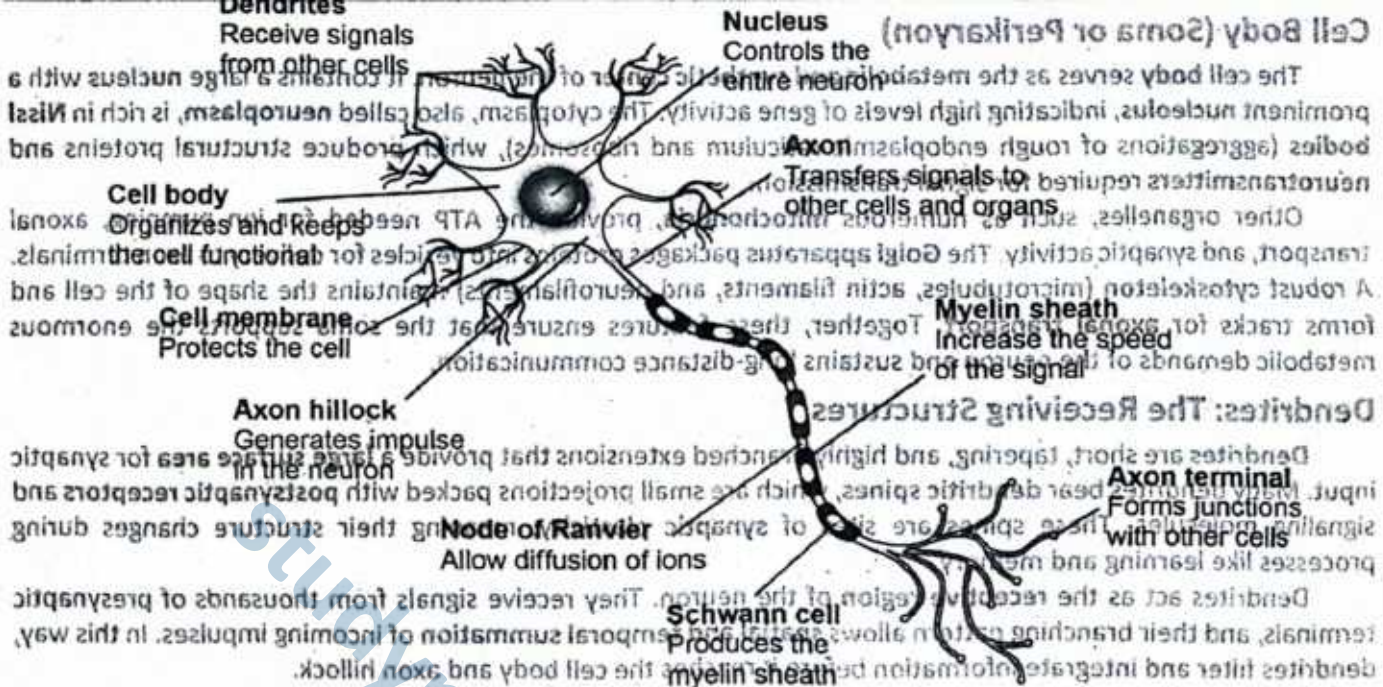


Figure: Parts of a neuron with functions

The relationship between structure and function in neurons is direct and essential. The soma sustains life and produces proteins, the dendrites receive and integrate inputs, the axon conducts impulses rapidly (especially if myelinated), and the terminals convert electrical signals into chemical communication. Variations in neuronal shape, such as pseudounipolar sensory neurons or multipolar motor neurons, show how form supports specialized roles. Ultimately, the efficiency and adaptability of the nervous system depend on this intimate link between neuronal structure and function.

Neuron Structure	Key Features	Main Function
Cell Body (Soma)	Contains nucleus, Nissl bodies, mitochondria, Golgi apparatus, cytoskeleton	Metabolic center; produces proteins and neurotransmitters; maintains cell life
Dendrites	Highly branched; may have dendritic spines	Receive signals from other neurons; provide large surface for synaptic input
Axon	Single long process; arises from axon hillock	Conducts nerve impulse away from soma
Axon Hillock & Initial Segment	High density of voltage-gated Na^+ channels	Trigger zone where action potential is initiated
Myelin Sheath	Formed by Schwann cells (PNS) or oligodendrocytes (CNS); interrupted by Nodes of Ranvier	Provides insulation; enables saltatory conduction for fast transmission
Axon Terminals (Synaptic Boutons)	Release neurotransmitters into synaptic cleft; contain synaptic vesicles and mitochondria	Release neurotransmitters into synaptic cleft
Synapse	Includes presynaptic terminal, cleft, postsynaptic membrane	Allows communication between neurons or with muscles/glands
Sensory Neuron	Usually pseudounipolar	Conducts impulses from receptors to CNS
Motor Neuron	Multipolar; long axon, many dendrites	Carries impulses from CNS to effectors
Interneuron	Short, multipolar, highly branched	Integrates signals, coordinates reflexes, supports higher processes
Myelinated Axon	Appears white; has myelin	Fast, efficient impulse conduction
Non-Myelinated Axon	Appears gray; lacks myelin	Slow conduction; common in autonomic system



Differentiate between myelinated and non-myelinated neurons.

Non-myelinated axons	Myelinated axons	Feature
(no myelin sheath)	(has myelin sheath)	Conduction speed
Lower (continuous ion exchange)	High (fewer ion fluxes)	Energy efficiency

Neurons transmit information through electrical impulses, and the speed of this conduction depends on whether the axons are **myelinated** or **non-myelinated**. Myelin is a multilayered lipid-rich sheath produced by specialized glial cells that provides insulation and enhances the efficiency of impulse conduction. The presence or absence of this sheath creates distinct structural and functional differences between these two types of neurons.

Structure of Myelinated and Non-Myelinated Axons

Myelinated axons are covered by a compact **myelin sheath**, which is formed by **Schwann cells** in the peripheral nervous system (PNS) and **oligodendrocytes** in the central nervous system (CNS). These cells wrap around the axon in multiple layers, leaving small unmyelinated gaps called **nodes of Ranvier**. These nodes are rich in voltage-gated sodium channels and play a vital role in nerve impulse propagation.

In contrast, **non-myelinated axons** lack a compact myelin sheath. Instead, small axons may be loosely surrounded by Schwann cells without forming tightly wrapped layers. These axons do not have nodes of Ranvier, and their conduction depends on depolarization of the entire axonal membrane.

Mechanism of Impulse Conduction

In myelinated axons, impulses travel by **saltatory conduction**. This means the action potential appears to "jump" from one node of Ranvier to the next. Because the myelin sheath prevents ion exchange along the internodal regions, depolarization is confined to the nodes. This results in **rapid conduction velocity** and greater **energy efficiency**, as fewer ions need to be restored by sodium-potassium pumps.

In non-myelinated axons, conduction occurs by **continuous conduction**, where the action potential is regenerated at every part of the axonal membrane. This process is much **slower** and **requires more energy**, since ion exchange happens along the entire length of the axon.

Functional Implications

Myelinated neurons conduct impulses at very high speeds, ranging from **tens to over 100 m/s** in large motor fibres. They are essential for rapid reflexes, voluntary muscle control, and fast sensory transmission such as touch and proprioception. The myelin sheath also reduces the metabolic cost of conduction by limiting ion exchange to nodes.

On the other hand, **non-myelinated neurons** conduct impulses much more slowly, at about **0.5 to 2 m/s**. They are typically involved in functions where rapid transmission is not essential, such as carrying signals for slow pain, temperature changes, and autonomic regulation.

Clinical Relevance

The importance of myelin is highlighted by disorders that damage it. In the CNS, **Multiple Sclerosis (MS)** causes demyelination and results in slowed or blocked conduction, leading to sensory and motor impairments. In the PNS, conditions such as **Guillain-Barré Syndrome (GBS)** involve acute inflammatory demyelination, producing weakness and altered sensation. Diagnostic nerve conduction studies show markedly **slowed conduction velocity** in demyelinating disorders compared to normal myelinated fibres.

► Key Differences Between Myelinated and Non-Myelinated Neurons

- **Myelin sheath:** Present in myelinated axons, absent in non-myelinated axons.
- **Nodes of Ranvier:** Regularly spaced in myelinated axons, absent in non-myelinated axons.
- **Conduction:** Saltatory in myelinated axons, continuous in non-myelinated axons.
- **Speed:** Myelinated axons conduct much faster (up to 120 m/s) than non-myelinated ones (0.5-2 m/s).
- **Energy efficiency:** Myelinated axons are more energy-efficient because fewer ions cross the membrane.
- **Function:** Myelinated neurons are responsible for rapid signalling (motor, proprioceptive, fast sensory), whereas non-myelinated neurons handle slower signals (pain, temperature, autonomic regulation).

The structural adaptations of myelinated and non-myelinated axons directly determine their functional roles in the nervous system. Myelination enables rapid and efficient conduction suited for quick responses and complex coordination, while non-myelinated axons provide slower conduction suitable for sustained and

regulatory functions. Understanding these differences is not only essential for learning neuronal physiology but also for appreciating the basis of many neurological diseases that result from damage to the myelin sheath.

Myelinated vs Non-myelinated Axons

Feature	Myelinated axons	Non-myelinated axons
Conduction speed	Fast (saltatory conduction)	Slow (continuous conduction)
Energy efficiency	High (fewer ion fluxes)	Lower (continuous ion exchange)
Typical locations	White matter, motor pathways	Autonomic fibres, many pain fibres
Diameter	Usually larger	Usually smaller
Nodes of Ranvier	Present, regularly spaced	Absent or sparse
Clinical vulnerability	Demyelination → conduction block	Axon loss affects function directly

SLOs Based (MCQs)

FUNCTION AND STRUCTURE BASED TYPES OF NEURONS

- What is the primary function of a neuron?
 - Generating energy for muscle contraction
 - Producing hormones for metabolic control
 - Receiving, processing, and transmitting information
 - Storing calcium for neurotransmission
- Which structure of a neuron contains Nissl bodies?
 - Axon
 - Dendrite
 - Cell body
 - Axon terminal
- Which part of the neuron receives synaptic input from other neurons?
 - Axon terminal
 - Axon hillock
 - Dendrites
 - Node of Ranvier
- Which type of neuron is typically pseudo-unipolar in structure?
 - Motor neuron
 - Sensory neuron
 - Interneuron
 - Bipolar neuron
- What type of neurons are located entirely within the CNS and serve to integrate information?
 - Motor neurons
 - Interneurons
 - Sensory neurons
 - Bipolar neurons
- Where are the cell bodies of motor neurons generally located?
 - In muscles
 - In dorsal root ganglia
 - In sensory organs
 - In the brain or spinal cord
- Which of the following is a key function of multipolar neurons?
 - Transmitting signals from sensory receptors
 - Stimulating endocrine glands only
 - Integrating and relaying neural information
 - Receiving signals from motor neurons
- What type of neuron is most commonly found in sensory pathways like the retina and olfactory epithelium?
 - Multipolar
 - Bipolar
 - Unipolar
 - Pseudounipolar
- Which part of a motor neuron transmits impulses to effector organs?
 - Axon terminal
 - Nissl bodies
 - Dendrites
 - Soma
- What are muscles and glands collectively referred to in the nervous system?
 - Neurons
 - Transducers
 - Effectors
 - Modulators

Answer Key

1. C 2. C 3. C 4. B 5. B 6. D 7. C 8. B 9. A 10. C

Short Questions

Q.1 What is a neuron and what is its primary role in the nervous system?

Ans. A neuron is a highly specialized cell that serves as the structural and functional unit of the nervous system. Unlike most other cells, neurons are excitable and can generate electrical impulses called action potentials. Each neuron has the unique ability to receive, integrate, and transmit information in the form of electrical and chemical signals. Neurons form complex communication networks that coordinate voluntary actions, such as muscle movement, and involuntary processes, such as breathing and heartbeat. Their primary role is to ensure rapid communication between different parts of

MARKING SCHEME (3 Marks)

- Definition (1), function (2).

the body and the central nervous system, thereby maintaining coordination, regulation, and integration of physiological processes. Without neurons, higher functions such as memory, learning, reflexes, and sensation would not be possible.

Q.2 Briefly describe the structure and function of dendrites.

Ans. Dendrites are short, branched extensions that project outward from the cell body of a neuron. Their structure is tree-like, which increases the overall surface area available to form synaptic connections with neighboring neurons. These numerous branches allow a single neuron to receive inputs from multiple presynaptic cells simultaneously. Functionally, dendrites act as the main receptive regions of the neuron. They collect incoming chemical signals released by neurotransmitters at synapses, convert them into small electrical changes, and conduct these impulses toward the cell body for processing. In this way, dendrites play a critical role in integrating synaptic inputs, determining whether a neuron will reach the threshold to fire an action potential, and ensuring effective communication within neural circuits.

MARKING SCHEME (3 Marks)

- Structure (1), function (2).

Q.3 What is the role of the axon in a neuron?

Ans. The axon is a long, slender projection that extends from the neuron's cell body, often several millimeters to over a meter in length. Structurally, it is designed to carry electrical impulses away from the cell body efficiently. Many axons are covered with a myelin sheath, which insulates the fiber and enables faster transmission of action potentials by saltatory conduction. The axon terminates in multiple axon terminals or synaptic boutons, where neurotransmitters are released into the synaptic cleft. Functionally, the axon ensures that the electrical signal generated in the neuron is delivered to the appropriate target cell, such as another neuron, a muscle fiber, or a gland. This transmission allows for communication, coordination of movements, and regulation of vital body functions.

MARKING SCHEME (3 Marks)

- Structure (1), function (2).

Q.4 What are sensory neurons and where are they located?

Ans. Sensory neurons, also called afferent neurons, are specialized nerve cells that detect changes in the environment and transmit this information to the central nervous system for interpretation. They are responsible for perceiving stimuli such as light, sound, temperature, touch, pressure, and chemical changes. Structurally, most sensory neurons are pseudounipolar, meaning they have a single process that splits into two branches—one connecting to the periphery and the other to the spinal cord. Their cell bodies are located in dorsal root ganglia of the spinal nerves or in cranial nerve ganglia. Functionally, sensory neurons form the first step in sensory pathways by converting external or internal stimuli into nerve impulses, thereby allowing the brain and spinal cord to generate appropriate responses.

MARKING SCHEME (3 Marks)

- Definition (2), location (1).

Q.5 Describe the function and location of motor neurons.

Ans. Motor neurons, also known as efferent neurons, are responsible for carrying signals from the central nervous system (CNS) to effectors such as skeletal muscles, smooth muscles, and glands. Their primary function is to initiate a response—such as muscle contraction, glandular secretion, or organ regulation—based on commands from the brain and spinal cord. Structurally, they are multipolar neurons, having a single long axon and multiple dendrites that allow them to integrate signals and coordinate precise outputs. The cell bodies of motor neurons are mainly located in the anterior horn of the spinal cord and certain cranial nerve nuclei in the brainstem. Their axons extend outward to synapse directly with effector tissues, thus serving as the final pathway for motor control.

MARKING SCHEME (3 Marks)

- Definition (2), location (1).

Q.6 What is the role of interneurons in neural communication?

Ans. Interneurons, also called relay neurons or association neurons, form the essential link between sensory and motor neurons within the central nervous system. Unlike sensory neurons, which bring in external information, and motor neurons, which send signals outward, interneurons integrate, process, and interpret the incoming sensory data before passing on commands. Structurally, most interneurons are multipolar, enabling them to form complex networks with thousands of

MARKING SCHEME (3 Marks)

- Unipolar characteristics (1.5), pseudounipolar characteristics (1.5).



to synapses. Functionally, they play a key role in reflex actions, by directly connecting sensory input to motor output, allowing rapid, automatic responses without conscious thought. In higher brain regions, interneurons are also critical for memory, learning, reasoning, and decision-making, as they facilitate communication between different neural circuits. This makes them central to both basic reflexes and advanced cognitive functions.

Q.7 Differentiate between unipolar and pseudounipolar neurons.

Ans. Unipolar neurons possess a single process that extends from the cell body, which often branches into dendritic and axonal ends. These neurons are commonly found in invertebrates and represent a simple structural type specialized for basic signal transmission. In contrast,

MARKING SCHEME (3 Marks)

- Definition (1), function (2).

pseudounipolar neurons—found in vertebrates—appear to have a single process that quickly bifurcates into two branches: one extending toward the periphery to receive sensory input and the other directed toward the spinal cord or brainstem to deliver impulses. This design allows rapid and efficient transmission of sensory signals, particularly in touch, pain, and proprioception pathways. While unipolar neurons serve more primitive nervous systems, pseudounipolar neurons are adapted for complex vertebrate sensory processing, ensuring quick relay of information from the body to the CNS.

Q.8 What structural type of neuron is most abundant in the human nervous system? Why?

Ans. The most abundant structural type of neuron in the human nervous system is the multipolar neuron. These neurons are characterized by a single long axon and multiple dendrites branching from the cell body.

MARKING SCHEME (3 Marks)

- Name of neuron type (1), reason (2).

This structure provides a large surface area for synaptic input, allowing multipolar neurons to integrate signals from many different sources simultaneously. They are especially abundant in the cerebral cortex and spinal cord, where they serve as motor neurons, interneurons, and projection neurons. The reason for their abundance is that the complexity of human behavior and thought requires neurons capable of handling multiple inputs and coordinating sophisticated responses. Their structural design makes them ideal for higher-order processing, motor coordination, and communication across extensive neural networks.

Q.9 Why are muscles and glands referred to as effectors in the nervous system?

Ans. Muscles and glands are referred to as effectors because they are the structures that carry out the final responses to neural commands. When stimulated by motor neurons, muscles contract and generate movement, enabling locomotion, posture maintenance, and other physical activities. Similarly, glands act as effectors by secreting

MARKING SCHEME (3 Marks)

- Muscles as effectors (1.5), glands as effectors (1.5).

chemical substances such as enzymes, sweat, saliva, or hormones in response to neural impulses. For example, salivary glands secrete saliva when triggered by parasympathetic signals, and endocrine glands release hormones under hypothalamic control. In this way, effectors translate electrical signals of the nervous system into actual physiological actions. Without effectors, neural activity would remain incomplete, as they are the organs that produce observable responses essential for survival and homeostasis.

SLO (B-12-G-08)

Explain the function of the three types of neurons with the help of a reflex arc.

REFLEX ARC

The nervous system functions through specialized cells called neurons, which transmit information rapidly as electrical impulses. In order to coordinate body activities, three main types of neurons work together: sensory neurons, interneurons, and motor neurons. Their coordinated activity is best understood through the reflex arc, a rapid, automatic response to a stimulus that helps protect the body and maintain homeostasis.

Reflex Action and Reflex Arc

A **reflex action** is a rapid, automatic, and involuntary response to a stimulus. It serves as a protective mechanism, enabling the body to respond swiftly to potentially harmful changes in the internal or external environment without conscious control.

The **reflex arc** refers to the specific neural pathway followed by a nerve impulse during a reflex action. It ensures the transmission of signals from the sensory organ to the spinal cord and then to the appropriate effector, bypassing the brain for immediate response.

► Example of a Reflex Arc: Withdrawal from a Hot Surface

1. **Receptors in the skin** (thermoreceptors and nociceptors) detect the excessive heat stimulus. They function as **biological transducers**, converting the physical energy of heat into an **electrical impulse** (action potential). This is the first step in transforming environmental change into a neural signal.
2. The **sensory neuron** carries this impulse from the receptor toward the **spinal cord** through the **dorsal root** of the spinal nerve. This neuron serves as the communication bridge between the periphery and the central nervous system (CNS).
3. Within the **spinal cord**, the **sensory neuron** synapses with an **interneuron**. The interneuron acts as a **processing center**, evaluating the danger and preparing a rapid response. Importantly, this happens without waiting for the brain, ensuring the speed that reflexes require.
4. The **interneuron** relays the impulse to a **motor neuron** through another synapse. This step demonstrates the integration role of interneurons, which connect sensory input to motor output and allow immediate reaction.
5. The **motor neuron** transmits the impulse outward from the spinal cord through the **ventral root**. This electrical command is directed to the **skeletal muscles** of the arm and hand.
6. The **skeletal muscles** act as **effectors**, contracting strongly and pulling the hand away from the hot surface. This protective response occurs within a fraction of a second, preventing tissue damage such as burns.
7. Simultaneously, impulses travel upward to the **brain**, where they are processed to create **conscious awareness of pain and heat**. However, by the time the brain interprets the sensation, the withdrawal action has already occurred, highlighting the efficiency of spinal reflexes.

This entire pathway—**receptor → sensory neuron → interneuron → motor neuron → effector**—constitutes the **reflex arc**.

Difference Between Reflex Action and Reflex Arc

Feature	Reflex Action	Reflex Arc
Definition	An immediate, automatic, and involuntary response to a stimulus.	The specific neural pathway followed by an impulse during a reflex action.
Function	Enables rapid response to potentially harmful stimuli.	Transmits nerve impulses from sensory receptor to effector without involving the brain.
Involvement	Involves receptors, sensory and motor neurons, and effectors.	Involves all five components: receptor, sensory neuron, interneuron, motor neuron, and effector.
Nature	A physiological response or outcome.	A structural and functional arrangement of neurons.
Example	Withdrawing a hand from a hot surface.	The neural pathway involved in the hand withdrawal reaction.
Conscious Control	Not under conscious control.	Operates independently of conscious brain activity.

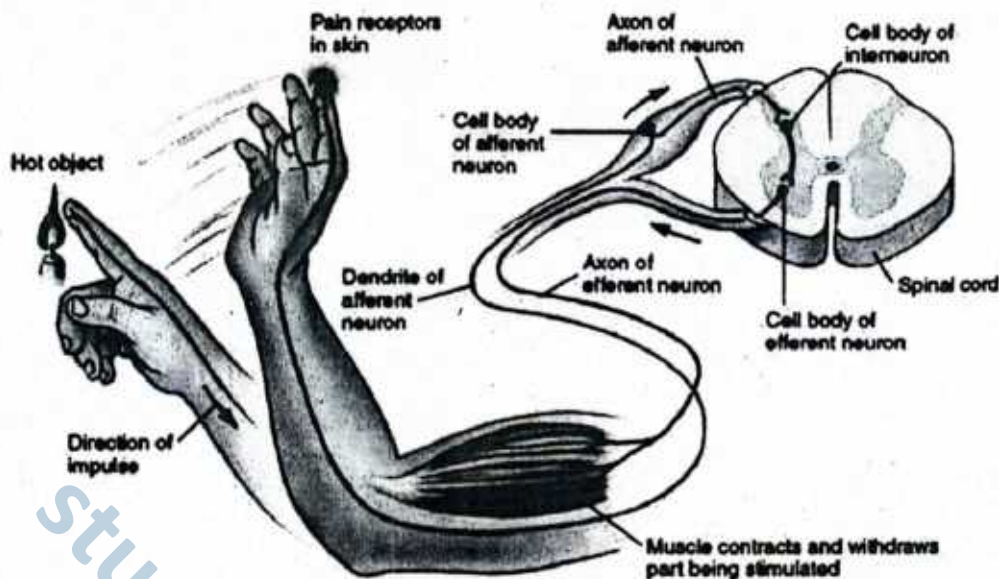


Figure: A reflex arc

SLOs Based (MCQs)

FUNCTIONS OF NEURONS, MYELINATED AND NON-MYELINATED NEURONS & REFLEX ARC

- Which structural feature of motor neurons allows them to transmit signals over long distances quickly?
 - Short axons and unmyelinated fibres
 - Long myelinated axons
 - Multiple cell bodies
 - Absence of dendrites
- Sensory neurons are suited for detecting stimuli and transmitting signals to the CNS due to:
 - Short unmyelinated axons
 - Lack of dendrites
 - Presence of long axons and dendrites
 - Location in muscles
- Which of the following neurons has short axons and highly branched dendrites to facilitate local processing?
 - Motor neuron
 - Interneuron
 - Sensory neuron
 - Bipolar neuron
- What is the functional role of myelin sheath in neurons?
 - Absorbs neurotransmitters
 - Slows down signal transmission
 - Insulates axons to speed up impulses
 - Stores ATP for axonal transport
- During a reflex arc, which neuron type directly activates the effector organ?
 - Interneuron
 - Sensory neuron
 - Bipolar neuron
 - Motor neuron

Answer Key

1. B 2. C 3. B 4. C 5. D

Short Questions

Q.1 How is the structure of motor neurons suited to their function?

Ans. Motor neurons, also called efferent neurons, are structurally adapted to carry signals from the central nervous system to effectors like muscles and glands. They are multipolar, possessing a single long axon and numerous dendrites. The dendrites enable them to receive and integrate multiple inputs from interneurons or other neurons. Their long axons, often covered with a myelin sheath, ensure that electrical impulses travel rapidly over large distances from the spinal cord or brain to peripheral effectors. The axon terminals form neuromuscular junctions, releasing neurotransmitters that trigger muscle contraction or glandular secretion. This structural design—large dendritic branching, myelination, and long axonal projections—makes motor neurons perfectly suited for rapid, efficient, and coordinated transmission of commands to initiate movement or physiological responses.

MARKING SCHEME (3 Marks)

- Definition (1), functions (2).

Q.2 In what way does the structure of sensory neurons support their role in signal transmission?

Ans. Sensory neurons, also called afferent neurons, are designed to detect environmental changes and relay signals to the central nervous system. They are usually pseudounipolar in vertebrates, which means a single process emerges from the cell body and quickly divides into two branches: one extending toward sensory receptors in the periphery and the other projecting into the spinal cord or brainstem. This design ensures rapid and direct signal conduction without delay through the cell body. Their dendritic endings are often specialized to detect specific modalities like touch, pain, temperature, or pressure. The axons of sensory neurons are frequently myelinated, which increases conduction velocity and allows for quick responses to stimuli. This structural adaptation ensures accurate detection, transmission, and processing of sensory information.

MARKING SCHEME (3 Marks)

- Definition (1), functions (2).

Q.3 Why are interneurons structurally suited for processing information in the CNS?

Ans. Interneurons, or association neurons, are located entirely within the central nervous system, serving as links between sensory and motor neurons. Their structure makes them well suited for information processing. They have short axons and extensively branched dendrites, which allow them to receive inputs from multiple neurons at once. Many interneurons are multipolar, giving them the ability to integrate and modify a wide range of signals. This dense synaptic connectivity makes them crucial in forming complex neural circuits for reflexes, motor coordination, and higher cognitive functions like memory and decision-making. Their short axons are particularly suited for local communication within the spinal cord and brain. Thus, the structural design of interneurons ensures rapid, localized integration and relaying of information.

MARKING SCHEME (3 Marks)

- Definition (1), functions (2).

Q.4 Compare the conduction speed between myelinated and non-myelinated neurons.

Ans. The conduction speed of nerve impulses depends on whether an axon is myelinated or not. Myelinated neurons have axons wrapped in a myelin sheath formed by Schwann cells (in the peripheral nervous system) or oligodendrocytes (in the CNS). This sheath insulates the axon and allows impulses to "jump" between nodes of Ranvier in a process called saltatory conduction. As a result, conduction in myelinated axons is extremely rapid, reaching speeds up to 100 m/s. In contrast, non-myelinated neurons lack this insulation, and impulses must travel continuously along the axon membrane. This results in slower transmission speeds, usually less than 2 m/s. While slower, non-myelinated neurons are still effective for autonomic functions where speed is less critical.

MARKING SCHEME (3 Marks)

- Myelinated conduction speed (1.5), non-myelinated conduction speed (1.5).

Q.5 Describe the role of three neuron types in a reflex arc.

Ans. A reflex arc is a simple, rapid, and involuntary response to a stimulus, involving three main types of neurons. First, the sensory neuron detects the stimulus through specialized receptors and transmits the impulse toward the central nervous system. Next, an interneuron located in the spinal cord processes this information and serves as the link between sensory input and motor output, often bypassing the brain for faster response. Finally, the motor neuron carries the processed signal from the CNS to the effector, such as a muscle or gland, which produces the response. For example, in the knee-jerk reflex, the sensory neuron detects stretch, the interneuron relays the message, and the motor neuron stimulates muscle contraction.

MARKING SCHEME (3 Marks)

- Sensory neuron (1), interneuron (1), motor neuron (1).

SLO [B-12-G-09]

Define nerve impulse.

NERVE IMPULSE

A **nerve impulse** is the **electrical signal** that travels along the length of a neuron. It is generated when a neuron is stimulated by a change in the environment (stimulus) and represents the way information is transmitted within the nervous system. In simple terms, it is the rapid movement of an electrical disturbance across the neuron's membrane, allowing messages to be carried from one part of the body to another.



The nerve impulse is the fundamental unit of communication in the nervous system. It enables sensory neurons to carry information from receptors to the CNS, interneurons to process and relay information within the CNS, and motor neurons to transmit commands to effectors like muscles and glands. Without nerve impulses, coordination, movement, and response to stimuli would not be possible.

► Nature of a Nerve Impulse

A nerve impulse is not electricity in the usual sense but rather a wave of depolarization and repolarization caused by the movement of sodium (Na^+) and potassium (K^+) ions across the neuron's membrane.

The impulse is self-propagating: once initiated at one point of the neuron, it continues to travel along the entire length of the axon without losing strength.

► Example of Nerve Impulse in Action

- When you touch a hot object, receptors in the skin generate a nerve impulse that travels via sensory neurons to the spinal cord.

- The spinal cord processes the signal and sends impulses through motor neurons to the muscles of the hand, causing immediate withdrawal.

- This rapid flow of impulses explains how reflexes protect the body in fractions of a second.

► Significance of Nerve Impulses

- Communication:** They enable fast transmission of information between different parts of the body.
- Coordination:** They link receptors and effectors, allowing the nervous system to coordinate activities.
- Survival:** Reflexes, thought processes, and voluntary actions all depend on nerve impulses.
- Clinical Relevance:** Damage to nerve pathways (e.g., in multiple sclerosis, where the myelin sheath is destroyed) slows or blocks impulses, leading to loss of function.

A nerve impulse is the fundamental unit of communication in the nervous system. It is an electrochemical wave of depolarization and repolarization that travels along a neuron's membrane. This process allows rapid and efficient transmission of information, forming the basis of all nervous system activity — from simple reflexes to complex thought.

SLO [B-12-G-10]

Describe the generation and transmission of nerve impulse.

SLO [B-12-G-11]

Name the factors responsible for the resting membrane potential of neuron.

GENERATION AND TRANSMISSION OF NERVE IMPULSE

Resting Membrane Potential

A neuron that is not actively transmitting a signal is said to be in a state of resting membrane potential (RMP). In this state, the inside of the axon is negatively charged relative to the outside, with a potential difference of about -70 millivolts (mV). The minus sign indicates that the neuron's interior is electrically negative compared to its exterior. This polarized condition is not random but the result of carefully regulated processes, and it plays a vital role in preparing the neuron to generate and transmit nerve impulses.

► Characteristics of the Resting Membrane Potential

- The outer surface of the neuron membrane is relatively positive, while the inner surface is negative.
- The typical potential difference is -70 mV.

- This electrical gradient is essential for excitability and for the rapid response of neurons to stimuli.

► Mechanisms Establishing Resting Membrane Potential

1. Selective Ion Distribution

- Potassium ions (K^+)** are present in much higher concentration inside the neuron (around 30 times more than outside).

- Sodium ions (Na^+)** are more concentrated outside the neuron (about 10 times more than inside).

- This unequal distribution of ions contributes to the electrical imbalance across the membrane.

2. Sodium-Potassium Pump (Na⁺/K⁺ ATPase)

- An active transport mechanism powered by ATP.
- For every cycle, it pumps three Na⁺ ions out of the neuron and two K⁺ ions in.
- This unequal exchange results in a net loss of positive charge inside, maintaining the negativity of the axon's interior.

3. Leaky Potassium Channels

- The neuron membrane is more permeable to K⁺ than to Na⁺ at rest.
- K⁺ ions continuously leak outward through passive channels, while fewer Na⁺ ions move inward.
- This further enhances the positive charge on the outside relative to the inside.

4. Negatively Charged Organic Ions

- Large anionic proteins, amino acids, and RNA molecules remain trapped inside the neuron.
- These immobile negative charges add to the overall negativity of the axoplasm.

Together, these mechanisms—selective ion distribution, the sodium–potassium pump, potassium leakage, and trapped negative ions—create and maintain the resting membrane potential. This polarized state is critical because it keeps the neuron in an excitable condition, allowing it to respond quickly and effectively when a stimulus is received.

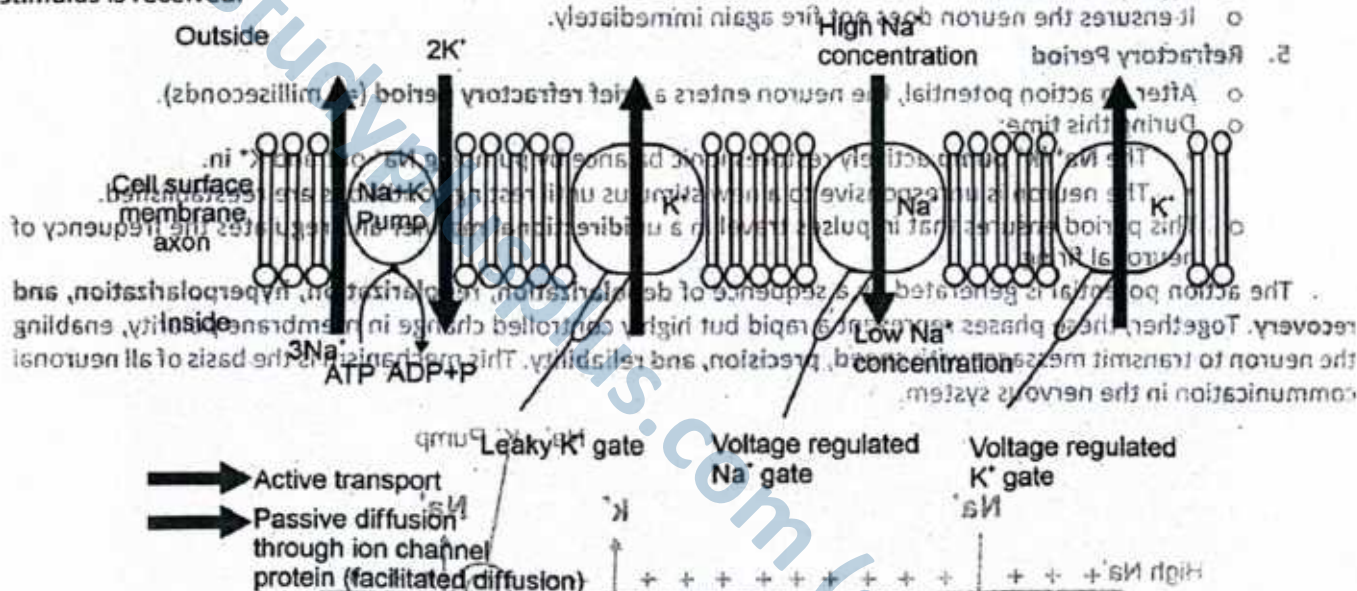


Figure: Ionic movement across neuron membrane

Generation of a Nerve Impulse (Action Potential)

When a neuron receives a stimulus strong enough to exceed the **threshold level**, its stable resting membrane potential is disturbed. This disturbance leads to a rapid and temporary reversal of polarity across the membrane, known as the **action potential** or **active membrane potential**. It is the fundamental unit of nerve impulse transmission, allowing signals to travel swiftly along the neuron.

► Characteristics of the Action Potential

- The membrane becomes **depolarized**: the inside becomes more positive, and the outside becomes negative.
- The potential reverses from about **-70 mV (resting)** to approximately **+50 mV**.
- The entire event is extremely rapid, lasting only a few milliseconds.
- Once initiated, the action potential is **self-propagating**, moving along the length of the axon without decreasing in strength.

► Phases of the Action Potential

1. Threshold Stimulus

- For an action potential to occur, the incoming stimulus must reach a **minimum strength** called the **threshold**.

- A stimulus below this level (**subthreshold**) fails to generate a response, whereas a threshold stimulus triggers depolarization.

2. Depolarization

- **Voltage-gated sodium (Na^+) channels open** in response to the threshold.
- Na^+ ions rapidly diffuse inward, causing the **membrane potential to rise from -70 mV to 0 mV**, and further up to about $+50$ mV.
- This reversal of polarity makes the **inside of the membrane electropositive** and the outside electronegative.

3. Repolarization

- Shortly after depolarization, Na^+ channels close while **voltage-gated potassium (K^+) channels open**.
- K^+ ions diffuse outward, restoring the **negative internal environment** of the neuron.
- The membrane potential begins to fall back toward the resting state.

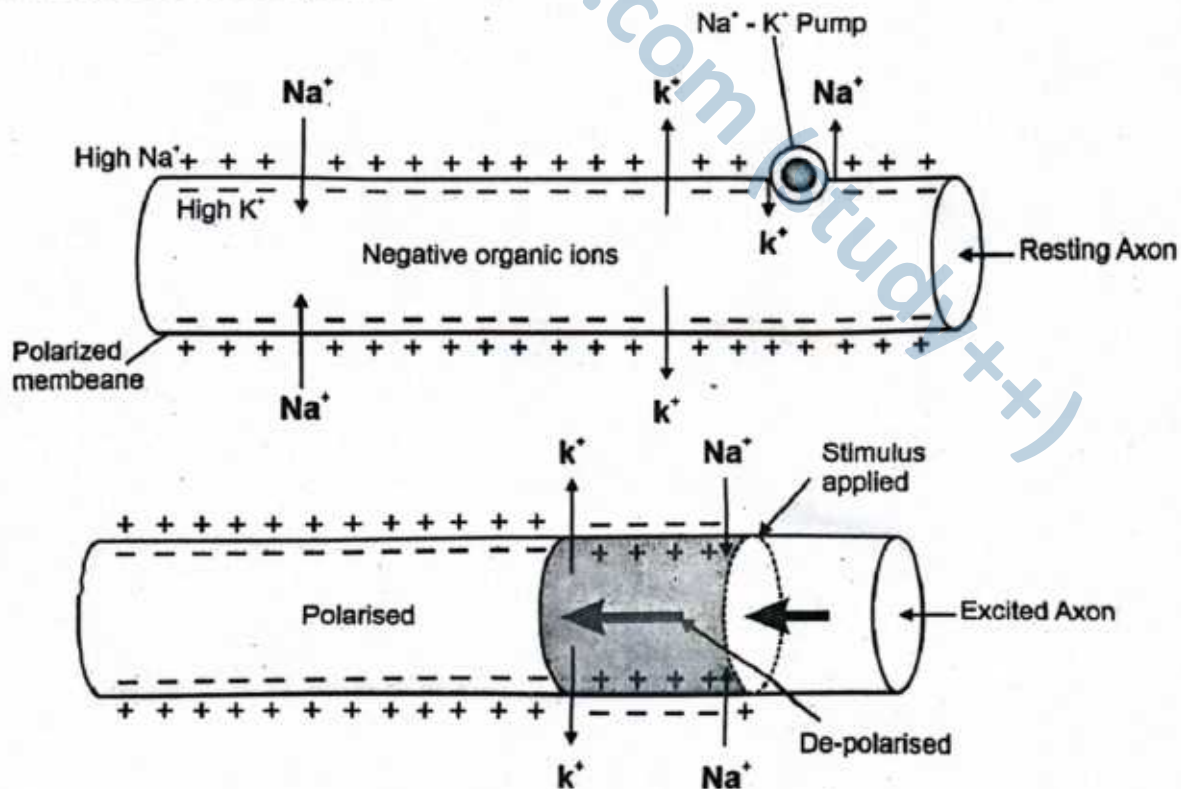
4. Hyperpolarization

- Closure of K^+ channels is slightly delayed, so **extra K^+ ions escape**.
- This causes the membrane potential to dip **below -70 mV** for a brief moment, a stage called hyperpolarization.
- It ensures the neuron does not fire again immediately.

5. Refractory Period

- After an action potential, the neuron enters a **brief refractory period (≈ 4 milliseconds)**.
- During this time:
 - The **Na^+/K^+ pump** actively restores ionic balance by pumping Na^+ out and K^+ in.
 - The neuron is unresponsive to a new stimulus until resting conditions are reestablished.
- This period ensures that impulses travel in a **unidirectional manner** and regulates the frequency of neuronal firing.

The action potential is generated by a sequence of **depolarization, repolarization, hyperpolarization, and recovery**. Together, these phases represent a rapid but highly controlled change in membrane polarity, enabling the neuron to transmit messages with **speed, precision, and reliability**. This mechanism is the basis of all neuronal communication in the nervous system.



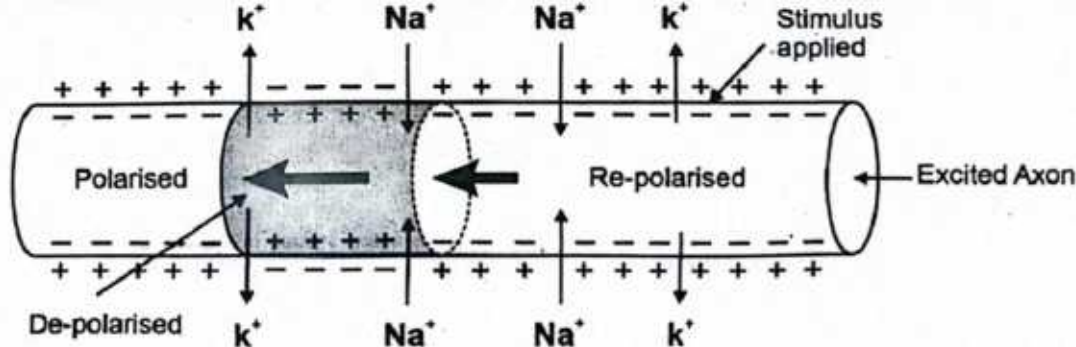


Figure: Conduction of nerve impulse

► Transmission of the Nerve Impulse

Once generated, the action potential travels along the axon as a **wave of depolarization**.

- In **unmyelinated neurons**, the impulse propagates continuously along the entire axonal membrane.
- In **myelinated neurons**, the impulse leaps from one **node of Ranvier** to the next. This mode of conduction, called **saltatory conduction**, greatly increases the speed of transmission while conserving energy.

► Significance

The generation and transmission of nerve impulses form the basis of rapid communication in the nervous system. The **resting potential** establishes the excitable state of neurons, while the **action potential** ensures the rapid flow of information. Together, these mechanisms allow the body to detect stimuli, process information, and produce quick responses — functions essential for survival.

SLO [B-12-G-12]

Evaluate from a graph the phenomena of polarization, depolarization and hyperpolarization of membrane.

EVALUATING MEMBRANE POTENTIALS FROM THE GRAPH

The graph shows the rapid changes in **membrane potential (in millivolts)** over time (in milliseconds) during the generation of an action potential. By analyzing this curve, students can identify the three key electrical phenomena of the neuron membrane: **polarization, depolarization, and hyperpolarization**.

1. Polarization (Resting Potential)

- At rest, the neuron is **polarized**, meaning the inside of the axon is **negative** (about -70 mV) compared to the outside.
- This phase appears as a **stable, flat line** before stimulation (0–1 ms).
- It is maintained by the sodium–potassium pump, selective permeability to K^+ , and trapped negative proteins inside the cell.
- Polarization prepares the neuron to respond rapidly when a stimulus is applied.

2. Depolarization

- When a **threshold stimulus** is applied, sodium (Na^+) channels open and Na^+ rushes into the neuron.
- This causes a **sharp upward spike** on the graph, where the membrane potential shifts from -70 mV to about $+50$ mV (1–3 ms).
- During depolarization, the inside becomes **positive** and the outside becomes **negative**, reversing the resting polarity.
- Depolarization is the **core event of the action potential**, initiating nerve impulse transmission.

3. Repolarization

- Immediately after depolarization, sodium channels close and potassium (K^+) channels open.
- K^+ ions move outward, restoring the negative charge inside.
- On the graph, this is shown as the **downward slope** returning toward -70 mV (3–5 ms).

4. Hyperpolarization

- Sometimes, K^+ continues to leave the neuron even after the resting level is reached.
- This causes the potential to **dip below -70 mV**, making the membrane even more negative than normal.



- On the graph, hyperpolarization appears as a small undershoot below the resting potential (around 5-6 ms).
- Soon after, the sodium-potassium pump restores the resting state.

5. Refractory Period

- The interval covering repolarization and hyperpolarization is known as the refractory period.
- During this time, the neuron cannot immediately fire again, ensuring that impulses travel only in one direction.

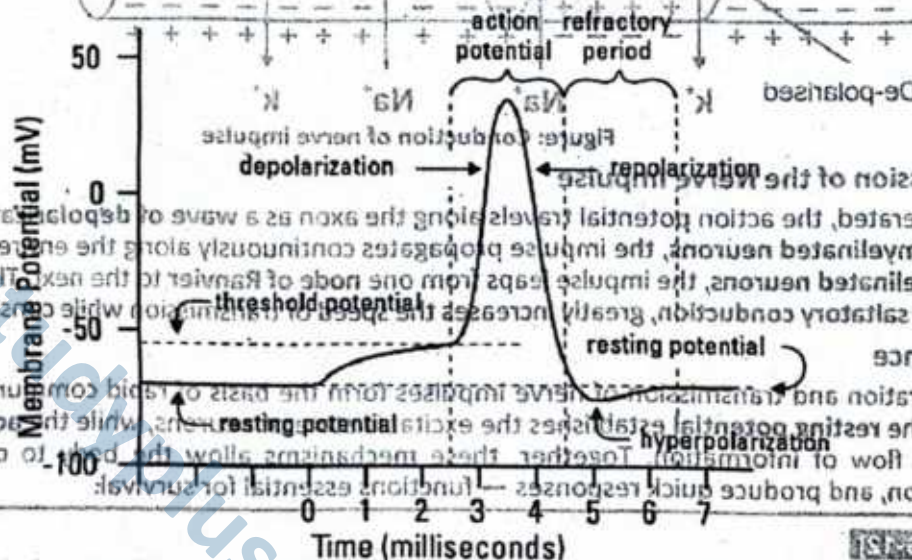


Figure: Action potential in a neuron

- Polarization:** Flat baseline at -70 mV before stimulation.
- Depolarization:** Sharp rise to $+50$ mV as Na^+ enters.
- Repolarization:** Decline back toward resting potential as K^+ exits.
- Hyperpolarization:** Small dip below -70 mV before returning to rest.

Thus, the graph provides a clear visualization of the dynamic electrical changes in a neuron during the action potential cycle.

Comparison between Resting Membrane Potential (RMP) and Active Membrane Potential (AMP)

Feature	Resting Membrane Potential (RMP)	Active Membrane Potential (AMP) / Action Potential
Definition	Electrical state of a neuron at rest, with a stable potential difference across the membrane.	A temporary reversal of membrane polarity triggered by a threshold stimulus.
Polarity	Inside of membrane is negative relative to outside.	Inside becomes positive relative to outside (depolarized state).
Typical Value	Approximately -70 mV.	Rises from -70 mV to $+50$ mV during depolarization.
Ion Movement	Maintained by active pumping of Na^+ out and K^+ in, and passive leakage of K^+ .	Characterized by Na^+ influx (depolarization) and K^+ efflux (repolarization).
Channels Involved	Na^+/K^+ pump, leaky K^+ channels.	Voltage-gated Na^+ and K^+ channels.
Energy Requirement	Requires ATP to maintain ion gradients.	Does not directly use ATP, but restoration via Na^+/K^+ pump post-impulse requires energy.
Functional Role	Prepares neuron to respond to stimuli by maintaining readiness.	Initiates and propagates the nerve impulse along the axon.
Duration	Continuous, as long as the neuron is unstimulated.	Very brief (a few milliseconds per action potential).
State of Neuron	Polarized, electrically stable.	Depolarized $\rightarrow$ Repolarized $\rightarrow$ Hyperpolarized, dynamically changing state.

MARKING SCHEME (3 Marks)	
1. Which of the following best explains why a neuron cannot respond to a second stimulus immediately after firing an action potential?	6. A student suggests that ATP is directly used during depolarization. Which correction is most accurate?
A. Depolarization phase is too fast	A. ATP is used to open Na^+ channels
B. Na^+/K^+ pump is inactivated	B. ATP is not required during depolarization
C. Sodium channels are still inactivated during refractory period	C. ATP fuels the influx of Na^+
D. Potassium ions are fully inside the neuron	D. ATP activates neurotransmitters
2. If a poison blocks the Na^+/K^+ pump in a neuron, which of the following would occur first?	7. How does hyperpolarization contribute to unidirectional nerve impulse transmission?
A. Depolarization fails to occur	A. It amplifies Na^+ influx
B. Resting potential gradually diminishes	B. It slows down the impulse
C. Action potentials become faster	C. It resets the ionic distribution immediately
D. Threshold is reached more quickly	D. It delays immediate re-excitation of the membrane
3. A neuron is stimulated but fails to generate an action potential. The most likely explanation is:	8. If voltage-gated K^+ channels close too quickly, which of the following may not occur?
A. Repolarization is too rapid	A. Depolarization
B. Stimulus was below the threshold	B. Repolarization
C. Too much K^+ entered the cell	C. Hyperpolarization
D. Na^+/K^+ pump was hyperactive	D. Sodium channel inactivation
4. Why is action potential described as an "all-or-none" event?	9. What would most likely happen if K^+ channels opened during the depolarization phase?
A. Only certain neurons can produce it	A. Amplified depolarization
B. It varies in strength depending on stimulus	B. Faster Na^+ influx
C. It either occurs fully or not at all	C. Reduced peak of action potential
D. It can be partially depolarized	D. No refractory period
5. During which part of the action potential do K^+ ions move most rapidly out of the neuron?	10. Which property of leaky K^+ channels is essential for maintaining the resting membrane potential?
A. Threshold phase	A. They block Na^+ ions
B. Depolarization	B. They help in equalizing charge
C. Repolarization	C. They allow continuous efflux of K^+
D. Hyperpolarization	D. They attract negative proteins

Answer Key

1. C 2. B 3. B 4. C 5. C 6. B 7. D 8. C 9. C 10. C

Short Questions

Q.1 What is meant by resting membrane potential and how is it maintained?

Ans. The resting membrane potential is the electrical difference across a neuron's plasma membrane when the cell is not transmitting an impulse. In most neurons, it is about -70 mV , meaning the inside of the cell is negatively charged compared to the outside. This difference is primarily maintained by three factors: the sodium-potassium pump actively transports 3 sodium ions (Na^+) out of the cell and 2 potassium ions (K^+) in, creating a gradient. Secondly, leaky potassium channels allow K^+ ions to diffuse out more easily than Na^+ enters, contributing to the negative interior. Finally, large negatively charged organic ions, such as proteins, remain trapped inside the cytoplasm, further lowering the internal charge. Together, these mechanisms maintain a stable resting potential essential for action potential generation.

Q.2 How do ion concentrations differ across the membrane during the resting state?

Ans. During the resting state, the ionic composition of the intracellular and extracellular fluids differs significantly, producing the resting membrane potential. Inside the neuron, potassium ions (K^+) are present in high concentration, around 140 mM, compared to much lower levels outside. This abundance of K^+ within the cell allows outward diffusion through leaky potassium channels. Conversely, sodium ions (Na^+) are concentrated outside the neuron at about 145 mM, whereas their concentration inside is much lower, approximately 15 mM. This steep sodium gradient favors the inward flow of Na^+ whenever sodium channels open. The imbalance of these ion distributions, established and maintained by the sodium-potassium pump, ensures the negative resting potential. This separation of charges across the membrane is the basis of neuronal excitability.

MARKING SCHEME (3 Marks)
• Na^+ concentration (1.5), K^+ concentration (1.5).

Q.3 Describe the sequence of events that occur during depolarization.

Ans. Depolarization occurs when a neuron is stimulated strongly enough to reach threshold potential, typically around -55 mV. At this point, voltage-gated sodium (Na^+) channels open suddenly. The influx of positively charged Na^+ ions causes the inside of the cell to become less negative, rapidly shifting the membrane potential from about -70 mV toward $+50$ mV. This reversal of polarity is the key characteristic of depolarization. As more sodium channels open in a positive feedback manner, the membrane potential rises sharply, creating the rising phase of the action potential. Depolarization allows the neuron to transmit electrical signals along its axon and initiate communication with target cells. This rapid shift in charge is crucial for nerve impulse conduction.

MARKING SCHEME (3 Marks)
• Three characteristics of depolarization (1 each).

Q.4 What causes repolarization after an action potential?

Ans. Repolarization is the process by which a neuron restores its resting negative membrane potential after depolarization. Once the action potential peak is reached (around $+50$ mV), the voltage-gated sodium (Na^+) channels close, stopping further sodium entry. At the same time, voltage-gated potassium (K^+) channels open, allowing potassium ions to flow out of the cell. This outward movement of positively charged K^+ ions reduces the positivity inside the neuron and restores the negative internal environment. The membrane potential thus begins to return toward -70 mV. Repolarization is essential for resetting the electrical state of the neuron so that it can fire another action potential if needed, ensuring proper sequential transmission of nerve impulses.

MARKING SCHEME (3 Marks)
• Definition of repolarization (1), cause (2).

Q.5 Explain the role of hyperpolarization in impulse conduction.

Ans. Hyperpolarization occurs when the membrane potential temporarily becomes more negative than the normal resting level, often dropping below -70 mV. This happens because potassium (K^+) channels close slowly, leading to an excessive outflow of K^+ ions after repolarization. Hyperpolarization creates a refractory period, during which the neuron is less likely to fire another action potential. This serves two key roles in impulse conduction: first, it prevents the immediate re-excitation of the neuron, ensuring that signals do not overlap or fuse. Second, it guarantees the one-way transmission of nerve impulses along the axon by preventing backflow of action potentials. Thus, hyperpolarization contributes to the reliability, precision, and directional flow of neural communication.

MARKING SCHEME (3 Marks)
• Explanation of hyperpolarization (1.5), role in impulse conduction (1.5).

Q.6 What is the importance of the refractory period in nerve impulse transmission?

Ans. The refractory period is the short interval following an action potential during which a neuron cannot immediately fire another impulse. It has two phases: the absolute refractory period, where no stimulus can trigger another action potential because sodium channels are inactivated, and the relative refractory period, where only a stronger-than-usual stimulus can initiate firing due to ongoing potassium efflux. This mechanism is critical for two reasons. First, it ensures unidirectional propagation of impulses along the axon by preventing action potentials from moving backward. Second, it provides time for the sodium-potassium pump to restore ionic gradients, reestablishing the resting membrane potential. Without this period, signals could overlap, leading to chaotic or ineffective nerve transmission.

MARKING SCHEME (3 Marks)
• Definition (1), importance (2).



Q.7 Why is the action potential considered an "all-or-none" response?

Ans. The action potential is described as an "all-or-none" phenomenon because it only occurs when the threshold stimulus, typically around -55 mV, is reached. If the stimulus is below threshold, no action potential is generated. Once threshold is crossed, however, the action potential always occurs with the same magnitude and intensity, regardless of whether the stimulus is weak or strong. This consistency arises because voltage-gated sodium channels open fully once activated, leading to rapid depolarization. Stronger stimuli do not produce a larger action potential but may increase the frequency of impulses. This principle ensures uniformity in signal transmission, preventing graded or variable responses that could distort communication in the nervous system.

MARKING SCHEME (3 Marks)

- Definition (1), reason/explanation (2).

Q.8 Describe the function of the Na^+/K^+ pump during the resting and recovery phases.

Ans. The sodium-potassium (Na^+/K^+) pump is an ATP-driven transporter that plays a crucial role in maintaining and restoring ion gradients across the neuronal membrane. During the resting phase, the pump maintains the resting membrane potential by continuously moving 3 sodium ions (Na^+) out of the cell and 2 potassium ions (K^+) in. This activity sustains the negative internal charge relative to the outside. During the recovery phase following an action potential, the pump restores disrupted ionic distributions, as excess Na^+ has entered and K^+ has left the cell during depolarization and repolarization. By re-establishing normal ion concentrations, the pump ensures the neuron is ready to fire another action potential, preserving the neuron's excitability and overall stability.

MARKING SCHEME (3 Marks)

- Resting role (1.5), recovery role (1.5).

Q.9 What would happen if the sodium channels failed to close after depolarization?

Ans. If sodium (Na^+) channels did not close after depolarization, the neuron would remain abnormally depolarized, as sodium ions would continuously flow into the cell. This failure would prevent repolarization, meaning the neuron could not return to its resting membrane potential. Consequently, subsequent action potentials could not be generated, effectively blocking signal transmission. Persistent depolarization might also lead to excitotoxicity, damaging or killing the neuron. On the other hand, one minor advantage of prolonged depolarization might be sustained activation of certain cellular responses, but this is generally harmful rather than beneficial. Overall, the inability to close sodium channels disrupts the precise control of nerve impulses essential for normal nervous system functioning.

MARKING SCHEME (3 Marks)

- Prevents repolarization (2), possible consequence/advantage (1).

Q.10 How does ion permeability differ between RMP and AMP?

Ans. During the resting membrane potential (RMP), the neuron's plasma membrane is much more permeable to potassium (K^+) than to sodium (Na^+). This is due to the presence of numerous leaky K^+ channels, which allow potassium to diffuse out, while sodium entry is limited. This differential permeability contributes to the negative internal charge of the neuron. In contrast, during an action potential (AMP), permeability shifts dramatically. At the onset of depolarization, voltage-gated sodium channels open, increasing Na^+ permeability and causing rapid inward flow. Later, during repolarization, sodium channels close while voltage-gated potassium channels open, increasing K^+ permeability and allowing potassium to exit the cell. These dynamic changes in ion permeability drive the rapid voltage shifts underlying impulse conduction.

MARKING SCHEME (3 Marks)

- RMP permeability (1.5), AMP permeability (1.5).

SLO [B-12-G-13]

Compare the velocities of nerve impulse in the axon membrane and in the synaptic cleft.

VELOCITIES OF NERVE IMPULSE IN AXON MEMBRANE AND SYNAPTIC CLEFT

The conduction of a nerve impulse occurs in two distinct ways:

1. Along the axon membrane (intra-neuronal conduction).
2. Across the synaptic cleft (inter-neuronal transmission).

These processes differ significantly in **speed**, **mechanism**, and **efficiency**, mainly due to the structural and physiological features of neurons.



Conduction along the Axon Membrane

The movement of a nerve impulse along the axon depends on ionic changes across the axon membrane. Several factors influence its velocity:

1. Myelin Sheath and Saltatory Conduction

- In **non-myelinated axons**, impulses move continuously along the membrane, with a relatively slow speed of **1–3 m/s**.
- In **myelinated axons**, the **myelin sheath** acts as an insulator and prevents ion leakage.
- Here, impulses "jump" from one **node of Ranvier** to the next, in a process called **saltatory conduction**.
- This mode of transmission is up to **50 times faster** than conduction in non-myelinated fibres.
- In humans, myelinated axons can transmit impulses at speeds up to **120 m/s**, making them rapid and energy-efficient.

2. Axon Diameter

- The **thicker the axon**, the faster the conduction.
- Larger diameters reduce internal resistance to the flow of ions, thereby increasing velocity.
- This principle is based on the inverse relationship between resistance and the cross-sectional area of a conductor.

3. Energy Efficiency

- Myelinated axons conserve energy because only the **nodes of Ranvier** require active ion exchange.
- This reduces metabolic cost while still maintaining high transmission speed.

Transmission across the Synaptic Cleft

When an impulse reaches the end of the axon (axon terminal), conduction switches from electrical to chemical transmission:

- The arrival of the action potential triggers the release of **neurotransmitters** (e.g., acetylcholine) into the **synaptic cleft**.
- These neurotransmitters **diffuse** across the cleft (20–40-nm wide) and bind to specific receptors on the postsynaptic membrane.
- This initiates a new action potential in the postsynaptic neuron.

Synaptic Delay

- Transmission across the synapse introduces a delay of approximately **0.5–1.0 milliseconds**, called the **synaptic delay**.
- This delay is due to the time required for:
 - Release of neurotransmitters,
 - Diffusion across the cleft,
 - Binding to receptors,
 - Activation of ion channels in the postsynaptic neuron.
- Although short, the synaptic delay is still longer than the time needed for the impulse to travel the same distance along an axon.

Comparison of Velocities

Feature	Axon Membrane	Synaptic Cleft
Nature of Transmission	Electrical (ionic changes)	Chemical (neurotransmitters)
Speed	Fast (1–3 m/s in non-myelinated; up to 120 m/s in myelinated axons)	Slow (synaptic delay = 0.5–1.0 ms)
Key Mechanism	Saltatory conduction in myelinated fibres; ion flow across membrane	Neurotransmitter release, diffusion, receptor binding
Efficiency	Highly efficient, rapid, energy-saving	Relatively slower, with risk of fatigue or failure

The velocity of impulse conduction along the axon membrane is far greater than across the synaptic cleft.

- **Axons (especially myelinated and large-diameter fibres)** conduct impulses at very high speeds due to saltatory conduction and reduced internal resistance.
- **Synapses**, however, introduce a necessary but slower chemical step, ensuring flexibility, regulation, and integration of signals within the nervous system.

Thus, while axonal conduction ensures speed, synaptic transmission ensures **control and coordination** in neural communication.



ROLE OF LOCAL CIRCUITS IN SALTATORY CONDUCTION

Saltatory Conduction

In **myelinated axons**, nerve impulses are transmitted by a special process called **saltatory conduction**. In this mode, the action potential does not travel continuously along every part of the axon membrane. Instead, it appears to "**jump**" from one **node of Ranvier** to the next, bypassing the myelinated regions. This adaptation makes conduction both **much faster** and **more energy-efficient** compared to unmyelinated fibres.

Mechanism: Local Circuits of Ionic Current

The key mechanism underlying saltatory conduction is the formation of **local circuits** within the axon:

- At an active **node of Ranvier**, **voltage-gated sodium channels** open, allowing a rapid influx of **Na^+ ions**, which depolarizes that region.
- These sodium ions spread **longitudinally inside the axoplasm**, creating a **local circuit of current** that flows from the depolarized node toward the adjacent resting node.
- When this current reaches the **next node**, where sodium channels are densely concentrated, it depolarizes the membrane to **threshold potential**.
- A new **action potential** is then generated at that node, while the previous one enters its refractory phase, ensuring **forward movement only**.

Thus, the impulse **does not regenerate** along the entire length of the axonal membrane, but is renewed only at the nodes, giving the appearance of "jumping."

Significance of Local Circuits

Local circuits make saltatory conduction possible and provide several advantages:

- **Rapid transmission:** The action potential leaps between nodes, greatly accelerating the speed of conduction (up to 120 m/s in humans).
- **Energy efficiency:** Only nodal regions undergo depolarization, so fewer ions cross the membrane and the **Na^+/K^+ pump** expends less ATP.
- **Directional propagation:** The refractory period behind each impulse prevents backward spread, ensuring **unidirectional flow** of nerve signals.

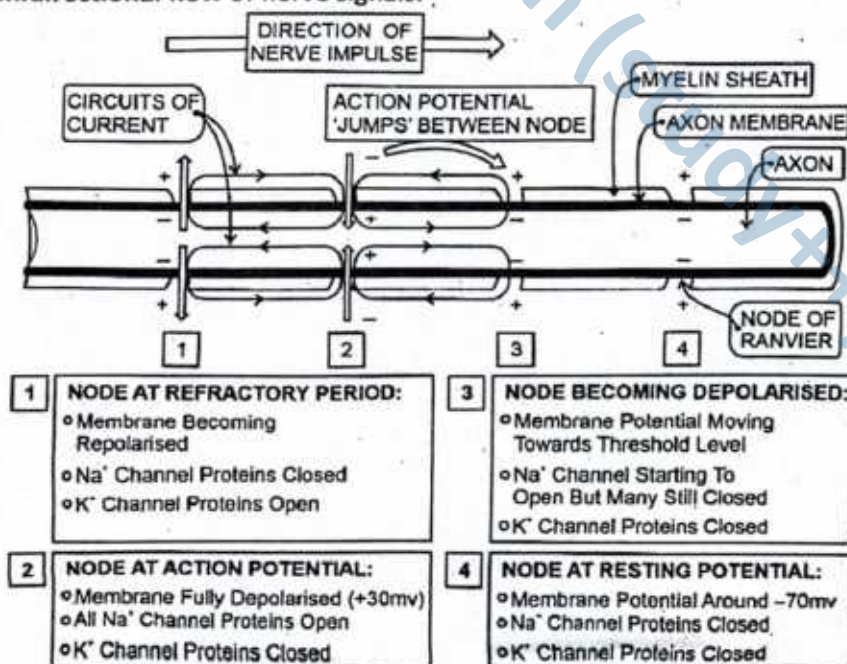


Figure: Synaptic transmission

STRUCTURE OF A SYNAPSE

A **synapse** is the specialized junction between two neurons, or between a neuron and an effector (muscle or gland). It serves as the site where **nerve impulses are transmitted** from one cell to another. Synapses ensure that communication within the nervous system is highly organized and directional.

Major Components of a Synapse

1. Presynaptic Neuron

- The neuron that sends the message.
- Its terminal end (synaptic knob or bouton) contains numerous **synaptic vesicles** filled with neurotransmitters such as acetylcholine, dopamine, or norepinephrine.
- It also contains **mitochondria** to supply ATP required for neurotransmitter release.

2. Synaptic Cleft

- A narrow gap (~20–40 nanometers wide) that separates the presynaptic terminal from the postsynaptic membrane.
- The nerve impulse cannot cross this gap electrically; instead, **neurotransmitters diffuse** across the cleft to carry the signal chemically.

3. Postsynaptic Membrane

- Located on the dendrite, soma, or effector surface of the next cell.
- It contains **specific receptor proteins** that bind neurotransmitters.
- This binding generates a new electrical signal in the postsynaptic neuron or triggers a response in an effector cell (e.g., muscle contraction).

SYNAPTIC TRANSMISSION OF NERVE IMPULSE

The **synapse** is not just a structural junction but a functional site where the **nerve impulse is transmitted** from one neuron to another, or from a neuron to an effector cell. This process, called **synaptic transmission**, converts an electrical signal (action potential) in the presynaptic neuron into a chemical signal in the synaptic cleft, and then back into an electrical response in the postsynaptic cell.

Sequence of Synaptic Transmission:

1. **Arrival of Action Potential:** When an **action potential** reaches the synaptic knob of the presynaptic neuron, it triggers the **opening of voltage-gated calcium channels** in the presynaptic membrane.
2. **Influx of Calcium Ions:** Due to a steep concentration gradient, **Ca²⁺ ions enter the synaptic knob**, increasing intracellular calcium levels.
3. **Vesicle Fusion and Neurotransmitter Release:** Elevated calcium concentration stimulates **synaptic vesicles to migrate** and fuse with the presynaptic membrane. As a result, **neurotransmitters are released** into the synaptic cleft via **exocytosis**.
4. **Diffusion and Receptor Binding:** Neurotransmitter molecules **diffuse across the synaptic cleft** and bind to **specific receptors** on the **postsynaptic membrane**.
5. **Generation of Postsynaptic Potential:** Binding of neurotransmitters causes **ion channels (e.g., Na⁺ channels)** to open, allowing the **influx of sodium ions**. This **depolarizes the postsynaptic membrane** and may initiate a new action potential. If this depolarization is sufficient to reach the threshold, it is termed an **Excitatory Postsynaptic Potential (EPSP)**.

6. **Neurotransmitter Breakdown or Reuptake:** Neurotransmitters do not remain in the cleft indefinitely. They are either **enzymatically degraded** or **reabsorbed** by the presynaptic neuron. For example:
- Acetylcholine is broken down by acetylcholinesterase.
 - Adrenaline (epinephrine) is degraded by monoamine oxidase (MAO).

Significance of Synaptic Transmission

- **Unidirectional** signal transmission from presynaptic to postsynaptic neurons.
- Amplification and modulation of neural signals.
- **Plasticity** in learning and memory formation due to variability in synaptic strength.

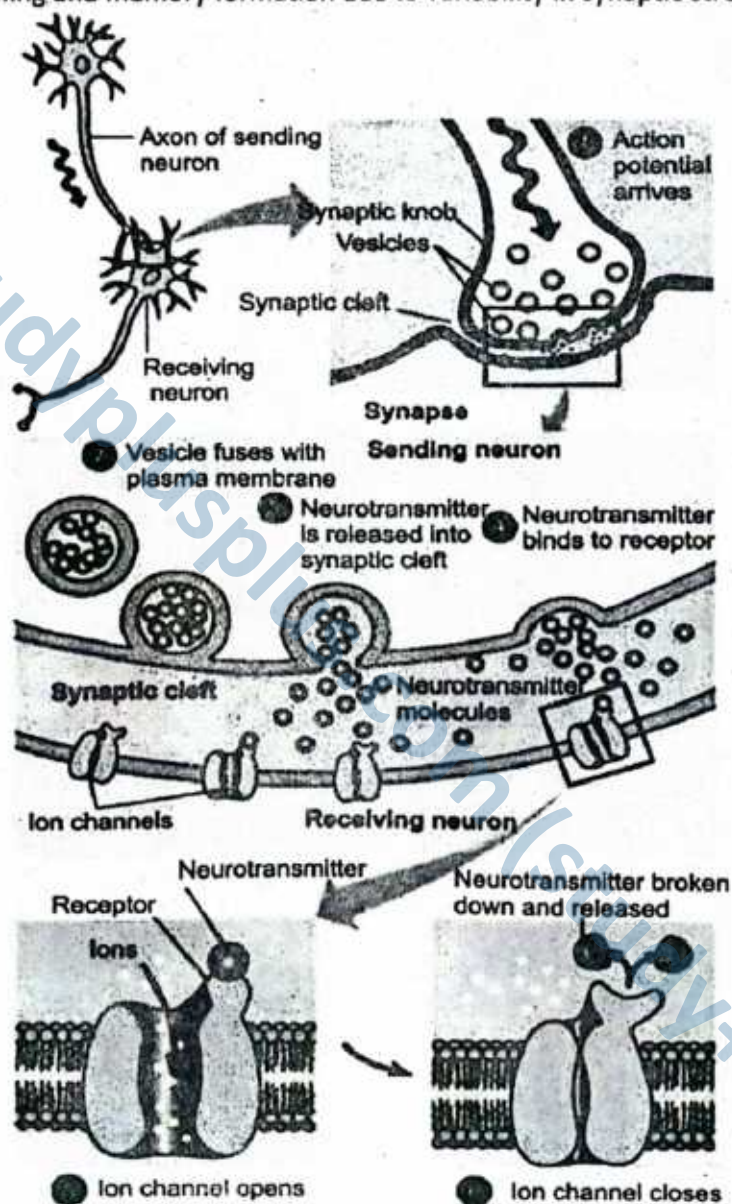


Figure: Synaptic transmission

SLOs Based (MCQs)

VELOCITIES OF NERVE IMPULSE, ROLE OF LOCAL CIRCUITS IN SALTATORY CONDUCTION AND SYNAPSE

- | | |
|--|---|
| 1. Which of the following contributes most to the high velocity of impulse conduction in myelinated axons? | <p>A. Continuous conduction</p> <p>B. Greater number of synapses</p> <p>C. Saltatory conduction</p> <p>D. Increased neurotransmitter production</p> |
|--|---|



2. What is the main reason thick axons conduct impulses faster than thin axons?
 - A. Presence of more mitochondria
 - B. Greater resistance
 - C. Lower internal resistance to ion flow
 - D. Increased synaptic vesicles
3. In myelinated axons, the action potential "jumps" between which of the following?
 - A. Synaptic clefts
 - B. Axon terminals
 - C. Nodes of Ranvier
 - D. Dendrites
4. Which statement best describes the role of local circuits in saltatory conduction?
 - A. They regenerate the neurotransmitters at each synapse
 - B. They allow Na^+ to move along axon, depolarizing next node
 - C. They inhibit the action potential in unmyelinated axons
 - D. They reverse polarity of synaptic vesicles
5. Why is saltatory conduction considered energy-efficient?
 - A. It uses multiple synapses
 - B. It avoids depolarizing the entire axon membrane
 - C. It requires more ATP
 - D. It increases neurotransmitter release
6. Synaptic delay refers to:
 - A. Time taken for action potential to reach axon terminal
 - B. Time for neurotransmitter synthesis
 - C. Time taken for transmission across synapse
 - D. Delay due to depolarization in axon
7. Which of the following ions initiates the release of neurotransmitters at synapse?
 - A. Sodium (Na^+)
 - B. Potassium (K^+)
 - C. Calcium (Ca^{2+})
 - D. Chloride (Cl^-)
8. The insulating property of the myelin sheath primarily functions to:
 - A. Increase membrane permeability
 - B. Prevent neurotransmitter diffusion
 - C. Reduce energy loss and increase speed
 - D. Decrease axonal resistance
9. Which component of the neuron directly receives neurotransmitters?
 - A. Axon hillock
 - B. Synaptic knob
 - C. Presynaptic membrane
 - D. Postsynaptic membrane
10. The neurotransmitter acetylcholine is inactivated by:
 - A. Monoamine oxidase
 - B. Cholinesterase
 - C. ATPase
 - D. Kinase

Answer Key

1. C	2. C	3. C	4. B	5. B	6. C	7. C	8. C	9. D	10. B
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Short Questions

Q.1 Why do myelinated axons conduct nerve impulses faster than non-myelinated axons?

Ans. Myelinated axons conduct impulses faster than non-myelinated axons due to three key factors. First, the myelin sheath acts as an electrical insulator, preventing leakage of ions and conserving the depolarization signal. Second, action potentials are generated only at the nodes of Ranvier, skipping the insulated regions in a process called saltatory conduction, which reduces the number of membrane segments that must depolarize. Third, energy consumption is minimized, as fewer ion channels open along the length of the axon. These features collectively increase conduction velocity, allowing rapid and efficient communication between neurons. In contrast, non-myelinated axons propagate impulses continuously along the entire membrane, which is slower and less energy-efficient.

MARKING SCHEME (3 Marks)

- Insulation (1), saltatory conduction (1), energy efficiency/increased speed (1).

Q.2 What is saltatory conduction and how does it enhance nerve impulse velocity?

Ans. Saltatory conduction is the mechanism by which nerve impulses "jump" from one node of Ranvier to the next in myelinated axons. In this process, the myelin-insulated regions of the axon do not depolarize; only the nodes, where voltage-gated sodium channels are concentrated, generate action potentials. This significantly reduces the number of depolarization events along the axon, allowing impulses to travel much faster than in non-myelinated fibers, where each membrane segment must depolarize. By minimizing the distance over which ions must flow, saltatory conduction increases conduction velocity by up to 50 times and reduces energy consumption because fewer ions are exchanged across the membrane. This efficiency is essential for rapid reflexes and coordinated nervous system responses.

MARKING SCHEME (3 Marks)

- Definition (1), mechanism/reason for enhanced speed (2).

Q.3 Describe the role of local circuits in saltatory conduction.

Ans. Local circuits are the small flows of ions, primarily sodium (Na^+), along the interior and exterior of a myelinated axon. When an action potential occurs at a node of Ranvier, sodium ions enter the axon, creating a localized depolarization. This depolarization spreads passively through the cytoplasm to the next node, where it triggers the opening of voltage-gated sodium channels and generates a new action potential. These local circuits allow the nerve impulse to move efficiently without depolarizing the myelinated internodal regions, conserving energy and increasing conduction speed. Additionally, local circuits maintain the integrity of the signal over long distances, ensuring rapid and faithful transmission of information from the CNS to effectors or other neurons.

MARKING SCHEME (3 Marks)

- Ion movement (1), depolarization triggering next node (1), efficiency/speed (1).

Q.4 How does axon diameter affect the speed of nerve impulse transmission?

Ans. Axon diameter directly influences conduction velocity. Larger diameter axons have lower internal resistance to the flow of ions, allowing depolarization currents to spread more quickly along the cytoplasm. This faster propagation shortens the time it takes for the action potential to reach the next node, increasing overall nerve impulse velocity. In contrast, smaller diameter axons resist ion flow more, slowing the rate of depolarization. The effect of axon diameter is particularly pronounced in unmyelinated fibers, where continuous conduction occurs along the entire length. In combination with myelination, large-diameter axons achieve the highest conduction speeds, which is crucial for rapid reflexes, precise motor coordination, and the efficient transmission of sensory information.

MARKING SCHEME (3 Marks)

- Relation diameter-speed (1), effect on depolarization/conduction (2).

Q.5 What is synaptic transmission and where does it occur?

Ans. Synaptic transmission is the process by which a neuron communicates with another neuron, muscle, or gland. When an action potential reaches the presynaptic terminal, it triggers the opening of voltage-gated calcium channels. Calcium influx causes synaptic vesicles to release neurotransmitters into the synaptic cleft. These chemical messengers bind to receptors on the postsynaptic membrane, leading to depolarization or hyperpolarization depending on the neurotransmitter type, and thus conveying the signal. Synaptic transmission occurs at synapses—the specialized junctions between neurons or between neurons and effectors. This process ensures that signals are transmitted accurately, unidirectionally, and can be modulated, allowing complex functions such as learning, memory, and coordinated movement.

MARKING SCHEME (3 Marks)

- Definition/process (2), location (1).

Q.6 Describe the process of neurotransmitter release at the synapse.

Ans. When an action potential reaches the synaptic knob of a presynaptic neuron, it triggers the opening of voltage-gated calcium channels. Calcium ions (Ca^{2+}) enter the synaptic terminal, causing synaptic vesicles to move toward and fuse with the presynaptic membrane. This fusion releases neurotransmitters into the synaptic cleft via exocytosis. The neurotransmitters then diffuse across the cleft and bind to specific receptors on the postsynaptic membrane. Binding of neurotransmitters opens ion channels in the postsynaptic neuron, initiating depolarization or hyperpolarization depending on the neurotransmitter type. This sequence of events allows the electrical signal from the presynaptic neuron to be converted into a chemical signal and then back into an electrical signal in the postsynaptic neuron, enabling communication between cells.

MARKING SCHEME (3 Marks)

- Calcium channel (1), synaptic vesicles (1), neurotransmitter binding (1).

Q.7 What is synaptic delay and what causes it?

Ans. Synaptic delay is the short time interval, typically around 0.5 to 1 millisecond, that occurs between the arrival of an action potential at the presynaptic terminal and the initiation of a postsynaptic response. It arises due to multiple steps in chemical synaptic transmission: the opening of voltage-gated calcium channels, the influx of Ca^{2+} ions, movement and fusion of synaptic vesicles, release of neurotransmitters into the synaptic cleft, diffusion of neurotransmitters across the cleft, and binding to postsynaptic receptors. Each of these processes requires a finite amount of time, collectively contributing to the synaptic delay. Although brief, this delay is critical in regulating the speed of neural circuits and ensuring proper temporal coordination of signal transmission.

MARKING SCHEME (3 Marks)

- Definition (1), cause (2).



Q.8 Why is the transmission across a synapse unidirectional?

Ans. Synaptic transmission is unidirectional because of the structural organization of neurons at the synapse. Neurotransmitters are released only from the presynaptic neuron, which contains synaptic vesicles and calcium channels, and not from the postsynaptic neuron. Conversely, receptors for neurotransmitters are located exclusively on the postsynaptic membrane and are absent on the presynaptic side. This arrangement ensures that the signal flows from the presynaptic to the postsynaptic neuron only. The unidirectional flow prevents feedback loops or reverse signaling that could disrupt neural circuit function. Structural polarity at both presynaptic and postsynaptic sites guarantees precise communication between neurons, maintaining the fidelity of signal transmission across complex neural networks.

MARKING SCHEME (3 Marks)

- Presynaptic structure (1.5), postsynaptic structure (1.5).

Q.9 How is the neurotransmitter action terminated in the synaptic cleft? Give an example.

Ans. Termination of neurotransmitter action is essential to prevent continuous stimulation of the postsynaptic neuron. Neurotransmitters can be removed from the synaptic cleft in three main ways: enzymatic degradation, reuptake by the presynaptic neuron, or diffusion away from the synapse. For example, acetylcholine (ACh) released at cholinergic synapses is hydrolyzed by the enzyme acetylcholinesterase into acetate and choline. This stops its action on postsynaptic receptors and allows the postsynaptic membrane to return to its resting state, ready for the next signal. Termination ensures precise timing of neural signals and prevents overstimulation, which could disrupt neural circuit function or lead to excitotoxicity in sensitive tissues.

MARKING SCHEME (3 Marks)

- Method of termination (1), example (2).

Q.10 Differentiate between electrical conduction along the axon and chemical conduction across the synapse.

Ans. Electrical conduction along the axon involves the propagation of action potentials due to ion movement across the axonal membrane. Depolarization and repolarization occur sequentially along the axon, enabling rapid, continuous, or saltatory conduction. In contrast, chemical conduction across a synapse relies on neurotransmitter release from the presynaptic neuron, diffusion across the synaptic cleft, and binding to postsynaptic receptors to generate a new electrical signal. Electrical conduction is fast and continuous, whereas chemical conduction is slower due to synaptic delay and is modifiable, allowing excitatory or inhibitory modulation of signals. The combination of electrical axonal conduction and chemical synaptic transmission allows complex integration and regulation of neural networks.

MARKING SCHEME (3 Marks)

- Electrical conduction (1.5), chemical conduction (1.5).

SLO [B-12-G-17]

Classify neurotransmitters as inhibitory and excitatory and list some common examples.

CLASSIFICATIONS OF NEUROTRANSMITTERS

Neurotransmitters are chemical messengers that mediate communication between neurons or between neurons and effectors across synapses. Depending on their action on the postsynaptic membrane, they are classified as **excitatory** or **inhibitory**.

Excitatory Neurotransmitters

Excitatory neurotransmitters increase the likelihood of generating an action potential in the postsynaptic neuron. They achieve this by enhancing the membrane's permeability to sodium ions (Na^+), thereby depolarizing the membrane and bringing it closer to the threshold potential.

A prominent example is **acetylcholine (ACh)**, which acts as a key excitatory neurotransmitter in the **peripheral nervous system**, particularly at neuromuscular junctions. In the **central nervous system (CNS)**, several **biogenic amines** (derived from amino acids) serve excitatory roles.



These include:

- **Epinephrine (adrenaline) and norepinephrine (noradrenaline):** These neurotransmitters increase the heart rate and prepare the body for stress responses (fight-or-flight).
- **Serotonin and dopamine:** These play critical roles in regulating sleep-wake cycles, mood, emotional behavior, attention span, and learning. Deficiencies or imbalances in these neurotransmitters are implicated in various neuropsychiatric disorders.

Inhibitory Neurotransmitters

Inhibitory neurotransmitters reduce the probability of action potential generation in the postsynaptic neuron. They do so by decreasing the membrane's permeability to Na^+ ions, often making the interior of the cell more negative (hyperpolarization), and raising the threshold for stimulation.

Key inhibitory neurotransmitters include:

- **Gamma-aminobutyric acid (GABA) and glycine:** These amino acids function as the primary inhibitory neurotransmitters in the CNS. GABA, in particular, exerts widespread inhibitory control, helping prevent overstimulation and maintaining neural stability.
- **Endorphins:** These are neuropeptides that act as both neurotransmitters and hormones. They bind to opioid receptors and modulate pain perception by inhibiting the release of pain-signaling neurotransmitters.

Comparative Table: Excitatory vs Inhibitory Neurotransmitters

Feature	Excitatory Neurotransmitters	Inhibitory Neurotransmitters
Effect on Postsynaptic Neuron	Increase likelihood of action potential by depolarizing membrane	Decrease likelihood of action potential by hyperpolarizing membrane
Mechanism of Action	Increase permeability to sodium ions (Na^+)	Decrease permeability to Na^+ or increase Cl^- or K^+ permeability
Membrane Potential Change	Moves closer to threshold (more positive)	Moves further from threshold (more negative)
Examples (Peripheral/Central)	Acetylcholine (PNS), epinephrine, norepinephrine, serotonin, dopamine (CNS)	GABA, glycine (CNS), endorphins (act as both neurotransmitters and hormones)
Physiological Roles	Muscle contraction, arousal, mood, attention, stress response	Pain modulation, relaxation, inhibition of overstimulation
Associated Disorders	Excess: seizures, anxiety; Deficit: depression, attention disorders	Deficit: epilepsy, anxiety; Excess: sedation, motor impairment

SLO [B-12-G-18]

Identify the main components of the nervous system.

MAIN COMPONENTS OF NERVOUS SYSTEM

The **human nervous system** is a highly organized and complex communication network that governs body coordination, regulation, and response to internal and external stimuli. It is structurally divided into two principal components:

- Central Nervous System (CNS)
- Peripheral Nervous System (PNS)

The **CNS** functions as the main control and integration center. It is located along the **midline of the body** and is composed of the **brain and spinal cord**. The **PNS**, on the other hand, connects the CNS to sensory receptors and effectors throughout the body. It **relays sensory information to the CNS** and **transmits motor commands from the CNS** to muscles and glands.



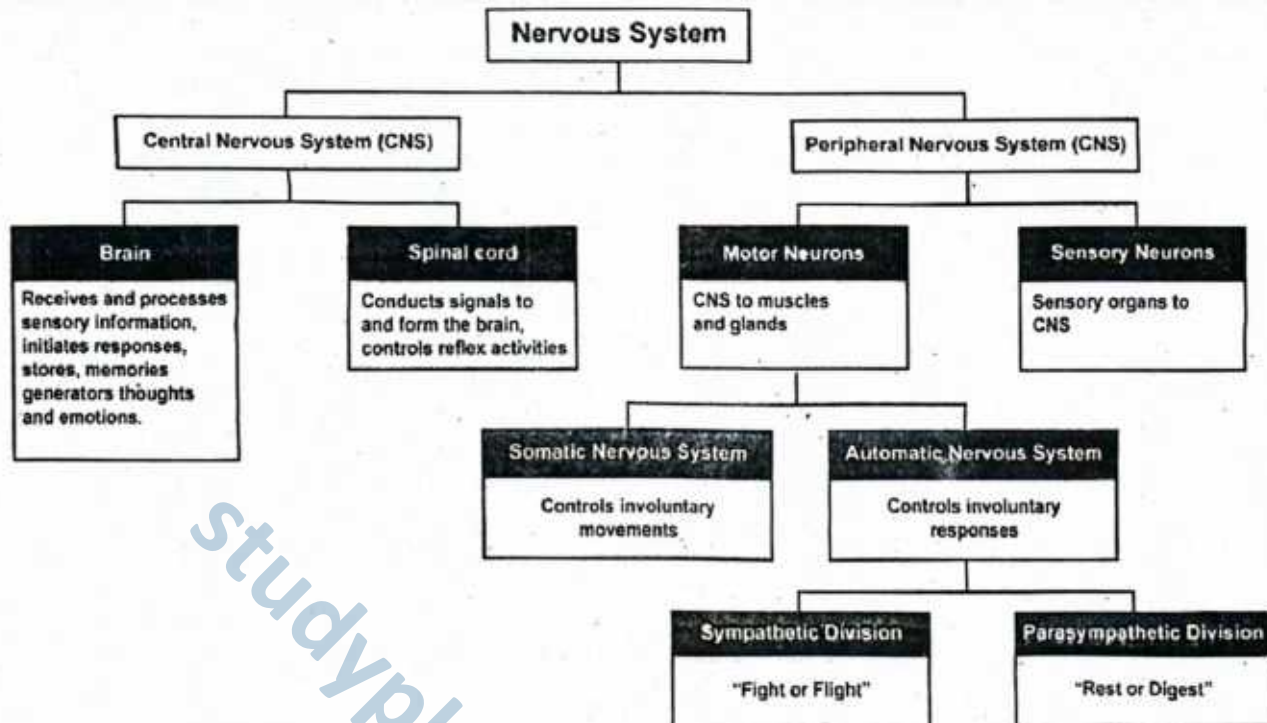


Figure: Organization of human nervous system

SLO [B-12-G-20]

Describe the architecture of human brain.

ARCHITECTURE OF HUMAN BRAIN

The human brain is the central organ of the nervous system, responsible for controlling and coordinating body activities, processing sensory information, and enabling cognition, memory, emotions, and behavior. It is protected by the skull and surrounded by cerebrospinal fluid (CSF) within the meninges.

1. General Structure

The brain is divided into three main parts:

(i) Forebrain (Prosencephalon)

- **Cerebrum:** The largest part; controls voluntary actions, reasoning, memory, speech, and senses.
 - **Cerebral hemispheres:** Left and right hemispheres connected by the **corpus callosum**.
 - **Cortex:** Outer layer of gray matter; site of higher mental functions.
 - **Subcortical structures:** Include **basal ganglia** (movement regulation) and **limbic system** (emotions, memory).
- **Diencephalon:** Connects the cerebrum with the brainstem.
 - **Thalamus:** Relay station for sensory information.
 - **Hypothalamus:** Controls homeostasis, endocrine functions, and autonomic activities.

(ii) Midbrain (Mesencephalon)

- Small region connecting forebrain and hindbrain.
- Involved in visual and auditory reflexes, motor coordination, and movement control.
- Contains **cerebral aqueduct**, connecting the third and fourth ventricles.

(iii) Hindbrain (Rhombencephalon)

- **Cerebellum:** Coordinates voluntary movements, balance, and posture.
- **Pons:** Bridges cerebrum and cerebellum; regulates breathing and sensory information.
- **Medulla oblongata:** Controls involuntary actions such as heartbeat, respiration, and digestion.

2. Protective Structures

(i) **Skull:** Bony casing protecting the brain.

(ii) **Meninges:** Three layers of connective tissue:

- Dura mater (outer tough layer)
- Arachnoid mater (middle web-like layer)
- Pia mater (inner delicate layer adhering to the brain surface)

(iii) **Cerebrospinal fluid (CSF):** Cushions the brain and maintains chemical stability.

3. Internal Organization

- **Gray matter:** Contains neuron cell bodies; forms the cerebral cortex and basal nuclei.
- **White matter:** Made of myelinated axons; connects different brain regions.
- **Ventricular system:** Four interconnected cavities (lateral, third, and fourth ventricles) filled with CSF.

SLO [B-12-G-19]

Explain briefly the major parts, functions of major divisions of the brain and its functions of brain.

MAJOR PARTS, DIVISIONS, AND FUNCTIONS OF THE HUMAN BRAIN

The human brain is the most intricate organ of the nervous system and serves as the **central command center** for bodily functions, perception, memory, emotion, and cognition. It is **divided into three major regions:** the **forebrain**, **midbrain**, and **hindbrain**, each containing specialized structures responsible for specific physiological and cognitive functions.

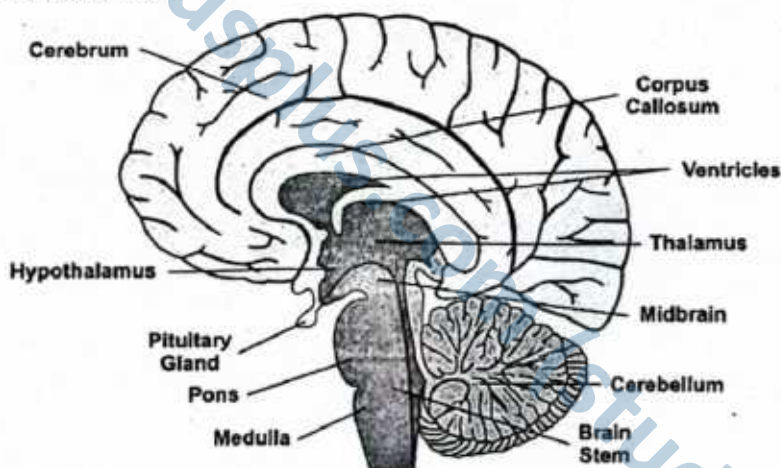


Figure: Structure of human brain

1. Forebrain (Prosencephalon)

The **forebrain** is the largest and most developed part of the human brain. It comprises the **cerebrum**, **thalamus**, and **limbic system**, and it is primarily responsible for higher-order functions, sensory perception, voluntary movements, and emotional responses.

Cerebrum:

The **cerebrum** is the largest structure of the brain and is divided into two **cerebral hemispheres**, connected by the **corpus callosum**, which allows communication between the hemispheres. Each hemisphere is further divided into four lobes: **frontal**, **parietal**, **temporal**, and **occipital**. The surface is marked by **sulci** (shallow grooves) and **fissures** (deep grooves), with the **longitudinal fissure** separating the two hemispheres.

Each lobe contains specialized functional areas:

- **Sensory areas** – receive and interpret sensory information from the body.
- **Association areas** – integrate and process data to form perceptions and ideas.
- **Motor areas** – initiate voluntary muscle movements.

The cerebrum also oversees higher-order brain functions, including memory, reasoning, judgment, thought, consciousness, and dreaming.



Thalamus:

Located beneath the cerebrum, the **thalamus** acts as a **central relay station** for most sensory information, excluding olfactory signals. It directs impulses to the appropriate areas of the **cerebral cortex** and **limbic system** for processing.

Limbic System:

The **limbic system** is a network of interconnected structures beneath the cerebrum, surrounding the thalamus. It is essential for **emotions, motivation, and memory**. Major components include:

- **Hypothalamus** – regulates **homeostasis**, including body temperature, hunger, thirst, water balance, sleep-wake cycles, menstrual rhythms, and blood pressure. It also controls the **pituitary gland**, linking the nervous and endocrine systems.
- **Amygdala** – involved in processing emotions such as fear, anger, love, rage, and sexual arousal.
- **Hippocampus** – critical for converting **short-term memory** into **long-term memory**, supporting learning and memory consolidation.

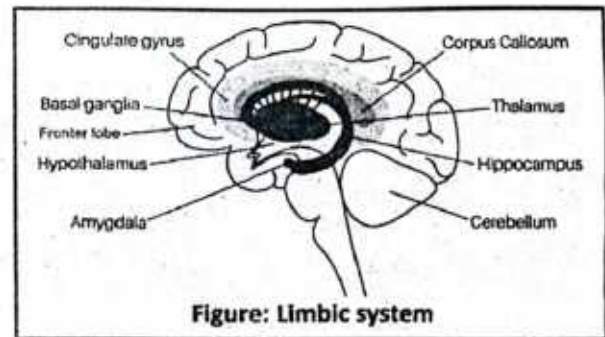


Figure: Limbic system

2. Midbrain (Mesencephalon)

The **midbrain** is a small but vital part of the brain located between the forebrain and hindbrain. It serves as a **relay center**, transmitting signals between the forebrain, spinal cord, and hindbrain. The midbrain also houses the **reticular formation**, a network of neurons involved in **arousal, consciousness, and attention**, and contains the **cerebral aqueduct**, which connects the third and fourth ventricles for **cerebrospinal fluid (CSF)** circulation.

3. Hindbrain (Rhombencephalon)

The **hindbrain** regulates vital body functions and coordinates movement. It consists of the **cerebellum, pons, and medulla oblongata**.

- **Cerebellum** – located beneath the cerebrum at the back of the brain, the cerebellum is responsible for **balance, equilibrium, and fine coordination of voluntary muscle movements**. It enables complex motor skills such as walking, writing, and playing instruments, and contributes to **motor learning and behavioral memory**.
- **Pons** – acts as a **bridge** connecting the cerebellum, medulla, and cerebrum. It regulates **breathing rhythms** and contributes to **heart rate control**.
- **Medulla Oblongata** – controls vital involuntary functions, including **heartbeat, blood pressure, respiration, swallowing, and reflexes** such as coughing and sneezing.

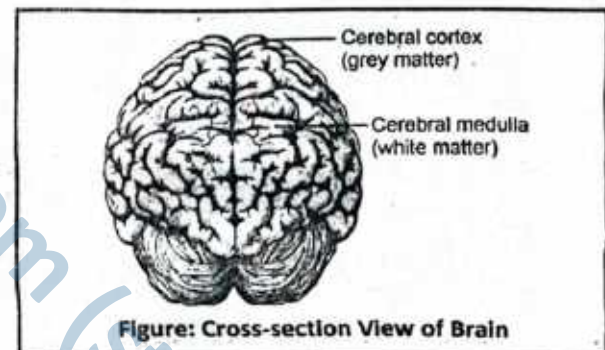


Figure: Cross-section View of Brain

4. Ventricular System and CSF

The brain contains **four interconnected ventricles** filled with **cerebrospinal fluid (CSF)**. These cavities **cushion the brain**, protect it from mechanical shocks, assist in **nutrient transport**, and help maintain **intracranial pressure**.

Structure and Functions of Human Brain

Part of Brain	Sub-structure	Key Features	Functions
Forebrain	Cerebrum	Largest part; divided into two cerebral hemispheres by longitudinal fissure	Controls voluntary actions, memory, reasoning, judgement, emotions, sensory interpretation, and motor responses
		Hemispheres interconnected by corpus callosum	

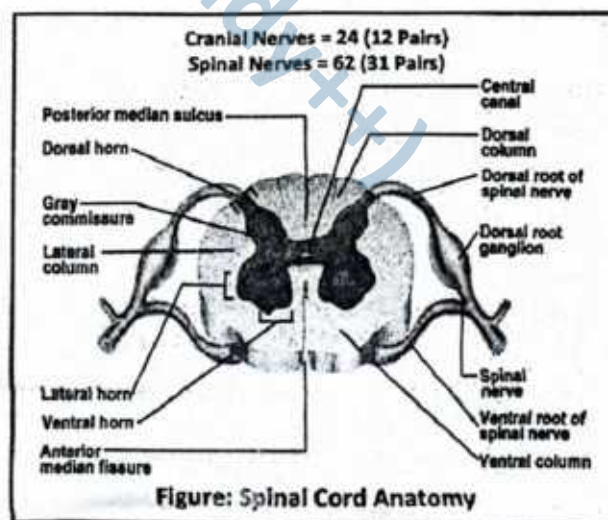
	Lobes of Cerebrum	Frontal, Parietal, Temporal, Occipital	Contain functional areas: sensory (input), association (interpretation), and motor (response initiation)
	Thalamus	Lies below cerebrum	Relays all sensory signals (except smell) to appropriate cerebral regions
	Hypothalamus	Below thalamus; part of limbic system	Regulates homeostasis, hunger, thirst, body temperature, BP, menstrual cycle, and links nervous and endocrine systems
	Limbic System	Includes amygdala, hippocampus, hypothalamus	Involved in emotions, motivation, memory processing
	Amygdala	Almond-shaped masses of neurons	Controls emotions: love, anger, fear, rage, sexual arousal
	Hippocampus	Curved horns connected to amygdala	Converts short-term memory into long-term memory
Midbrain	Reticular Formation	Small, relay centre connecting forebrain and hindbrain	Coordinates communication between spinal cord, cerebrum, and cerebellum
Hindbrain	Cerebellum	Located at back of brain, highly folded	Maintains posture, balance, coordination of voluntary movement; involved in motor learning
	Pons	Bridge-like structure	Connects cerebellum, medulla, and cerebrum; regulates heart rate and breathing pattern
	Medulla Oblongata	Lowest part of brainstem	Controls involuntary actions: heartbeat, breathing, swallowing, BP regulation
Brain Ventricles	Four interconnected cavities	Filled with cerebrospinal fluid (CSF)	Cushion brain, maintain chemical environment, aid in nutrient and waste exchange

Spinal Cord

The **spinal cord** is a cylindrical structure of the central nervous system that serves as the **primary neural pathway** connecting the **brain** with the **rest of the body**. It extends from the **medulla oblongata** at the base of the brain down to approximately the level of the **second lumbar vertebra**. Encased within the **vertebral column** and cushioned by **cerebrospinal fluid (CSF)**, the spinal cord is protected by the **meninges** and is a vital conduit for both **sensory** and **motor** information.

► Structure of the Spinal Cord

In **cross-section**, the spinal cord displays a characteristic arrangement of **grey** and **white matter**. The **grey matter** lies centrally and is shaped like an **"H"** or a **butterfly**, consisting of **neuron cell bodies**, **dendrites**, and **unmyelinated fibers**. It forms **anterior (ventral) horns**, which contain **motor neurons** sending signals to **skeletal muscles**, and **posterior (dorsal)**



horns, which receive sensory input from the body. The lateral horns, present in certain thoracic and lumbar segments, contain autonomic neurons controlling internal organs.

Surrounding the grey matter is the **white matter**, composed primarily of **myelinated axons** organized into **ascending tracts** (carrying sensory information to the brain) and **descending tracts** (conveying motor commands from the brain to the body). This **segregation of fibers** allows the spinal cord to efficiently transmit signals over long distances.

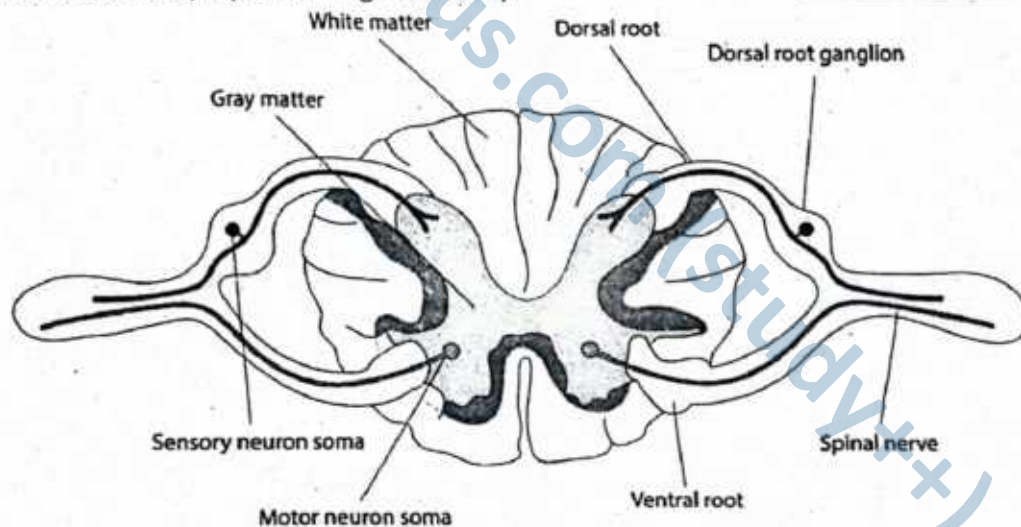
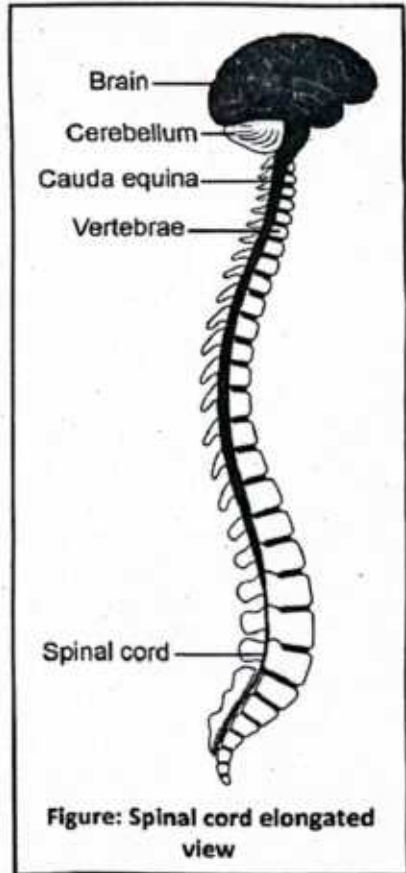
At the center of the spinal cord runs the **central canal**, a narrow cavity filled with **cerebrospinal fluid**, which provides cushioning, nutrient transport, and chemical stability.

► Spinal Nerves and Roots

From the spinal cord, **31 pairs of spinal nerves** emerge, each attached via a **dorsal (posterior) root** and a **ventral (anterior) root**. The **dorsal roots** carry sensory information from the body to the spinal cord, while the **ventral roots** transmit motor commands from the spinal cord to muscles and glands. **Dorsal root ganglia**, located just outside the spinal cord, house the **cell bodies of sensory neurons**. This arrangement ensures that sensory and motor pathways remain organized and functionally distinct.

► Architectural Highlights

An **important architectural distinction** is that the **arrangement of grey and white matter in the spinal cord is inverted** compared to the brain: in the spinal cord, **grey matter is central** and white matter is peripheral, whereas in the brain, **grey matter forms the outer cortex** and white matter lies deeper. This structural organization reflects the spinal cord's role as a **conduit for signal transmission**, integrating reflex arcs, motor commands, and sensory input with high efficiency.



Differences between Brain and Spinal Cord

Feature	Brain	Spinal Cord
Main Function	Coordination of higher functions, reflexes, and sensory interpretation	Transmission of signals between brain and body
Grey Matter Location	Outer cortex	Inner core (H-shaped)
White Matter Location	Inner region	Outer region
Protective Cavity	Enclosed in the cranium	Enclosed in the vertebral column
Presence of Ventricles	Four ventricles filled with CSF	Central canal filled with CSF
Role in Reflexes	Indirect or higher-level integration	Direct role in reflex arcs

CRANIAL AND SPINAL NERVES IN MAN

The **peripheral nervous system (PNS)** consists of **cranial and spinal nerves**, which connect the **central nervous system (CNS)**—the brain and spinal cord—to the rest of the body. These nerves transmit **sensory information** to the CNS and **motor commands** from the CNS to muscles and glands, enabling voluntary and involuntary responses. Architecturally, the nerves are bundles of **axons** (nerve fibers) enclosed in **connective tissue sheaths**, allowing rapid signal conduction.

► Cranial Nerves

Humans have **12 pairs of cranial nerves** that arise directly from the **brain** (primarily the brainstem) rather than the spinal cord. Each nerve is specialized for specific sensory, motor, or mixed functions:

1. **Olfactory (I)** – sensory; responsible for **smell**.
2. **Optic (II)** – sensory; carries **visual information** from the retina to the brain.
3. **Oculomotor (III)** – motor; controls most **eye movements** and **pupil constriction**.
4. **Trochlear (IV)** – motor; innervates the **superior oblique muscle** of the eye.
5. **Trigeminal (V)** – mixed; provides **sensory input** from the face and **motor control** for mastication.
6. **Abducens (VI)** – motor; controls the **lateral rectus muscle** of the eye.
7. **Facial (VII)** – mixed; governs **facial expressions**, **taste** from the anterior tongue, and **salivary glands**.
8. **Vestibulocochlear (VIII)** – sensory; responsible for **hearing and balance**.
9. **Glossopharyngeal (IX)** – mixed; involved in **taste**, **swallowing**, and **salivary secretion**.
10. **Vagus (X)** – mixed; innervates **viscera**, regulating **heart rate**, **digestion**, and other **autonomic functions**.
11. **Accessory (XI)** – motor; controls **neck and shoulder muscles**.
12. **Hypoglossal (XII)** – motor; manages **tongue movements**.

Cranial nerves are structurally distinct in that some are **purely sensory**, **purely motor**, or **mixed**, depending on whether they carry afferent (sensory) or efferent (motor) fibers. They emerge from the **brainstem** or **forebrain**, pass through **cranial foramina**, and branch extensively to reach their target organs.

► Spinal Nerves

Humans have **31 pairs of spinal nerves**, each arising from the **spinal cord** via a **dorsal (posterior) root** and a **ventral (anterior) root**. These nerves are **mixed**, carrying both **sensory fibers** (from the dorsal root ganglia) and **motor fibers** (from ventral horn neurons).

The spinal nerves are grouped based on the **region of the spinal cord** they emerge from:

- **Cervical (8 pairs)** – neck, shoulders, arms.
- **Thoracic (12 pairs)** – chest and upper abdomen.
- **Lumbar (5 pairs)** – lower back and legs.
- **Sacral (5 pairs)** – pelvis, lower limbs, and genital region.
- **Coccygeal (1 pair)** – tailbone area.

After emerging from the vertebral column, spinal nerves branch into **dorsal rami**, which innervate **back muscles and skin**, and **ventral rami**, which supply **limbs and the anterior body wall**. Spinal nerves are organized into **plexuses** (e.g., cervical, brachial, lumbar, sacral) in certain regions, allowing **redundancy and coordination** of motor and sensory innervation.

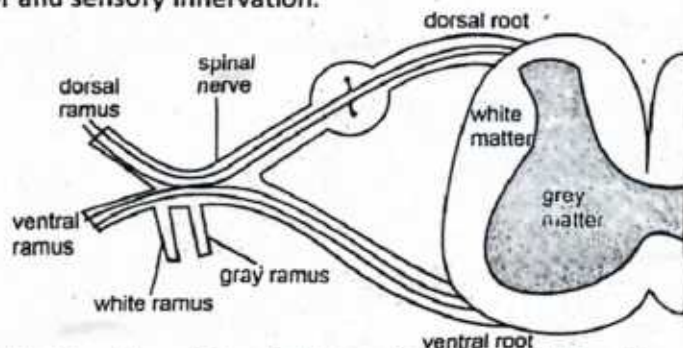


Figure: The formation of the spinal nerve from the ventral and dorsal roots

Functional Significance and Architectural Highlights

- Cranial nerves mainly handle special senses (smell, vision, hearing, taste) and head and neck motor control, with some exceptions (vagus nerve) extending to the thoracic and abdominal organs.
- Spinal nerves provide a structural network linking the CNS with the rest of the body, enabling reflex actions, voluntary movement, and sensory perception.
- Architecturally, both cranial and spinal nerves are composed of axon bundles, with connective tissue sheaths (endoneurium, perineurium, epineurium) ensuring protection, support, and efficient signal conduction.

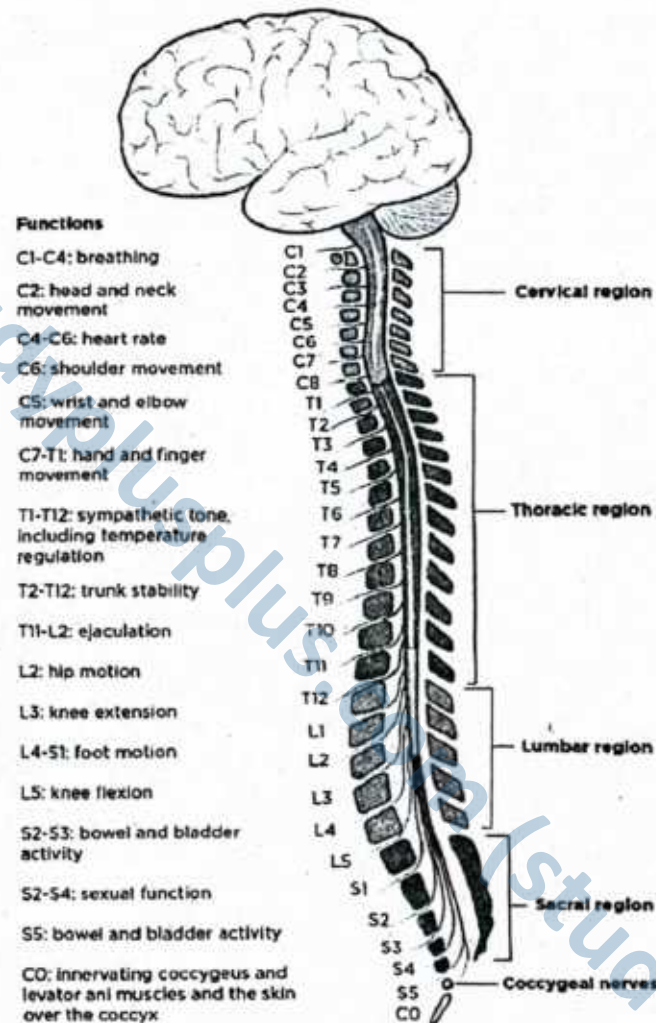


Figure: Spinal nerves

SLOs Based (MCQs)

CNS

- Which part of the brain serves as the relay station for all sensory impulses except smell?
 - Cerebellum
 - Thalamus
 - Pons
 - Medulla oblongata
- The corpus callosum connects:
 - Cerebrum with cerebellum
 - Thalamus with hypothalamus
 - Left and right cerebral hemispheres
 - Spinal cord with medulla
- Which lobe of the cerebrum is primarily involved in vision?
 - Temporal
 - Parietal
 - Frontal
 - Occipital
- Which structure of the limbic system is responsible for converting short-term memory into long-term memory?
 - Amygdala
 - Hippocampus
 - Hypothalamus
 - Thalamus



5. The medulla oblongata controls all of the following except:
 A. Heartbeat B. Respiration
 C. Blood pressure D. Memory storage
- 15 6. Which part of the brain controls body equilibrium and coordination of voluntary movement?
 A. Medulla B. Pons
 C. Cerebellum D. Midbrain
7. The meninges surrounding the brain do not include:
 A. Dura mater B. Arachnoid mater
 C. Choroid mater D. Pia mater
8. The bridge between the cerebellum and cerebrum is provided by:
 A. Thalamus B. Pons
 C. Midbrain D. Hippocampus
9. In a transverse section of the spinal cord, the central canal is filled with:
 A. Plasma B. Lymph
 C. Synovial fluid D. Cerebrospinal fluid
10. Which of the following pairs is correctly matched?
 A. Frontal lobe – Visual input
 B. Amygdala – Long-term memory
 C. Pons – Involuntary swallowing
 D. Hypothalamus – Endocrine and autonomic regulation

Answer Key

1. B 2. C 3. D 4. B 5. D 6. C 7. C 8. B 9. D 10. D

Short Questions

Q.1 Why is the cerebrum considered the most advanced part of the human brain?

Ans. The cerebrum is the largest and most highly developed part of the brain, making it the center of advanced human abilities. It is divided into two hemispheres and four major lobes, each with specialized functions. The cerebrum is responsible for higher-order thinking such as reasoning, problem-solving, decision-making, and creativity. It also plays a vital role in voluntary motor control, allowing precise movements of the body. Furthermore, it integrates sensory information from vision, hearing, touch, and other senses, enabling complex perception of the environment. Language processing, learning, and memory also rely heavily on cerebrum function. Because of its involvement in intelligence, self-awareness, and voluntary activity, it is considered the most advanced region of the human brain.

MARKING SCHEME (3 Marks)

- Three main functions (1 mark each).

Q.2 How does the hypothalamus link the nervous and endocrine systems?

Ans. The hypothalamus is a small but critical part of the brain that acts as the primary connection between the nervous and endocrine systems. Functioning within the nervous system, it continuously monitors internal conditions such as body temperature, hunger, thirst, and circadian rhythms, ensuring homeostasis. At the same time, it regulates the endocrine system by controlling the pituitary gland, often called the "master gland." Through releasing and inhibiting hormones, the hypothalamus directs the pituitary to regulate thyroid, adrenal, and reproductive glands. This dual role allows the hypothalamus to translate nervous signals into hormonal responses, ensuring that the body's physiological and behavioral activities remain balanced and coordinated.

MARKING SCHEME (3 Marks)

- Nervous system function (1.5 marks), endocrine system function (1.5 marks).

Q.3 What is the significance of the hippocampus in learning and memory?

Ans. The hippocampus is a seahorse-shaped structure located within the temporal lobe and is vital for memory and learning. It plays a key role in converting short-term memory into long-term storage, enabling individuals to retain and recall important information. Without proper hippocampal function, a person may be able to learn new facts but fail to retain them for future use. Additionally, the hippocampus contributes to spatial memory and navigation, often referred to as "cognitive mapping," which allows individuals to remember locations and directions. Its role in organizing and retrieving stored experiences makes it central to both academic learning and everyday decision-making. Damage to the hippocampus, such as in Alzheimer's disease, severely impairs memory formation and recall.

MARKING SCHEME (3 Marks)

- Role in learning (1.5 marks), role in memory (1.5 marks).

Q.4 Explain the role of the limbic system in emotional regulation.

Ans. The limbic system is a group of interconnected brain structures, including the amygdala, hippocampus, and hypothalamus, that collectively regulate emotions and behaviors. The amygdala is central to processing strong emotions such as fear, aggression, and pleasure,

MARKING SCHEME (3 Marks)

- Amygdala (1 mark), hippocampus (1 mark), hypothalamus (1 mark).



while also linking emotions to memories. The hippocampus works alongside the amygdala in associating emotions with specific experiences, allowing learning through emotional memory. The hypothalamus translates emotional states into physiological responses, such as increased heart rate during fear or excitement. Together, these structures allow humans to process, express, and regulate emotions, influencing motivation, behavior, and social interactions. This system is also involved in emotional learning, making it essential for survival and adaptation.

Q.5 How does the medulla oblongata contribute to survival?

Ans. The medulla oblongata, located at the base of the brainstem, is a vital structure that ensures survival by controlling essential involuntary processes. It functions as the autonomic control center, regulating heartbeat, breathing rate, and blood pressure without conscious effort. The medulla also acts as the body's respiratory center, adjusting breathing patterns based on oxygen and carbon dioxide levels, which is crucial for maintaining life. In addition, it serves as a reflex center, governing vital protective reflexes such as swallowing, sneezing, coughing, and vomiting. These reflexes safeguard the body against harmful substances and ensure proper functioning of the respiratory and digestive systems. Because of these life-sustaining roles, damage to the medulla is often fatal.

MARKING SCHEME (3 Marks)

- Autonomic functions (1 mark), respiratory center (1 mark), reflex center (1 mark).

Q.6 Differentiate between the structural organization of grey and white matter in the brain and spinal cord.

Ans. In the brain, grey matter forms the outer surface called the *cerebral cortex* and contains neuron cell bodies, dendrites, and synapses responsible for information processing, decision-making, and memory. Beneath the cortex lies white matter, composed mainly of myelinated axons, which provides communication pathways connecting different brain regions. In the **spinal cord**, however, the arrangement is reversed. Here, grey matter lies centrally in a butterfly or "H-shaped" structure and is surrounded by white matter on the outside. The central grey matter functions mainly in reflex processing and integration, while the surrounding white matter serves as long tracts that transmit sensory information upward to the brain and motor commands downward to effectors. This structural difference reflects their specialized roles.

MARKING SCHEME (3 Marks)

- Brain organization (1.5 marks), spinal cord organization (1.5 marks).

Q.7 Describe how damage to the cerebellum affects motor function.

Ans. The cerebellum is primarily responsible for maintaining posture, balance, and coordination of voluntary muscle movements. It ensures that motor actions are smooth, precise, and well-timed by integrating sensory feedback with motor commands. When the cerebellum is damaged due to trauma, stroke, or disease, its functions become impaired. As a result, individuals often experience loss of balance, difficulty walking in a straight line, and problems with coordinated movement. Tremors of the hands, slurred speech, and reduced muscle tone are also common symptoms. Tasks requiring fine motor control, such as writing or buttoning clothes, become especially difficult. Thus, cerebellar damage does not cause paralysis but severely disrupts the accuracy and fluidity of movements.

MARKING SCHEME (3 Marks)

- Functions of cerebellum (1 mark), effects after damage (2 marks).

Q.8 Why are meninges and cerebrospinal fluid critical for CNS protection?

Ans. The central nervous system (CNS) is delicate and requires multiple layers of protection. **Meninges** are three connective tissue membranes—dura mater, arachnoid mater, and pia mater—that enclose the brain and spinal cord. They provide physical protection, act as a barrier against pathogens, and maintain structural stability. **Cerebrospinal fluid (CSF)**, which circulates within the ventricles and subarachnoid space, cushions the CNS against shocks and sudden movements. It also helps maintain chemical stability by supplying nutrients, removing waste products, and regulating ionic balance. Together, meninges and CSF prevent mechanical injury, reduce the risk of infection, and support the CNS in functioning efficiently within the rigid cranial and vertebral cavities.

MARKING SCHEME (3 Marks)

- Meninges (1.5 marks), CSF (1.5 marks).

Q.9 What is the functional role of the midbrain in humans, despite its reduced size?

Ans. The midbrain, located between the forebrain and hindbrain, is relatively small in humans but plays several crucial roles. Structurally, it houses the tectum and tegmentum, containing nuclei that regulate visual and auditory reflexes, such as turning the head toward a sound or tracking

MARKING SCHEME (3 Marks)

- Location (1 mark), functions (2 marks).



moving objects. Functionally, it acts as a relay center that connects higher brain centers to motor pathways, contributing to voluntary motor control. The midbrain also contains part of the reticular formation, which regulates alertness and arousal, and pathways for dopamine production important in movement regulation. Thus, despite its reduced size in humans compared to other animals, the midbrain remains essential for reflex integration, sensory processing, and motor coordination.

Q.10 How do spinal nerves originate from the spinal cord, and what is their significance?

Ans. Spinal nerves originate as paired structures from the spinal cord, with each nerve formed by the fusion of a **dorsal root** and a **ventral root**. The dorsal root carries sensory input from receptors into the spinal cord, while the ventral root carries motor commands from the spinal cord to effectors. When these roots merge, they form a mixed spinal nerve that contains both sensory and motor fibers. Humans have 31 pairs of spinal nerves that branch out to different body regions. These nerves are significant because they ensure two-way communication between the brain and the rest of the body—transmitting sensory feedback for processing and carrying motor instructions for voluntary and involuntary actions.

MARKING SCHEME (3 Marks)

- Origin (1 mark), functions (2 marks).

SLO [B-12-G-22]

Explain the structure types and functions of the autonomic nervous system.

STRUCTURE TYPES AND FUNCTIONS OF THE AUTONOMIC NERVOUS SYSTEM

Peripheral Nervous System: A Brief Overview

The **peripheral nervous system (PNS)** links the **central nervous system (CNS)** with the rest of the body. It is divided into two main parts:

1. The **somatic nervous system**, which controls **voluntary activities** through skeletal muscles.
2. The **autonomic nervous system (ANS)**, which regulates **involuntary functions** of internal organs, smooth muscles, and glands.

► Structure of the Autonomic Nervous System

The **autonomic nervous system** operates largely beyond conscious control and ensures the **maintenance of homeostasis**. Structurally, it consists of **motor pathways** with two neurons in sequence:

- A **preganglionic neuron**, whose cell body lies in the **CNS** and whose axon extends to a **ganglion**.
- A **postganglionic neuron**, which arises from the **ganglion** and innervates the target organ (smooth muscle, cardiac muscle, or gland).

Autonomic ganglia serve as relay centers, while autonomic nerves are distributed widely to thoracic, abdominal, and pelvic organs, as well as blood vessels and glands.

► Types of Autonomic Nervous System

The **autonomic nervous system (ANS)** has two major divisions: the **sympathetic** and the **parasympathetic nervous systems**. Though they often act in opposite ways, they work together to maintain **homeostasis**, adjusting the body's functions according to stress, rest, or energy demands.

1. Sympathetic Nervous System (SNS)

The sympathetic division is sometimes referred to as the **thoracolumbar outflow**, because its preganglionic neurons arise from the **thoracic (T1–T12)** and **lumbar (L1–L2)** regions of the spinal cord.

Structural Features:

- **Preganglionic fibers** are **short** and synapse in sympathetic ganglia located near the spinal cord (paravertebral or prevertebral ganglia).
- **Postganglionic fibers** are **long**, enabling the sympathetic system to reach many organs simultaneously, producing a rapid, body-wide response.
- **Neurotransmitters:** **Noradrenaline (norepinephrine)** is the main transmitter at effector organs (except sweat glands, which use acetylcholine).

Functions:

The **SNS** prepares the body for **"fight or flight"** situations—sudden stress, danger, or intense physical activity. Its effects include:



- **Cardiovascular system:** Increases heart rate, force of contraction, and blood pressure.
- **Respiratory system:** Dilates bronchioles, allowing greater oxygen intake.
- **Eyes:** Dilates pupils (mydriasis) to enhance vision.
- **Digestive and urinary systems:** Inhibits digestive activity and reduces urinary output to conserve energy.
- **Musculoskeletal system:** Diverts blood flow to skeletal muscles for immediate action.
- **Metabolism:** Stimulates glucose release from the liver and increases metabolic activity to provide energy.

This system equips the body for quick reactions, survival, and high-energy demands.

2. Parasympathetic Nervous System (PNS)

The parasympathetic division is known as the **craniosacral outflow**, because its preganglionic neurons arise from the **brainstem (cranial nerves III, VII, IX, X)** and **sacral spinal cord (S2–S4)**.

Structural Features:

- **Preganglionic fibers** are **long** and extend almost to the target organ.
- **Postganglionic fibers** are **short**, with ganglia located close to or within the effector organs.
- **Neurotransmitter:** **Acetylcholine** is used at both preganglionic and postganglionic synapses.

Functions:

The PNS is active during restful states and is often described as the **“rest and digest”** system. Its functions promote conservation and restoration of energy:

- **Cardiovascular system:** Slows down **heart rate** and reduces blood pressure.
- **Respiratory system:** Constricts bronchioles to a resting state.
- **Eyes:** Constricts pupils (**miosis**) and adjusts the lens for near vision.
- **Digestive system:** Stimulates salivation, gastric secretions, peristalsis, and nutrient absorption.
- **Urinary and reproductive systems:** Promotes urination and facilitates reproductive functions.

This system restores the body to baseline conditions after stress and supports long-term health by aiding digestion, repair, and energy storage.

Functional Balance

The sympathetic and parasympathetic divisions are antagonistic yet complementary. For example:

- During exercise, the **sympathetic system** raises heart rate and suppresses digestion.
- Afterward, the **parasympathetic system** slows the heart rate and restores digestive activity.

This balance ensures the body can adapt to both sudden challenges and periods of rest.

Functional Balance of the ANS

The **sympathetic and parasympathetic divisions** usually act in **antagonistic yet complementary ways**, maintaining a balance according to the body's needs. For instance, while the sympathetic system accelerates the heart during stress, the parasympathetic system restores it to a resting rhythm afterward.

SLO [B-12-G-29]

Explain the structure and functioning of the receptors for smell, taste, and touch/pain.

SENSORY RECEPTORS AND THEIR WORKING

The human body perceives the external world through **specialized sensory receptors** that detect specific types of stimuli and convert them into nerve impulses, which are then interpreted by the brain. Among these, the receptors for **smell (olfaction)**, **taste (gustation)**, and **touch/pain (somatosensation)** play a vital role in survival, behaviour, and quality of life.

1. Receptors for Smell (Olfactory Receptors)

Structure:

- The sense of smell originates in the **olfactory epithelium**, located in the upper region of the nasal cavity.
- This epithelium contains **olfactory receptor neurons**, which are specialized **chemoreceptors**.
- Each olfactory neuron has **cilia (olfactory hairs)** that project into the mucus lining the nasal cavity.
- Odour molecules dissolve in the mucus and bind to **receptor proteins** on the cilia.

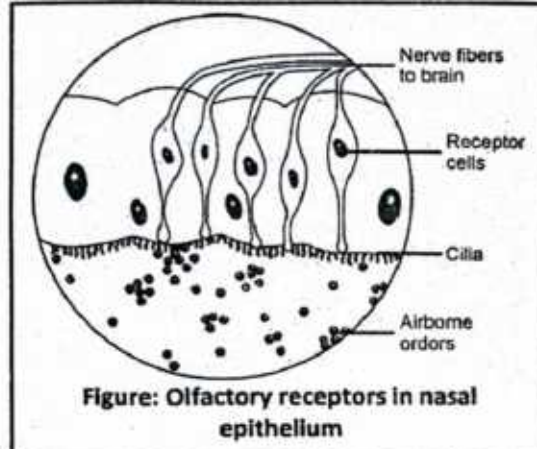


Functioning:

- (i) When an odorant binds to the receptor proteins, it triggers a **signal transduction pathway** that generates an electrical impulse.
- (ii) These impulses are transmitted via the **olfactory nerve (cranial nerve I)** to the **olfactory bulb**.
- (iii) From the olfactory bulb, signals are relayed to higher brain centers, including the **olfactory cortex, limbic system, and hypothalamus**, where smells are identified and linked to emotions and memories.

Significance:

- Smell contributes to flavour perception, detection of hazards (e.g., smoke, spoiled food), and emotional responses.



2. Receptors for Taste (Gustatory Receptors)

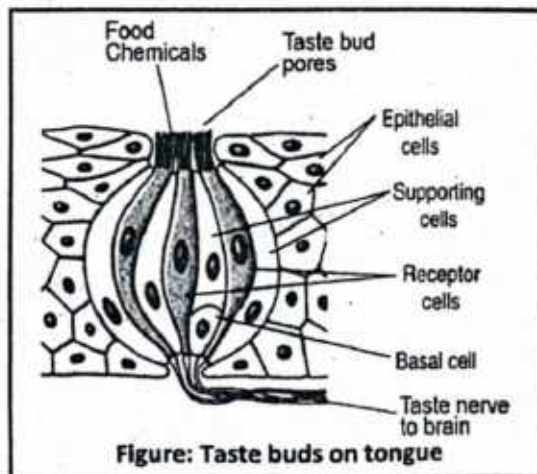
Structure:

- Taste is detected by **taste buds**, which are clusters of receptor cells located mainly on the **tongue**, but also on the soft palate and pharynx.
- Taste buds are embedded in specialized structures called **papillae** (fungiform, circumvallate, and foliate papillae).
- Each taste bud contains **gustatory receptor cells** with microvilli that project into a taste pore exposed to the oral cavity.

Functioning:

- (i) Dissolved food molecules interact with receptor proteins on the microvilli.
- (ii) This stimulates depolarization of the gustatory receptor cells, leading to the release of neurotransmitters.

- (iii) The signals are carried by **cranial nerves (VII – facial, IX – glossopharyngeal, and X – vagus)** to the **gustatory cortex** in the brain.



Types of Taste Sensations:

There are five primary tastes:

- **Sweet** (sugars, energy-rich compounds)
- **Sour** (acids, hydrogen ions)
- **Salty** (salts, sodium ions)
- **Bitter** (alkaloids, toxins)
- **Umami** (glutamate, amino acids, savory foods)

Significance:

- Taste helps regulate nutrition, detecting beneficial food and avoiding harmful substances (many toxins taste bitter).

3. Receptors for Touch and Pain (Somatosensory Receptors)

The skin, the body's largest sensory organ, contains **mechanoreceptors, thermoreceptors, and nociceptors** that detect physical contact, pressure, vibration, temperature, and pain.

a) Touch Receptors (Mechanoreceptors):

- **Meissner's corpuscles**: Sensitive to **light touch**, found in fingertips and lips.
- **Pacinian corpuscles**: Detect deep pressure and vibration.
- **Merkel discs**: Respond to sustained pressure and texture.
- **Ruffini endings**: Detect skin stretch and contribute to perception of object manipulation.



b) Pain Receptors (Nociceptors):

- Free nerve endings scattered throughout the skin, muscles, joints, and some internal organs.
- Respond to tissue damage, harmful chemicals, or extreme temperatures.
- Pain signals travel via A-delta fibers (fast, sharp pain) and C fibers (slow, dull pain) to the spinal cord and brain.

Functioning:

- Touch receptors generate nerve impulses in response to mechanical deformation of the skin.
- Pain receptors transmit protective signals to the brain, which interprets them as discomfort or injury.
- The brain integrates these signals in the somatosensory cortex of the parietal lobe, allowing precise localization of sensations.

Significance:

- Touch enables exploration, communication, and fine motor control.
- Pain acts as a warning system, preventing further tissue damage and triggering protective reflexes.

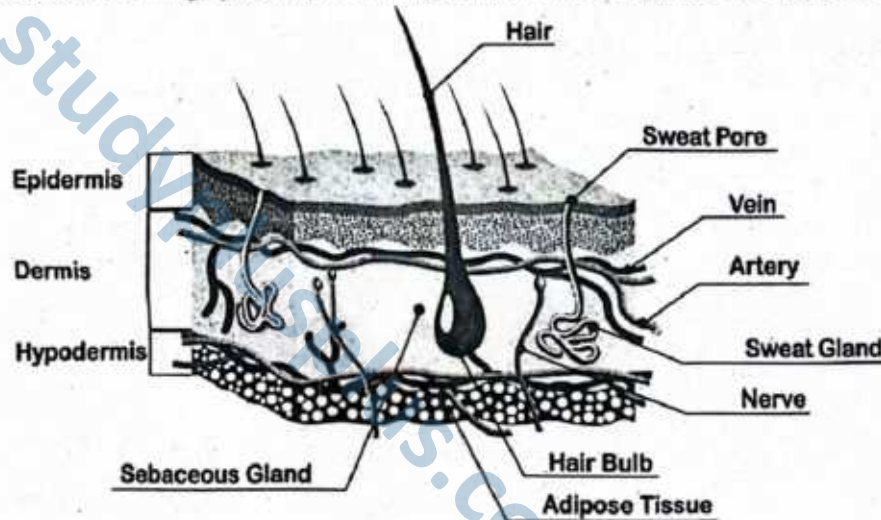


Figure: Receptors in human skin

Summary

- **Olfactory receptors** detect chemical molecules in the air, enabling the sense of smell.
- **Gustatory receptors** in taste buds perceive dissolved chemicals, allowing detection of sweet, sour, salty, bitter, and umami.
- **Somatosensory receptors** in the skin provide sensations of touch, pressure, temperature, and pain, essential for protection and interaction with the environment.

Together, these receptors form a critical sensory network that keeps humans aware, safe, and adapted to their surroundings.

SLOs Based (MCQs)

PERIPHERAL NERVOUS SYSTEM AND SENSORY RECEPTORS

- | | |
|---|--|
| 1. Which of the following is a conscious activity controlled by the somatic nervous system?
A. Digestion
B. Reflex action
C. Voluntary movement
D. Hormone secretion | 3. Meissner's corpuscles are primarily involved in detecting:
A. Pressure
B. Light touch
C. Pain
D. Temperature |
| 2. Which nerves are involved in the parasympathetic division of the autonomic nervous system?
A. Only spinal nerves
B. Thoracic and lumbar nerves
C. Cranial and sacral nerves
D. All cervical nerves | 4. Which structure is responsible for detecting chemicals in the nasal cavity?
A. Taste buds
B. Pacinian corpuscles
C. Olfactory receptors
D. Merkel disks |

5. Pain receptors in the skin are:
A. Encapsulated nerve endings

- B. Neurons with taste hairs
C. Free nerve endings
D. Modified epithelial cells

Answer Key

1. C 2. C 3. B 4. C 5. C

Short Questions

Q.1 How do the sympathetic and parasympathetic divisions of the autonomic nervous system function antagonistically?

Ans. The autonomic nervous system maintains internal balance by having two opposing divisions: the **sympathetic** and **parasympathetic** systems. The **sympathetic** division prepares the body for "fight-or-flight" situations. It increases heart rate, raises blood pressure, dilates pupils for better vision, and diverts blood flow away from digestive organs to muscles, enabling rapid physical responses. In contrast, the **parasympathetic division** promotes "rest-and-digest" functions. It slows down the heart rate, lowers blood pressure, stimulates digestive processes, and enhances glandular secretions for recovery and energy storage. Their antagonistic actions—one stimulating while the other calms—work together to ensure that the body adapts quickly to stress but can also return to normal once the emergency has passed.

MARKING SCHEME (3 Marks)

- Sympathetic division functions (1.5 marks), parasympathetic division functions (1.5 marks).

Q.2 Describe the structure and function of taste buds.

Ans. Taste buds are specialized sensory structures responsible for detecting taste stimuli. Structurally, they are groups of modified epithelial cells located within the papillae of the tongue, particularly on the sides and tip. Each taste bud contains a small opening called a **taste pore**, through which microscopic projections known as **taste hairs** extend. These hairs come in contact with dissolved food chemicals (tastants) present in saliva. Functionally, taste buds detect the four primary taste sensations—sweet, sour, salty, and bitter (and umami, considered the fifth). Once stimulated, they generate nerve impulses that travel via cranial nerves to the brain, where the signals are processed, allowing us to perceive and distinguish flavors.

MARKING SCHEME (3 Marks)

- Structure (1.5 marks), function (1.5 marks).

Q.3 Compare Meissner's corpuscles and Pacinian corpuscles in terms of their location and function.

Ans. Meissner's corpuscles are mechanoreceptors located close to the surface of the skin, especially in sensitive regions such as the fingertips, lips, and eyelids. They are small, encapsulated structures that respond to light touch, texture, and low-frequency vibrations, making them essential for tasks requiring fine tactile sensitivity. In contrast, **Pacinian corpuscles** are larger, onion-shaped mechanoreceptors situated deeper within the dermis and subcutaneous tissue. They are adapted to detect deep pressure and high-frequency vibrations rather than light touch. Both types of corpuscles work together to provide a full range of tactile sensations, ensuring the nervous system can detect both gentle contact and strong mechanical pressure.

MARKING SCHEME (3 Marks)

- Meissner's corpuscles (1.5 marks), Pacinian corpuscles (1.5 marks).

SLO [B-12-G-30]

Define narcotic drugs as agents that interact with normal nervous activity.

EFFECTS OF DRUGS ON NERVOUS COORDINATION

Narcotic drugs are chemical substances that act on the central nervous system (CNS) to alter normal nervous activity, usually by depressing brain function. They reduce the perception of pain, induce sleep or drowsiness, and may produce a sense of euphoria. Because of these effects, they are often prescribed for medical use (as painkillers or anesthetics) but are also prone to misuse and addiction.



Characteristics of Narcotic Drugs

1. CNS Depressants

Narcotic drugs primarily act as **central nervous system (CNS) depressants**. They reduce the excitability of neurons, thereby slowing down overall brain activity. This results in:

- **Sedation and relaxation**, which help patients with severe pain or anxiety.
- **Drowsiness or sleep induction**, often making them useful as pre-anesthetic agents.
- In higher doses, excessive CNS depression can lead to **respiratory suppression**, **reduced reflexes**, and even **coma**, which is why medical use requires careful supervision.

2. Analgesic Effect

One of the most important medical properties of narcotics is their ability to act as **potent analgesics** (pain relievers). They work by:

- Binding to **opioid receptors (mu, kappa, delta)** located in the brain and spinal cord.
- Blocking or reducing the transmission of pain signals from peripheral nerves to the CNS.
- Increasing the brain's tolerance to pain, making it feel less intense or distressing.
- Because of this, narcotics are often prescribed for **post-surgical pain**, **cancer pain**, or **severe injury-related pain** when other painkillers are insufficient.

3. Addictive Potential

Narcotics can cause **dependence** because of their effects on the brain's **reward system**:

- They stimulate the release of **dopamine**, creating feelings of pleasure, calmness, or euphoria.
- Repeated use can lead to **psychological addiction**, where a person craves the drug for emotional comfort.
- **Physical dependence** also develops, meaning the body adapts to the presence of the drug and cannot function normally without it.
- This addictive potential is the reason why many narcotics (e.g., heroin) are tightly controlled or illegal outside medical use.

4. Tolerance Development

With prolonged or repeated use, narcotic drugs often lead to **tolerance**, a condition in which:

- The body becomes less responsive to the same dose of the drug.
- Increasingly **higher doses** are required to achieve the desired effect (e.g., pain relief or sedation).
- This can escalate the risk of **overdose**, as the therapeutic window (safe range) becomes narrower with time.

Tolerance, dependence, and addiction together form the basis of **narcotic abuse disorders**, posing both medical and social challenges.

► Common Examples

- **Natural narcotics:** Morphine, codeine (from opium poppy)
- **Semi-synthetic narcotics:** Heroin (diacetylmorphine)
- **Synthetic narcotics / prescription drugs:** Tramadol, pethidine, methadone, fentanyl
- **Stimulants / party drugs:** ICE (crystal meth), Molly (MDMA/ecstasy)
- **CNS stimulants / everyday drugs:** Caffeine (tea, coffee, energy drinks)
- **CNS depressants / alcohol:** Ethanol (beer, spirits)

► Interaction with Nervous Activity

Narcotics bind to **opioid receptors** in the brain, spinal cord, and other tissues, which normally interact with the body's natural pain-relieving chemicals (**endorphins**). By mimicking or enhancing these signals, narcotics:

- Reduce the transmission of pain impulses.
- Produce sedation and sleep.
- Suppress reflexes, breathing rate, and sometimes even consciousness at high doses.

► Medical and Social Aspects

- **Medical Use:** Prescribed for **severe pain relief**, control of cough, and as anesthetics during surgery.
- **Abuse:** Non-medical use can impair judgment, cause dependence, and damage health.
- **Withdrawal symptoms:** Irritability, anxiety, muscle pain, insomnia, and craving when drug use is stopped abruptly.



COMPARE THE USE AND ABUSE OF DRUGS

Substance use and abuse are major public health concerns. While some drugs have **medical or socially accepted uses**, their misuse can lead to severe physical, psychological, and social consequences. This section compares the **use and abuse** of five major categories of drugs: **heroin, cannabis, nicotine, alcohol, and inhalants**.

1. Heroin

Use (Medical/Controlled Context):

- Heroin (diacetylmorphine) is a **semi-synthetic narcotic** derived from morphine.
- In Pakistan, it has **no accepted medical use** and is **completely illegal**. Some countries use it under strict medical supervision for **pain management**, but this is not practiced in Pakistan.

Abuse:

- Heroin is **highly addictive**. Even a single exposure can create dependence in vulnerable individuals.
- It produces **intense euphoria**, sedation, and a feeling of detachment from reality.
- **Health consequences**: Collapsed veins, infections, liver and kidney disease, respiratory depression, and overdose leading to death.
- **Social impact**: Crime, unemployment, family breakdown, and social stigma.

2. Cannabis (Marijuana)

Use (Medical/Traditional):

- Cannabis is sometimes used medically for pain relief or nausea control.
- In Pakistan, limited traditional or religious use exists, e.g., hashish (charas) in certain regions.

Abuse:

- Recreational abuse is widespread, especially among youth.
- **Short-term effects**: relaxation, altered perception, euphoria, increased appetite.
- **Long-term effects**: memory impairment, reduced concentration, anxiety, psychosis, and respiratory problems if smoked.
- **Legal status**: Cannabis is **illegal in Pakistan**, though enforcement varies regionally.

3. Nicotine (Tobacco Products)

Use (Legal / Socially Accepted):

- Found in cigarettes, shisha, and chewing tobacco (gutka, naswar).
- Nicotine is a stimulant and mild CNS depressant, widely used socially.

Abuse:

- **Highly addictive**; leads to physical dependence.
- **Health consequences**: lung cancer, oral cancer, cardiovascular disease, respiratory disorders, and pregnancy complications.
- **Social consequences**: Normalization of tobacco use among youth, economic burden, and second-hand smoke effects.

4. Alcohol (Ethanol)

Use (Medical / Social):

- In many countries, alcohol is used socially or medically (e.g., as a disinfectant).
- In Pakistan, alcohol is mostly **illegal**, except for limited medical or religiously permitted consumption by non-Muslims.

Abuse:

- **Recreational use** occurs illicitly.
- **Short-term effects**: CNS depression, impaired coordination, reduced judgment, and euphoria.
- **Long-term abuse**: Liver cirrhosis, neurological damage, addiction, domestic violence, and social problems.
- Risk of **methanol poisoning** is high due to illicitly brewed alcohol.

5. Inhalants (Nail Polish Remover, Glue, Petrol, Paint Thinner)

Use (Industrial / Household):

- Substances like nail polish remover (acetone), glue, petrol, and paint thinner have legitimate uses in industry, household, and crafts.

Abuse:

- Inhaled deliberately for **rapid intoxication**, particularly among street children and adolescents.
- Short-term effects: dizziness, euphoria, hallucinations, impaired judgment, and nausea.
- Long-term effects: brain damage, liver and kidney toxicity, heart failure, sudden sniffing death syndrome.
- Highly dangerous because **toxicity is immediate and sometimes irreversible**, even after a single use.

Comparative Summary

Drug Category	Legal/Accepted Use	Abuse Pattern	Short-Term Effects	Long-Term Effects	Social Impact
Heroin	None in Pakistan	Injected, smoked, snorted	Euphoria, sedation	Addiction, organ damage, overdose	Crime, unemployment, stigma
Cannabis	Traditional/limited medical	Smoked or ingested	Relaxation, altered perception	Memory issues, psychosis, respiratory problems	Youth abuse, legal issues
Nicotine	Socially accepted	Smoking, chewing	Mild stimulation, relaxation	Cancer, heart disease, addiction	Economic burden, normalization of use
Alcohol	Limited/medical	Drinking illicitly	Euphoria, sedation, impaired coordination	Liver disease, addiction, neurological damage	Domestic violence, social problems
Inhalants	Industrial/household	Sniffing, huffing	Euphoria, hallucinations, dizziness	Brain, liver, kidney damage, sudden death	

SLO [B-12-G-32]

Explain the terms drug addiction and drug tolerance with reference to caffeine and nicotine and their adverse effects.

DRUG ADDICTION AND DRUG TOLERANCE WITH REFERENCE TO CAFFEINE AND NICOTINE

Drugs are substances that interact with the **nervous system**, altering normal physiological or psychological activity. While some are used legally for stimulation or medical purposes, repeated use can lead to **drug addiction** and **drug tolerance**, even with everyday substances like **caffeine** and **nicotine**.

1. Drug Addiction

Definition:

Drug addiction is a **chronic, relapsing disorder** characterized by **compulsive drug use**, loss of control over intake, and continued consumption despite harmful consequences. It often involves **physical dependence** (the body requires the drug to function normally) and **psychological dependence** (cravings and emotional reliance).

Caffeine Addiction:

- Found in **tea, coffee, soft drinks, and energy drinks**, caffeine is the most commonly consumed psychoactive substance in Pakistan.



- Repeated caffeine intake can lead to **mild physical dependence**, resulting in:
 - Headaches, irritability, and fatigue when caffeine is suddenly reduced (withdrawal symptoms).
 - Compulsive consumption to maintain alertness.
- Long-term overuse can cause insomnia, anxiety, rapid heart rate, and digestive problems.

Nicotine Addiction:

- Present in **cigarettes, shisha, gutka, and naswar**, nicotine is highly **addictive** due to its effects on the **dopaminergic reward system** in the brain.
- Users develop **strong cravings** and continue consumption despite health risks.
- Health consequences include lung and oral cancers, cardiovascular disease, chronic respiratory disorders, and complications in pregnancy.

2. Drug Tolerance

Definition:

Drug tolerance is a phenomenon in which **repeated exposure to a substance reduces its physiological effect**, requiring **increasing doses** to achieve the same effect. Tolerance often precedes **physical dependence** and can increase the risk of adverse effects.

Caffeine Tolerance:

- Regular consumption of caffeine leads to **reduced sensitivity of adenosine receptors**, diminishing alertness effects.
- Habitual users may require **higher doses** to feel the same level of wakefulness or concentration.
- Excessive doses to overcome tolerance may cause restlessness, palpitations, and digestive disturbances.

Nicotine Tolerance:

- Chronic nicotine use leads to desensitization of nicotinic acetylcholine receptors in the brain.
- Smokers or chewers require higher amounts of tobacco products to achieve the same stimulant or euphoric effect.
- This escalation increases the risk of cancer, cardiovascular disease, and other serious health problems.

3. Adverse Effects of Caffeine and Nicotine

Substance	Type of Effect	Short-Term Effects	Long-Term / Chronic Effects
Caffeine	CNS stimulant	Alertness, mild euphoria, increased heart rate	Insomnia, anxiety, palpitations, digestive issues, mild dependence
Nicotine	CNS stimulant, addictive	Increased alertness, relaxation, mild euphoria	Lung/oral cancer, heart disease, chronic respiratory disorders, strong dependence

SLO [B-12-G-33]

Associate the effects of drug addiction and tolerance with the functioning of the nervous system.

EFFECTS OF DRUG ADDICTION AND TOLERANCE

Drug addiction and tolerance are phenomena that directly impact the **structure and function of the nervous system**. Repeated exposure to psychoactive substances alters **neuronal communication, receptor sensitivity, and neurotransmitter balance**, producing both short-term and long-term physiological and behavioral effects.

1. Effects of Drug Addiction on the Nervous System

Drug addiction is a chronic condition characterized by **compulsive substance use**, despite negative *consequences on health, social life, and cognitive functioning*. Addiction profoundly affects the **structure, chemistry, and signalling of the nervous system**, leading to both temporary and long-term alterations in brain function.

a) Neurotransmitter Alteration

- Addictive substances such as **heroin, nicotine, alcohol, cocaine, and cannabis** target the **brain's reward pathways**, primarily the **mesolimbic dopamine system**, which includes the **ventral tegmental area (VTA), nucleus accumbens, and prefrontal cortex**.
- These drugs **increase dopamine release** or inhibit its reuptake, producing intense **pleasure, euphoria, and reinforcement of drug-seeking behavior**.



- Over time, the brain's **natural reward system** is altered, reducing responsiveness to normally rewarding stimuli (e.g., food, social interaction) and increasing reliance on the drug for dopamine release.

b) Receptor Adaptation

- Chronic drug exposure leads to **receptor desensitization or downregulation**, reducing the sensitivity of neurons to neurotransmitters.
- Examples include:
 - **Nicotine:** Overstimulates **nicotinic acetylcholine receptors**, causing desensitization; this requires increased nicotine intake to achieve the same stimulant effect.
 - **Opioids (e.g., heroin, morphine):** Bind **mu-opioid receptors**, decreasing the activity of **endogenous endorphins**, which are natural pain-relieving and reward-mediating chemicals.
- These adaptations contribute to **tolerance** and the development of **physical dependence**, making cessation difficult.

c) Behavioural and Cognitive Effects

- Chronic drug use alters **prefrontal cortex function**, which is responsible for **decision-making, impulse control, planning, and judgment**.
- The **hippocampus**, essential for **learning and memory**, is also affected, leading to **memory deficits, impaired learning, and difficulty forming new information**.
- Other cognitive and emotional consequences include:
 - Reduced ability to resist cravings
 - Emotional instability, irritability, and heightened stress sensitivity
 - Difficulty in maintaining occupational and social responsibilities

d) Physical Dependence and Withdrawal

- The nervous system adapts to the **constant presence of the drug**, establishing a new homeostatic balance.
- When the drug is suddenly removed, **withdrawal symptoms** emerge due to the **temporary imbalance of neurotransmitters and autonomic nervous system activity**.
- **Common withdrawal symptoms** include:
 - Anxiety, irritability, and depression
 - Tremors and muscle aches
 - Sweating, nausea, and vomiting
 - Altered heart rate, blood pressure, and other autonomic functions
- Severe withdrawal from opioids or alcohol can be **life-threatening**, highlighting the deep integration of addiction into nervous system function.

e) Long-Term Neural Consequences

- Persistent drug addiction can lead to **structural changes in neurons**, including:
 - Loss of dendritic spines in the **prefrontal cortex and hippocampus**
 - Impaired synaptic plasticity, which diminishes the brain's ability to adapt or learn
 - Dysregulation of the **hypothalamic-pituitary-adrenal (HPA) axis**, increasing susceptibility to stress-related disorders
- These changes explain why addiction is considered a **chronic brain disease** with long-lasting effects, even after prolonged abstinence.

2. Effects of Drug Tolerance on the Nervous System

Drug tolerance is a phenomenon in which the nervous system becomes **less responsive to a substance over repeated exposure**, requiring progressively higher doses to achieve the same effect. Tolerance is a **neuroadaptive response** that involves both **cellular and systemic changes** in the central and peripheral nervous system.

a) Receptor Downregulation

- Repeated drug use often leads to **desensitization or downregulation of receptors**.
- Examples include:
 - **Dopamine receptors** in the brain's reward pathways: Downregulation reduces the pleasurable effects of drugs like cocaine or methamphetamine.
 - **Nicotinic acetylcholine receptors:** Chronic nicotine use decreases receptor sensitivity, requiring more tobacco products to achieve stimulation.



- **Opioid receptors:** Persistent opioid exposure reduces receptor activity, necessitating higher doses for pain relief or euphoria.
- These changes explain why individuals **increase dosage over time**, which increases the risk of **toxicity and overdose**.

b) Neuronal Adaptation

- Neurons adjust to chronic drug exposure through multiple mechanisms, including:
 - **Modulation of ion channel activity** to stabilize excitability.
 - Altered **neurotransmitter synthesis, release, and reuptake** to compensate for the drug's effects.
 - Changes in **synaptic plasticity**, affecting how neurons communicate.
- **Example:** Regular caffeine intake reduces the sensitivity of **adenosine receptors**, so habitual users require higher amounts to maintain alertness and avoid fatigue.
- These adaptations demonstrate the brain's **homeostatic attempts** to maintain normal function despite persistent chemical interference.

c) Impact on Central and Peripheral Nervous System Functioning

- **Central Nervous System (CNS):**
 - Altered neuronal excitability affects **mood, cognitive function, alertness, and motor coordination**.
 - Tolerance may blunt the desired effects of stimulants or depressants, causing **increased consumption** to achieve the original effect.
- **Autonomic Nervous System (ANS):**
 - Chronic drug use can modify **heart rate, blood pressure, and digestive functions**.
 - For example, nicotine tolerance may **reduce its stimulatory effects on cardiovascular parameters**, but dependence and withdrawal symptoms still affect ANS function.

d) Long-Term Consequences

- Tolerance contributes to **physical dependence**, where the nervous system requires the substance to maintain normal function.
- It increases **vulnerability to overdose**, as higher doses may be taken to overcome desensitized receptors.
- Long-term neuronal adaptations can result in **persistent changes in brain function**, making recovery from addiction more difficult.

3. Integrated View: Addiction, Tolerance, and Nervous System Function

Aspect	Nervous System Involvement	Outcome on Function
Addiction	Dopamine reward pathways, receptor overstimulation, prefrontal cortex, hippocampus	Compulsive behavior, impaired decision-making, emotional instability
Tolerance	Receptor desensitization, synaptic adaptation, neurotransmitter balance	Need for higher doses, reduced drug efficacy, increased risk of overdose
Withdrawal / Dependence	CNS and ANS affected	Anxiety, tremors, altered heart rate, sleep disturbances

SLO [B-12-G-34]

Describe the way how pain medicines can reduce or numb pain in the human body.

MECHANISM OF ACTION OF PAIN MEDICINES

Pain is an unpleasant sensory and emotional experience caused by **tissue damage, inflammation, or nerve injury**. The body perceives pain through **specialized nerve endings** called **nociceptors**, which transmit signals via the **peripheral nervous system (PNS)** to the **central nervous system (CNS)**, including the spinal cord and brain. Pain medicines (analgesics) **reduce or numb pain by interfering with these signals at various levels**.

1. Types of Pain Medicines

Pain-relieving drugs are classified based on their **mechanism of action**:



(i) Non-Opioid Analgesics

- **Examples:** Paracetamol (acetaminophen), ibuprofen, aspirin
- **Mechanism:**
 - Inhibit **prostaglandin synthesis** by blocking **cyclooxygenase (COX) enzymes**.
 - Prostaglandins are chemicals released at the site of injury that **sensitize nociceptors**, causing pain and inflammation.
 - By reducing prostaglandins, these drugs **diminish pain perception** and **lower inflammation**.

(ii) Opioid Analgesics (Narcotics)

- **Examples:** Morphine, codeine, tramadol, pethidine
- **Mechanism:**
 - Bind to **opioid receptors (mu, delta, kappa)** in the **CNS and spinal cord**.
 - Inhibit the transmission of pain signals from the spinal cord to the brain.
 - Increase **endogenous endorphin activity**, producing **analgesia** and a **sense of well-being**.
 - Effective for **moderate to severe pain**, including post-surgical pain, trauma, and cancer pain.

(iii) Local Anesthetics

- **Examples:** Lidocaine, bupivacaine
- **Mechanism:**
 - Block **voltage-gated sodium channels** in peripheral nerve fibers.
 - Prevent **action potential generation and propagation**, temporarily **numbing the targeted area**.
 - Used in dental procedures, minor surgeries, and regional anesthesia.

(iv) Adjuvant Analgesics (Supportive Drugs)

- **Examples:** Anticonvulsants (gabapentin), antidepressants (amitriptyline)
- **Mechanism:**
 - Alter **nerve excitability** or **neurotransmitter levels** in cases of **neuropathic pain**.
 - Enhance the effects of primary analgesics in chronic pain management.

2. Pathway of Pain Reduction

Pain medicines reduce or numb pain by **interfering with the transmission and perception of pain signals** at multiple levels of the nervous system, from the site of injury to the brain. This multi-level action explains the efficacy of different classes of analgesics.

a) Peripheral Level

- At the **site of tissue injury**, specialized nerve endings called **nociceptors** detect harmful stimuli such as mechanical pressure, heat, or chemical irritation.
- **Non-opioid analgesics** (e.g., paracetamol, ibuprofen, aspirin) act locally to **block the synthesis of prostaglandins**, which are chemical mediators released during injury or inflammation.
 - **Prostaglandins** sensitize nociceptors, making them more responsive to pain signals.
 - By reducing prostaglandin levels, non-opioid analgesics **diminish the sensitivity of nociceptors**, reducing the initiation of pain signals.
- Local anesthetics (e.g., lidocaine, bupivacaine) block voltage-gated sodium channels in peripheral nerve fibers, preventing the generation and propagation of action potentials.
 - This effectively **numbs the targeted area**, stopping pain signals from reaching the spinal cord.

b) Spinal Level

- Once pain signals reach the **dorsal horn of the spinal cord**, neurotransmitters such as **substance P** and **glutamate** relay information to ascending pathways toward the brain.
- **Opioid analgesics** act at this level by binding to **opioid receptors (mu, delta, kappa)** on spinal neurons.
 - They inhibit the release of excitatory neurotransmitters, reducing the **transmission of pain signals**.
 - This produces **segmental analgesia**, decreasing the intensity of pain before it reaches higher brain centers.
- Some **adjuvant drugs** (e.g., anticonvulsants or antidepressants) also act on spinal circuits to **modulate nerve excitability**, particularly in neuropathic pain.

c) Brain Level

- Pain signals that reach the **brainstem, thalamus, and cerebral cortex** are interpreted as **pain perception** and are accompanied by emotional and cognitive responses.



- **Opioids and adjuvant drugs** modulate several brain regions:
 - **Thalamus:** Filters and relays pain information to cortical areas.
 - **Limbic system:** Reduces the emotional and distressing aspects of pain.
 - **Cortex:** Alters the conscious perception of pain, including attention and awareness.
- By acting at these central sites, analgesics can decrease both the sensory intensity of pain and the associated emotional suffering, improving patient comfort.

d) Integrated View

- Pain reduction occurs through a **hierarchical mechanism**:
 1. **Peripheral:** Reduce nociceptor activation or block nerve conduction.
 2. **Spinal:** Inhibit neurotransmitter release in the dorsal horn.
 3. **Central (Brain):** Alter perception and emotional response to pain.
- The choice of analgesic depends on the type of pain (acute, chronic, neuropathic) and the desired site of action.

3. Clinical Considerations

- Pain medicines are selected based on pain type (acute vs. chronic), severity, and underlying cause.
- Overuse of opioids can lead to tolerance, dependence, and addiction.
- Non-opioid analgesics may cause liver or kidney damage if used in excessive doses.
- Local anesthetics are usually safe but require careful dosing to avoid systemic toxicity.

SLO [B-12-G-35]

Discuss that certain pain medications are addictive.

ADDICTIVE POTENTIAL OF CERTAIN PAIN MEDICATIONS

While pain medications are essential for **managing acute and chronic pain**, some drugs, particularly **opioid analgesics**, carry a **high risk of addiction**. Understanding the mechanisms behind their addictive potential is crucial for safe clinical use and public health awareness.

1. Opioid Analgesics and Addiction

Opioid drugs include morphine, codeine, tramadol, pethidine, and fentanyl. They are highly effective for moderate to severe pain but act directly on the central nervous system (CNS), producing both analgesia and euphoria.

Mechanism of Addiction:

- Opioids bind to mu-opioid receptors in the brain and spinal cord, blocking pain transmission.
- Activation of reward pathways in the mesolimbic system increases dopamine release, creating a pleasurable sensation.
- Repeated use can lead to physical dependence, psychological cravings, and compulsive drug-seeking behaviour, even when pain is no longer present.

Clinical Observations:

- Patients may require **increasing doses** to achieve the same pain relief due to **tolerance**.
- Long-term use can result in **addiction, withdrawal symptoms, and difficulty discontinuing the drug**.

2. Factors Contributing to Addiction

- **Duration and frequency of use:** Prolonged opioid therapy increases risk.
- **Dosage:** Higher doses enhance activation of the reward system.
- **Individual susceptibility:** Genetic, psychological, and social factors influence addiction risk.
- **Route of administration:** Intravenous or rapid-release forms are more addictive than slow-release oral forms.

3. Other Pain Medications and Addiction Risk

- **Tramadol:** A synthetic opioid with milder addictive potential, but misuse can still lead to **physical and psychological dependence**.
- **Codeine:** Found in some cough **syrups** and painkillers; repeated use can cause **habit formation**, especially in **adolescents**.



- **Non-opioid analgesics** (e.g., paracetamol, NSAIDs) **do not produce euphoria** and are generally **non-addictive**, though overuse can cause **organ toxicity**.

4. Physiological Basis of Addiction

- Chronic opioid use alters the nervous system:
 - **Receptor adaptation:** Downregulation of opioid receptors reduces drug sensitivity, leading to tolerance.
 - **Neurotransmitter imbalance:** Decreased natural endorphin production increases dependence on the drug for normal function.
 - **Reward system hijacking:** Dopamine-mediated reinforcement strengthens compulsive use patterns.

SLOs Based (MCQs)

- Which category of drugs primarily binds to opioid receptors to block pain signals?
 - NSAIDs
 - Opioids
 - Anticonvulsants
 - Caffeine
- Which of the following is a depressant that slows down central nervous system activity?
 - Nicotine
 - Alcohol
 - Caffeine
 - Cannabis
- Caffeine dependence is mainly characterized by:
 - Respiratory suppression
 - Hallucinations
 - Tolerance and mild withdrawal
 - Dopamine blockade
- The drug derived from *Cannabis sativa* that alters perception and induces mild euphoria is:
 - Heroin
 - Cannabis
 - Nicotine
 - Acetaminophen
- Which mechanism describes how local anesthetics block pain?
 - Blocking prostaglandin synthesis
 - Stimulating endorphin production
 - Blocking sodium ion channels in neurons
 - Enhancing serotonin uptake

Answer Key

1. B 2. B 3. C 4. B 5. C

Short Questions

Q.1 Differentiate between drug addiction and drug tolerance using caffeine and nicotine as examples.

Ans. Drug addiction is a compulsive physiological and psychological dependence on a substance, such as nicotine found in tobacco. Individuals feel a strong craving for the drug and may experience withdrawal symptoms like irritability, anxiety, and restlessness if the substance is unavailable. Addiction often leads to repeated use despite harmful consequences. **Drug tolerance**, on the other hand, occurs when the body gradually adapts to a substance, requiring increasing amounts to achieve the same effect. For example, habitual caffeine consumers may need higher doses over time to feel alert or awake. Both conditions encourage habitual use and can impact physical and mental health, but addiction involves compulsive behavior, whereas tolerance is primarily physiological adaptation.

MARKING SCHEME (3 Marks)

- Drug addiction (1 mark), drug tolerance (1 mark), examples (1 mark).

Q.2 How does heroin affect the central nervous system?

Ans. Heroin is an opioid drug that profoundly affects the central nervous system (CNS). It binds to opioid receptors in the brain and spinal cord, mimicking natural endorphins, which blocks pain signals and produces intense feelings of euphoria and relaxation. Short-term effects include drowsiness, confusion, and slowed reflexes. Chronic use can cause severe dependence, respiratory depression, impaired brain function, and increased risk of overdose and death. The drug alters neural pathways, reinforcing compulsive drug-seeking behavior and making cessation extremely challenging. Withdrawal symptoms may include anxiety, sweating, nausea, muscle pain, and insomnia, highlighting the dangerous impact of heroin on both physiological and psychological health.

MARKING SCHEME (3 Marks)

- Effect on CNS (1.5 marks), side effects (1.5 marks).

Q.3 Describe the harmful effects of alcohol on the nervous system.

Ans. Alcohol is a CNS depressant that interferes with normal neural communication. **Short-term effects** include slowed reflexes, impaired judgment, poor coordination, and delayed reaction times, increasing the risk of



accidents. Long-term consumption damages neurons and brain structures, reducing cognitive function and memory. Chronic alcohol abuse also affects other organs like the liver and pancreas, indirectly influencing CNS health through metabolic imbalances. **Sudden withdrawal** from heavy drinking may trigger anxiety, hallucinations, tremors, or seizures. Overall, alcohol disrupts both structural and functional integrity of the nervous system, impairing physical, cognitive, and emotional regulation.

MARKING SCHEME (3 Marks)
• Three harmful effects (1 mark each).

SLO [B-12-G-36]

Classify nervous disorders into vascular, infectious, structural, functional and degenerative disorders.

CLASSIFICATION OF NERVOUS DISORDERS

Nervous system disorders are conditions that impair the structure or function of the central or peripheral nervous system, leading to symptoms such as paralysis, sensory loss, cognitive impairment, or movement disorders. These disorders can be classified based on their underlying cause into five main categories: vascular, infectious, structural, functional, and degenerative.

1. Vascular Disorders

Definition: Disorders caused by impaired blood supply to the brain or spinal cord.

Examples and Features:

- **Stroke (Cerebrovascular Accident, CVA):**
 - Caused by blockage (ischemic stroke) or rupture (hemorrhagic stroke) of blood vessels.
 - Symptoms include sudden paralysis, facial drooping, speech difficulties, and loss of coordination.
- **Transient Ischemic Attack (TIA):**
 - A temporary blockage of blood flow causing brief neurological deficits.

Significance:

- Early diagnosis and treatment are critical to prevent permanent brain damage or death.

2. Infectious Disorders

Definition: Disorders caused by pathogens (bacteria, viruses, fungi, or parasites) affecting the nervous system.

Examples and Features:

- **Meningitis:** Infection of the meninges, leading to fever, severe headache, neck stiffness, and vomiting.
- **Encephalitis:** Viral infection causing brain inflammation, seizures, and altered consciousness.
- **Poliomyelitis:** Viral infection of the spinal cord causing paralysis; still reported in some regions of Pakistan.

Significance:

- Early medical intervention with antibiotics, antivirals, or vaccines can prevent severe complications.

3. Structural Disorders

Definition: Disorders resulting from physical changes, injury, or abnormal anatomy of the nervous system.

Examples and Features:

- **Traumatic Brain Injury (TBI):** Caused by accidents, falls, or head trauma, leading to cognitive, motor, or sensory deficits.
- **Herniated Intervertebral Disc:** Pressure on spinal nerves causing pain, weakness, or numbness in limbs.
- **Tumors (benign or malignant):** Can compress brain or spinal tissue, causing headaches, seizures, or neurological deficits.

Significance:

- Structural damage often requires surgery, rehabilitation, or long-term management.

4. Functional Disorders

Definition: Disorders where no structural damage is detectable, but normal nervous system function is impaired.

Examples and Features:

- **Epilepsy:** Abnormal electrical activity in the brain leading to seizures, loss of consciousness, or involuntary movements.

- **Migraine:** Functional disorder of cerebral blood vessels and neurons, causing severe headache, nausea, and visual disturbances.
- **Functional neurological disorder (FND):** Neurological symptoms without detectable structural abnormality, often linked to stress or psychological factors.

Significance:

- Functional disorders often require medications, therapy, and lifestyle management.

5. Degenerative Disorders

Definition: Disorders caused by **progressive loss of neurons or their functions**, often associated with aging or genetic factors.

Examples and Features:

- **Alzheimer's Disease:** Progressive memory loss, cognitive decline, and personality changes.
- **Parkinson's Disease:** Degeneration of dopaminergic neurons, leading to tremors, rigidity, and slow movements.
- **Amyotrophic Lateral Sclerosis (ALS):** Degeneration of motor neurons causing muscle weakness and paralysis.

Significance:

- Degenerative disorders are chronic, progressive, and currently incurable, but treatment can slow progression and improve quality of life.

Summary Table: Classification of Nervous Disorders

Category	Cause	Examples	Key Features
Vascular	Impaired blood supply	Stroke, TIA	Paralysis, speech loss, coordination problems
Infectious	Pathogens (bacteria, viruses, fungi)	Meningitis, Encephalitis, Polio	Fever, headache, seizures, paralysis
Structural	Physical injury or abnormal anatomy	TBI, Herniated disc, Tumors	Pain, weakness, sensory deficits
Functional	Impaired function, no structural damage	Epilepsy, Migraine, FND	Seizures, headache, neurological symptoms
Degenerative	Progressive neuronal loss	Alzheimer's, Parkinson's, ALS	Memory loss, tremors, paralysis

SLO [B-12-G-37]

Describe the causes, symptoms and treatment of one type of each category of disorders outlined above (e.g., stroke as vascular, meningitis as infectious, brain tumor as structural, headache as functional, and Alzheimer disease as degenerative disorder).

CAUSES, SYMPTOMS, AND TREATMENT OF REPRESENTATIVE NERVOUS DISORDER

To understand nervous disorders in practical terms, one can study a representative disorder from each category: vascular, infectious, structural, functional, and degenerative.

1. Vascular Disorder – Stroke (Cerebrovascular Accident, CVA)

Causes:

- **Ischemic stroke:** Blockage of a cerebral artery due to a blood clot or atherosclerosis.
- **Hemorrhagic stroke:** Rupture of a blood vessel causing bleeding into the brain.
- Risk factors include hypertension, diabetes, smoking, high cholesterol, and heart disease.

Symptoms:

- Sudden weakness or paralysis, often on one side of the body (hemiplegia).
- Facial drooping, difficulty speaking (aphasia), vision problems, and loss of balance or coordination.
- Severe headache in hemorrhagic stroke.

Treatment:

- **Emergency care** to restore blood flow or stop bleeding.
- **Medications:** Clot-busting drugs (thrombolytics) for ischemic stroke; antihypertensives for prevention.
- **Rehabilitation:** Physiotherapy, occupational therapy, and speech therapy.

2. Infectious Disorder – Meningitis

Causes:

- Infection of the meninges by bacteria (e.g., *Neisseria meningitidis*), viruses, or fungi.
- Spread can occur through respiratory droplets or contaminated objects.

Symptoms:

- Severe headache and neck stiffness.
- Fever, nausea, vomiting, sensitivity to light (photophobia).
- In severe cases: seizures, confusion, and loss of consciousness.

Treatment:

- Bacterial meningitis: Immediate intravenous antibiotics and supportive care.
- Viral meningitis: Mostly self-limiting; rest, hydration, and symptom management.
- Vaccination is key for prevention against common bacterial strains.

3. Structural Disorder – Brain Tumor

Causes:

- Abnormal growth of cells in the brain; may be **benign or malignant**.
- Can arise due to **genetic mutations, radiation exposure, or unknown factors**.

Symptoms:

- **Persistent headaches**, often worse in the morning.
- **Nausea, vomiting, and seizures**.
- Neurological deficits such as **weakness, sensory changes, or speech difficulties**, depending on tumor location.

Treatment:

- **Surgical removal** when feasible.
- **Radiotherapy and chemotherapy** for malignant tumors.
- Supportive care includes medications to reduce **intracranial pressure** and manage seizures.

4. Functional Disorder – Headache (Migraine as Example)

Causes:

- Functional disorder of brain neurons and cerebral blood vessels; may be triggered by **stress, hormonal changes, certain foods, or sensory stimuli**.

Symptoms:

- **Severe, pulsating headache**, usually on one side of the head.
- Nausea, vomiting, **sensitivity to light (photophobia) or sound (phonophobia)**.
- **Aura in some patients:** Visual disturbances or tingling sensations.

Treatment:

- **Acute relief:** Analgesics (paracetamol, NSAIDs), triptans for severe migraines.
- **Preventive therapy:** Beta-blockers, anticonvulsants, or lifestyle changes.
- Avoiding triggers and maintaining **regular sleep, hydration, and stress management**.

5. Degenerative Disorder – Alzheimer's Disease

Causes:

- Progressive degeneration of **neurons in the brain**, particularly in the **hippocampus and cerebral cortex**.
- Exact cause is unclear; **genetic factors, age, and abnormal protein accumulation (beta-amyloid plaques, tau tangles)** play roles.

Symptoms:

- Gradual **memory loss** and difficulty learning new information.
- **Disorientation, confusion, mood changes**, and difficulty performing daily activities.
- **Advanced stages:** loss of speech, recognition of family members, and physical dependence.



Treatment:

- **Medications:** Cholinesterase inhibitors (donepezil) and NMDA receptor antagonists (memantine) slow progression but do not cure.
- **Supportive care:** Cognitive therapy, structured routines, caregiver support.
- **Lifestyle interventions:** healthy diet, physical activity, and mental stimulation may delay symptom progression.

Summary Table

Category	Representative Disorder	Causes	Key Symptoms	Treatment
Vascular	Stroke	Blocked/ruptured cerebral arteries	Paralysis, speech problems, headache	Emergency care, meds, rehabilitation
Infectious	Meningitis	Bacteria, virus, fungi	Headache, neck stiffness, fever	Antibiotics/antivirals, vaccination
Structural	Brain Tumor	Abnormal cell growth	Headache, seizures, neurological deficits	Surgery, radiotherapy, chemotherapy
Functional	Migraine	Functional neuronal & vascular changes	Severe headache, nausea, photophobia	Analgesics, triptans, preventive therapy
Degenerative	Alzheimer's	Neuronal degeneration, protein accumulation	Memory loss, confusion, disorientation	Medications, supportive care, lifestyle interventions

SLO [B-12-G-38]

Explain the principles of important diagnostic tests for nervous disorders i.e., EEG, CT scan and MRI.

PRINCIPLES OF DIAGNOSTIC TESTS FOR NERVOUS DISORDERS

Diagnosis of nervous system disorders relies on a combination of clinical evaluation and specialized diagnostic tests. Among the most important tools are Electroencephalography (EEG), Computed Tomography (CT) scan, and Magnetic Resonance Imaging (MRI). These tests allow physicians to detect structural, functional, and electrical abnormalities in the brain and spinal cord.

1. Electroencephalography (EEG)

Principle:

- EEG measures the **electrical activity of the brain** using electrodes placed on the scalp.
- Neurons communicate via **electrical impulses**, and EEG detects **voltage fluctuations** generated by postsynaptic potentials.
- The recorded patterns, called **brain waves**, vary in frequency and amplitude, reflecting different states such as wakefulness, sleep, or seizure activity.

Clinical Uses:

- **Epilepsy:** Detects abnormal **spike-and-wave discharges** during seizures.
- **Sleep disorders:** Identifies sleep stages and abnormalities like **narcolepsy**.
- **Coma and brain death assessment:** Measures **baseline brain activity**.

Advantages:

- Non-invasive and safe.
- Provides **real-time functional information** about brain activity.

Limitations:

- Poor spatial resolution; cannot detect **structural abnormalities**.

2. Computed Tomography (CT) Scan

Principle:

- CT scan uses **X-rays and computer processing** to create **cross-sectional images** of the brain or spinal cord.



- X-rays pass through tissues at different densities; the computer constructs a detailed image showing bones, soft tissues, and blood vessels.

Clinical Uses:

- **Stroke:** Detects ischemic or hemorrhagic lesions quickly.
- **Trauma:** Identifies skull fractures, intracranial bleeding, and swelling.
- **Tumors and hydrocephalus:** Visualizes masses or enlarged ventricles.

Advantages:

- Fast and widely available.
- Good for **emergency assessment** of acute neurological conditions.

Limitations:

- Exposure to ionizing radiation.
- Limited soft tissue contrast compared to MRI.

3. Magnetic Resonance Imaging (MRI)

Principle:

- MRI uses strong magnetic fields and radiofrequency waves to align and detect the behavior of hydrogen nuclei (protons) in body tissues.
- Different tissues respond differently to magnetic pulses, producing high-resolution images of brain and spinal structures in multiple planes.

Clinical Uses:

- **Structural disorders:** Detects tumors, brain malformations, and demyelinating diseases like multiple sclerosis.
- **Vascular abnormalities:** Visualizes aneurysms and infarcts.
- **Functional MRI (fMRI):** Measures brain activity by detecting blood flow changes during tasks.

Advantages:

- Excellent **soft tissue contrast**.
- No ionizing radiation; safe for repeated use.

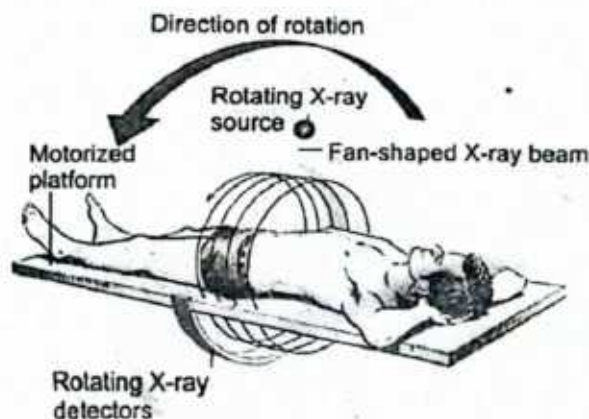
Limitations:

- Expensive and less accessible in some regions.
- Contraindicated in patients with **metal implants or pacemakers**.

CT scanners give doctors a 3-D view of the body. The images are exquisitely detailed but require a dose of radiation that can be 100 times that of a standard X-ray.



Figure: Receptors in human skin



Computed tomography scans are made by rotating an x-ray beam around the patient, imaging the body in a series of slices that a computer stitches together.

Figure: (a) Principle of CT Scan (b) CT Scan

Comparison of CT Scan, MRI, and ECG

Feature	CT Scan (Computed Tomography)	MRI (Magnetic Resonance Imaging)	ECG (Electrocardiography)
Full Form	Computed Tomography	Magnetic Resonance Imaging	Electrocardiography
Purpose	Produces cross-sectional images of internal body structures	Provides detailed images of soft tissues, brain, and spinal cord	Records electrical activity of the heart
Principle	Uses X-rays and computer processing	Uses strong magnetic fields and radio waves	Detects electrical impulses generated by heart muscle
Radiation Involved	Yes (ionizing radiation)	No	No
Best For	Visualizing bones, tumours, bleeding, and lung diseases	Imaging brain, spinal cord, joints, and soft tissues	Diagnosing heart rhythm abnormalities and cardiac health
Image Resolution	Moderate resolution	High-resolution images	Not applicable (graphical output, not an image)
Time Required	A few minutes	20-60 minutes	Less than 5 minutes
Invasiveness	Non-invasive	Non-invasive	Non-invasive
Cost	Generally lower than MRI	More expensive than CT scan	Relatively inexpensive
Limitations	Exposure to radiation; less detail in soft tissues	Not suitable for patients with metal implants or pacemakers	Limited to heart function; does not provide structural images

Do you know?

In CT Scan a large donut-shaped X-ray machine or scanner called tomograph takes X-ray images at many different angles around the body. These images are processed by a computer to produce cross-sectional pictures of the body. In each of these pictures the body is seen as an X-ray "slice" of the body, which is recorded on a film. This recorded image is called a tomogram.

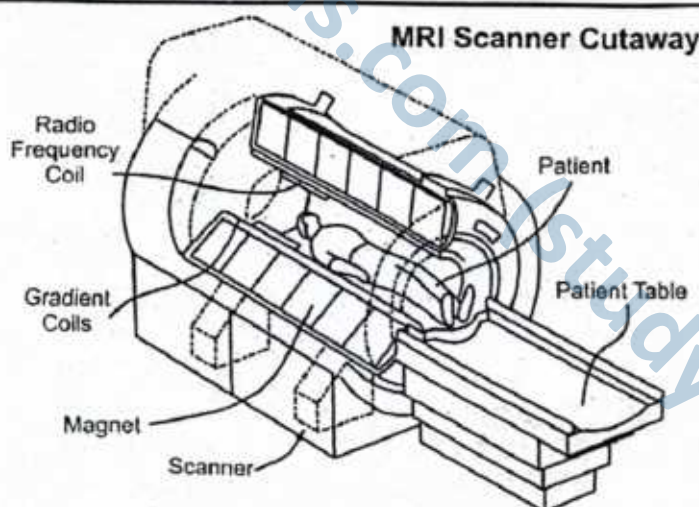


Figure: MRI scanner

SLOs Based (MCQs)

NERVOUS DISORDERS AND NEVOUS DISGNOSTIC TECHNIQUE

- Which of the following is a vascular disorder of the central nervous system?
 - Alzheimer's disease
 - Stroke
 - Meningitis
 - Brain tumour
- Which structure is primarily affected in meningitis?
 - Cerebral cortex
 - Meninges
 - Spinal cord
 - Cerebrospinal fluid
- What is the primary cause of bacterial meningitis?
 - Overactive immune response
 - Viral infection
 - Invasion of meninges by bacteria
 - Blocked blood vessels in the brain



4. Which class of disorders is Alzheimer's disease categorized under?
 A. Functional B. Infectious
 C. Structural D. Degenerative
5. Which diagnostic technique records the electrical activity of the cerebral cortex?
 A. MRI B. CT scan
 C. EEG D. PET scan
6. What is the main purpose of anticoagulants in the treatment of stroke?
 A. Reduce inflammation B. Inhibit pain signals
 C. Prevent blood clot formation
 D. Stimulate neuron regeneration
7. Which symptom is most typical of a brain tumour?
 A. Neck stiffness B. High fever
 C. Seizures and persistent headaches
 D. Short-term memory loss
8. Which imaging technique uses powerful magnets and radio waves to produce detailed images?
 A. EEG B. MRI
 C. CT scan D. X-ray
9. What is the function of prostaglandins at the site of injury or inflammation?
 A. Inhibit immune response
 B. Cause pain, fever, and swelling
 C. Transmit motor signals
 D. Regulate neuron firing rate
10. What pattern dominates brain activity during deep sleep (Stage N3)?
 A. Alpha waves B. Beta waves
 C. Delta waves D. Sleep spindles

Answer Key

1. B 2. B 3. C 4. D 5. C 6. C 7. C 8. B 9. B 10. C

Short Questions

Q.1 Describe the main cause and symptom of stroke.

Ans. A stroke occurs when the blood supply to a part of the brain is interrupted or reduced, either due to a **blockage (ischemic stroke)** caused by a thrombus or embolus, or due to **rupture of cerebral blood vessels (hemorrhagic stroke)**, often linked to hypertension or aneurysms. This sudden disruption prevents oxygen and nutrients from reaching brain tissue, leading to cellular death. **Symptoms** include sudden weakness or paralysis, usually on the side opposite the affected brain hemisphere, difficulty speaking or understanding speech, vision problems, dizziness, and severe headache. Rapid recognition and treatment are critical to minimize permanent brain damage.

MARKING SCHEME (3 Marks)

- Cause (1 mark), symptoms (2).

Q.2 Differentiate between viral and bacterial meningitis in terms of treatment.

Ans. **Viral meningitis** is usually mild and self-limiting; treatment focuses on relieving symptoms like fever, headache, and nausea using analgesics, hydration, and rest. No specific antiviral medication is typically required, and patients often recover within 7–10 days. **Bacterial meningitis**, however, is a medical emergency and requires immediate administration of intravenous antibiotics such as penicillin or ceftriaxone, along with corticosteroids to reduce inflammation and prevent complications like brain damage or sepsis. Rapid intervention is essential to reduce mortality and long-term neurological deficits.

MARKING SCHEME (3 Marks)

- Viral meningitis treatment (1.5), bacterial meningitis treatment (1.5).

Q.3 What is the role of EEG in diagnosing nervous system disorders?

Ans. **Electroencephalography (EEG)** is a non-invasive technique that records the brain's electrical activity through electrodes placed on the scalp. It is commonly used to **diagnose epilepsy** by identifying abnormal neural discharges, **monitor sleep patterns** in patients with sleep disorders, and **assess brain function** in cases of coma, encephalopathy, or brain injuries. EEG helps clinicians detect irregularities in brain rhythms, localize seizure activity, and monitor therapeutic interventions.

MARKING SCHEME (3 Marks)

- 1 mark each for three roles.

Q.4 Describe why MRI is preferred over CT scan in some brain investigations.

Ans. **Magnetic Resonance Imaging (MRI)** is preferred in certain brain investigations because it provides **superior soft tissue resolution**, making it easier to detect subtle structural changes such as early-stage

MARKING SCHEME (3 Marks)

- 1 mark each for three advantages.



tumors, demyelinating lesions, or brain edema. MRI also allows **detailed imaging of neural pathways** and vascular structures using specialized techniques like MR angiography. Unlike CT scans, MRI **does not use ionizing radiation**, reducing long-term risk to patients. Additionally, MRI can detect abnormalities in regions difficult to visualize on CT, such as the posterior fossa and brainstem.

Q.5 What are functional disorders of the nervous system? Give an example.

Ans. Functional disorders are conditions in which the nervous system appears structurally normal but fails to operate properly, causing abnormal symptoms. They differ from structural disorders, which involve identifiable lesions or damage. Functional disorders often result from over activation, dysregulation, or sensitivity of neural circuits. An example is **headache**, including **migraines** and tension-type headaches, where pain arises from abnormal sensory processing without visible structural damage in the brain. These disorders can affect daily life but may not be detected by conventional imaging.

MARKING SCHEME (3 Marks)

- Definition (2), example (1).

Exercise

I

Multiple Choice Questions (MCQs)

❖ **Select the correct Answer.**

- The cell transmits impulses from the
 - effector organ to the spinal cord
 - receptor cells to the effector organ
 - receptor cells to the spinal cord
 - spinal cord to the effector organ
- Depolarization of an axon is produced by the movement of:
 - Na^+ into the axon and K^+ out of the axon
 - Na^+ into the axon to bond with K^+
 - K^+ into the axon and Na^+ out of the axon
 - Na^+ and K^+ within the axon towards the axon terminal
- What will happen if the receptor sites on the post-synaptic membrane are blocked by a drug at the neuromuscular junction?
 - inhibition of acetylcholine
 - inhibition of cholinesterase
 - muscle contraction
 - muscle paralysis
- Which of these are the first and last elements in a spinal reflex?
 - axon and dendrite
 - sense organ and muscle effector
 - ventral horn and dorsal horn
 - motor neuron and sensory neuron
- Impulses travel very rapidly along nerves to the leg of a man. Which fact accounts for the speed at which they travel?
 - a nerve impulse is an all or none phenomenon
 - the nerves contain myelinated fibres
 - there is a high concentration of Na^+ ions inside the axons
 - there is a potential difference across the axon membranes
- Where are neurotransmitter receptors located?
 - on the nuclear membrane
 - at nodes of Ranvier
 - on the postsynaptic membrane
 - in the myelin sheath
- The human nervous system is capable of a wide range of functions. What is the basic unit of the nervous system?
 - glial cell
 - meninges
 - neuron
 - cerebrospinal fluid
- Which of the following is the correct order of meninges from the inner side?
 - Pia mater - arachnoid mater - durometer
 - Pericardium - myocardium - endocardium
 - Durometer - pia mater - arachnoid mater
 - Durometer - arachnoid mater - pia mater
- Which of the following structure at a synapse has the neurotransmitter?
 - schwan cells
 - synaptic cleft
 - synaptic knobs
 - synaptic vesicles
- The action potential while the propagation of a nerve impulse is due to the movement of _____
 - K^+ ions from intracellular to extracellular fluid
 - K^+ ions from extracellular to intracellular fluid
 - Na^+ ions from intracellular to extracellular fluid
 - Na^+ ions from extracellular to intracellular fluid



11. Which one is the ability of neurons to initiate nerve impulses?
A. myelination B. conductivity
C. irritability D. neurotransmission
12. Which of the following is one of the parts of the hindbrain?
A. hypothalamus B. cerebellum
C. corpus callosum D. spinal cord
13. If there is an injury in the hypothalamus region of the brain, it is most likely to affect
A. regulation of body temperature
B. decision making
C. co-ordination during locomotion
D. short-term memory
14. Afferent neurons carry nerve impulses from
A. CNS to muscles B. CNS to receptors
C. receptors to CNS D. effector organs to CNS
15. The cranial nerve that regulates the heartbeat
A. VII B. VIII
C. IX D. X
16. Nissl's granules present in the neurons are made up of
A. protein B. ribosome
C. RNA D. DNA
17. A neuron that carries information from the peripheral nervous system to the central nervous system is
A. afferent neuron B. efferent neuron
C. both D. none
18. Which of the following parts of the brain controls the body temperature and urge of eating?
A. thalamus B. cerebellum
C. pons D. hypothalamus
19. The central nervous system consists of the
A. brain and spinal cord
B. spinal nerves only
C. cerebrum and the spinal column
D. brain stem and the cerebellum
20. Which one of the following controls involuntary activities such as coughing and sneezing?
A. medulla B. cerebrum
C. pons D. cerebellum
21. Spinal cord and brain are wrapped in protective membranes known as
A. nodes of ranvier B. meninges
C. axomembranes D. myelin sheath

Answer Key

1. C	2. A	3. D	4. B	5. B	6. C	7. C	8. D	9. D	10. D	11. C
12. B	13. A	14. C	15. D	16. C	17. A	18. A	19. A	20. A	21. B	

II

Short Answer Questions

Q.1 Why is neuron coordination important?

Ans. Neuronal coordination is crucial for the proper functioning of the human body because it ensures the integration and timely response to both internal and external stimuli. It allows sensory information to be detected, processed, and acted upon efficiently, enabling reflexes, voluntary and involuntary movements, and higher cognitive functions such as reasoning, learning, and memory. Neuronal coordination also maintains homeostasis by regulating processes like heart rate, digestion, and hormone secretion. Without it, body systems would act independently, leading to disorganized responses and impaired survival.

MARKING SCHEME (3 Marks)

- Definition of neuronal coordination (1 mark), importance in body function and homeostasis (2 marks).

Q.2 Describe the receptors as transducers sensitive to various stimuli.

Ans. Receptors are specialized cells or structures that act as biological transducers by converting different types of environmental or internal stimuli into electrical signals (nerve impulses). They can respond to light (photoreceptors in the eye), sound (mechanoreceptors in the ear), pressure and touch (mechanoreceptors in skin), temperature (thermoreceptors), and chemical signals (chemoreceptors in taste buds and olfactory cells). These electrical signals are transmitted through sensory neurons to the central nervous system, where they are interpreted, allowing the body to react appropriately to changing conditions.

MARKING SCHEME (3 Marks)

- Definition of receptors as transducers (1.5 marks), examples of stimuli and their detection (1.5 marks).

Q.3 Name the five fundamental parts of the human reflex arc.

Ans. The human reflex arc is the simplest neural pathway enabling a rapid, involuntary response to a stimulus. It consists of:

1. **Receptor** – detects the stimulus in the body or environment.
2. **Sensory (afferent) neuron** – transmits the impulse from the receptor to the central nervous system.
3. **Relay (interneuron)** – located in the spinal cord or brain; processes the information and coordinates the response.
4. **Motor (efferent) neuron** – carries the processed impulse from the CNS to the effector.
5. **Effector** – a muscle or gland that executes the response, such as contraction or secretion.

This arrangement allows a reflex to occur rapidly without conscious thought.

MARKING SCHEME (5 Marks)

- 1 mark for each component (total 5 marks).

Q.4 What is the function of neurotransmitters?

Ans. Neurotransmitters are chemical messengers released from the axon terminals of presynaptic neurons into the synaptic cleft. They transmit signals to postsynaptic neurons or effector cells (muscles or glands). By binding to specific receptors on the postsynaptic membrane, neurotransmitters initiate or inhibit action potentials, allowing precise and rapid communication within the nervous system. They are essential for processes such as muscle contraction, sensation, mood regulation, memory formation, and learning. Examples include acetylcholine, dopamine, serotonin, and GABA. Their proper function ensures coordinated physiological and behavioral responses.

MARKING SCHEME (3 Marks)

- Definition (1.5 marks), role/functions (1.5 marks).

Q.5 What characteristics do the brain and spinal cord have in common?

Ans. Both the brain and spinal cord are central nervous system (CNS) components responsible for processing, integrating, and transmitting neural information. They share several characteristics:

- **Protection:** Both are enclosed by meninges and cushioned by cerebrospinal fluid (CSF) to prevent injury.
- **Composition:** Both contain grey matter, where processing occurs, and white matter, which transmits signals.
- **Functional role:** They coordinate reflexes, voluntary movements, sensory perception, and autonomic functions.
- **Connectivity:** Both serve as communication hubs, with the brain processing higher functions and the spinal cord relaying signals to and from the body.

These common features make them central to overall body regulation and survival.

MARKING SCHEME (3 Marks)

- Protection and structure (1 mark), functional role and connectivity (2 marks).

Q.6 What is the limbic system?

Ans. The limbic system is a group of interconnected structures in the brain, including the hypothalamus, amygdala, hippocampus, and cingulate gyrus. It is primarily involved in regulating emotions such as fear, anger, pleasure, and motivation. The hippocampus is essential for learning and memory formation, while the amygdala processes emotional responses and triggers appropriate physiological reactions via the autonomic nervous system. The hypothalamus coordinates emotional responses with endocrine and autonomic functions, such as heart rate, blood pressure, and hormone release. Overall, the limbic system links emotional states to behavior and physiological changes.

MARKING SCHEME (3 Marks)

- Definition and main components (1.5 marks), functions (1.5 marks).

Q.7 Name the sensory receptors of human skin and their functions.

Ans. Human skin contains specialized sensory receptors for detecting different types of stimuli:

- **Meissner's corpuscles:** Detect light touch and texture, mainly in fingertips and lips.
- **Pacinian corpuscles:** Respond to deep pressure and vibration, located in deeper dermis and joints.
- **Merkel discs:** Sense sustained touch and fine details.
- **Ruffini endings:** Detect skin stretch and help with grip control.

MARKING SCHEME (3 Marks)

- Receptor name (1 mark each for any three), functions (1 mark each for any three).



- **Free nerve endings:** Sense pain (nociception) and temperature changes. These receptors transmit information via sensory neurons to the CNS for interpretation and appropriate response.

Q.8 How do narcotic drugs interact with normal nervous activity?

Ans. Narcotic drugs, including **morphine and heroin**, bind to **opioid receptors** in the brain and spinal cord, which normally respond to natural endorphins. This binding **suppresses the transmission of pain signals**, produces intense euphoria, and alters mood. Prolonged use disrupts **normal neurotransmitter activity**, leading to **tolerance** (requiring higher doses for the same effect), **physical dependence**, and impaired neural function. Chronic use can also **affect cognition, emotion regulation, and autonomic responses**, making withdrawal and cessation challenging.

MARKING SCHEME (3 Marks)

- Mechanism of interaction (1.5 marks), effects on CNS and dependence (1.5 marks).

Q.9 Find out some of the common withdrawal symptoms of alcohol.

Ans. Alcohol withdrawal occurs when a dependent individual stops or reduces alcohol intake. Common symptoms include **anxiety, tremors, sweating, irritability, insomnia, nausea, and headaches**. Severe withdrawal can lead to **hallucinations, seizures, and delirium tremens**, a potentially life-threatening condition. These symptoms arise because chronic alcohol use **modifies neurotransmitter activity** in the CNS, especially GABA and glutamate systems, and sudden absence of alcohol disrupts the balance, causing hyperexcitability of neurons.

MARKING SCHEME (3 Marks)

- Common mild symptoms (1.5 marks), severe symptoms and explanation (1.5 marks).

Q.10 Write the differentiate between:

(a) Thermoreceptors and Nociceptors.

Ans. Thermoreceptors are specialized sensory receptors that detect **temperature changes** such as heat and cold. They are located in the **skin and hypothalamus** and help **regulate body temperature** through neural signaling. Nociceptors detect **pain from tissue damage** caused by mechanical, chemical, or thermal stimuli. They are found in **skin, muscles, joints, and viscera** and trigger **protective pain responses** to prevent injury. While thermoreceptors mainly maintain homeostasis, nociceptors alert the body to danger or harm.

MARKING SCHEME (3 Marks)

- Definition (1 mark), location (1 mark), function/role (1 mark).

(b) Axoplasm and Axolemma

Ans. Axoplasm is the cytoplasm inside the axon, containing **mitochondria, neurofibrils, and enzymes** that support the transport of substances along the axon. Axolemma is the plasma membrane surrounding the axon, composed of lipid bilayer and protein channels, which maintains the membrane potential and allows action potential conduction. Axoplasm is located internally, while axolemma forms the outer membrane. Together, they ensure the axon's structural integrity and efficient nerve impulse transmission.

MARKING SCHEME (3 Marks)

- Definition (1 mark), composition / location (1 mark), function (1 mark).

(c) Neuroglial cells and Schwann cells

Ans. Neuroglial cells are supporting cells of the CNS found in the brain and spinal cord. They provide structural support, nutrition, insulation, and homeostasis, and examples include **astrocytes and oligodendrocytes**. Schwann cells are myelin-forming cells of the PNS, located along peripheral nerves, that form the myelin sheath and assist in axon regeneration. While neuroglial cells support and maintain neurons in the CNS, Schwann cells are specialized for insulation and repair in the PNS.

MARKING SCHEME (3 Marks)

- Definition (1 mark), location (1 mark), function/examples (1 mark).

(d) Sensory Neurons and Motor Neurons

Ans. Sensory neurons carry impulses **from receptors to the CNS**, detecting and transmitting sensory stimuli. They usually have **long dendrites and short axons** and are located in the **dorsal root of spinal nerves**. Motor neurons carry impulses **from the CNS to effectors** (muscles or glands), initiating **motor responses**. They typically have **short dendrites and long axons** and are located in the **ventral root of spinal nerves**. Sensory neurons enable perception of stimuli, while motor neurons produce appropriate responses.

MARKING SCHEME (3 Marks)

- Direction/function (1 mark), structure / location (1 mark), role (1 mark).



(e) Reflex action and Reflex arc

Ans. A reflex action is an **involuntary, immediate response** to a stimulus, such as withdrawing a hand from a hot object. A reflex arc is the **neural pathway** that mediates this response, involving a **receptor, sensory neuron, interneuron, motor neuron, and effector**. Reflex actions **protect the body from harm**, while the reflex arc ensures **rapid transmission of signals** for these actions. Essentially, the reflex action is the observable response, and the reflex arc is the underlying neural mechanism.

MARKING SCHEME (3 Marks)

- Definition (1 mark), function (1 mark), components/examples (1 mark).

(f) Resting Membrane Potential (RMP) and Active Membrane Potential (AMP)

Ans. Resting membrane potential (RMP) is the **voltage difference across a neuron's membrane at rest**, typically around **-70 mV**, maintained by the **sodium-potassium pump** and selective ion permeability. Potassium (K^+) is more concentrated inside, while sodium (Na^+) is more concentrated outside. Active membrane potential (AMP) occurs **during impulse transmission**, when the membrane depolarizes to approximately **+40 mV** due to **Na^+ influx**, followed by **K^+ efflux**. RMP is a **polarized state**, while AMP involves **depolarization and repolarization**. RMP ensures the neuron is ready to fire, and AMP represents the **active signal conduction** along the neuron.

MARKING SCHEME (3 Marks)

- Definition (1 mark), ion distribution / membrane state (1 mark), functional role (1 mark).

(g) Depolarization and Repolarization

Ans. Depolarization is the **loss of resting membrane potential** when **Na^+ channels open** and **Na^+ ions flow into the neuron**, making the inside more positive. It initiates the action potential. Repolarization is the **restoration of resting membrane potential after depolarization**, caused by **K^+ exiting the neuron**, returning the membrane to a negative state. Depolarization allows signal propagation, while repolarization resets the membrane to prepare for the next impulse. Both are critical phases of the action potential cycle.

MARKING SCHEME (3 Marks)

- Definition (1 mark), ion movement/membrane potential (1 mark), role in impulse (1 mark).

(h) Repolarization and Hyperpolarization

Ans. Repolarization restores the membrane to its **resting potential** after depolarization through **K^+ efflux**, making the neuron ready for the next impulse. Hyperpolarization occurs when the membrane **becomes more negative than the resting state**, due to **excess K^+ leaving or Cl^- entering**. While repolarization resets the neuron, hyperpolarization **temporarily inhibits new impulses**, ensuring proper directionality of signal transmission. Both phases maintain neuronal excitability and prevent premature firing.

MARKING SCHEME (3 Marks)

- Definition (1 mark), cause (1 mark), result/function (1 mark).

(i) Refractory period and Absolute refractory period

Ans. The refractory period is a phase **after an action potential** when the neuron is **less responsive** and can generate another impulse only with a **stronger-than-normal stimulus**. The absolute refractory period occurs **during depolarization** and is a period when **no stimulus, however strong, can trigger another action potential**. These periods ensure **unidirectional propagation** of impulses and prevent overlap of action potentials. The absolute refractory period guarantees the **"all-or-none" principle** is maintained.

MARKING SCHEME (3 Marks)

- Definition (1 mark), timing (1 mark), impulse generation (1 mark).

(j) Presynaptic and Postsynaptic Neurons

Ans. The presynaptic neuron **transmits the nerve signal** and contains **synaptic knobs filled with neurotransmitter vesicles**. When an action potential arrives, neurotransmitters are released into the **synaptic cleft**. The postsynaptic neuron **receives the signal via receptors** on its membrane, which **bind neurotransmitters** and initiate a response. This structural arrangement ensures **unidirectional communication** between neurons, enabling rapid and precise signal transmission across synapses.

MARKING SCHEME (3 Marks)

- Role (1 mark), structure (1 mark), neurotransmitter action (1 mark).

(k) Axon and Dendrite

Ans. Axons are long, uniform extensions of a neuron that **carry impulses away from the cell body** toward other neurons, muscles, or glands.



Most neurons have **one axon**, which may be **myelinated** to increase conduction speed. Dendrites are **short, branched processes** that **receive impulses from other neurons** and conduct them toward the cell body. Neurons usually have **multiple dendrites**, which are typically **non-myelinated**, allowing for multiple synaptic connections and signal integration. Together, axons and dendrites enable **efficient input-output communication** in the nervous system.

MARKING SCHEME (3 Marks)
• Location (1 mark), function (2 marks).

(l) Synaptic knob and Synaptic vesicles

Ans. The synaptic knob is the **terminal swelling at the end of an axon** where neurotransmitters are released into the synaptic cleft. Inside the synaptic knob are **synaptic vesicles**, which **store neurotransmitters** such as acetylcholine or dopamine. When an action potential reaches the knob, these vesicles **fuse with the membrane** to release neurotransmitters, enabling communication with the postsynaptic neuron. This structural organization ensures **rapid and controlled signal transmission** across synapses.

MARKING SCHEME (3 Marks)
• Function (1 mark), number/structure (1 mark), myelination/length (1 mark).

(m) Pons and Medulla Oblongata

Ans. The pons is located above the medulla in the brainstem and acts as a relay center between the cerebrum and cerebellum. It also coordinates facial sensations, eye movements, and some autonomic functions. The medulla oblongata lies below the pons, continuous with the spinal cord, and controls vital involuntary actions like heartbeat, breathing, and blood pressure. It also mediates reflexes such as coughing, sneezing, swallowing, and vomiting. Both are integral for CNS communication and involuntary control.

MARKING SCHEME (3 Marks)
• Location (1 mark), function (1 mark), reflex role (1 mark).

(n) White matter and Grey matter

Ans. White matter is composed of myelinated axons, giving it a whitish appearance, and it conducts nerve impulses rapidly between different regions of the CNS. Grey matter contains neuronal cell bodies, dendrites, and unmyelinated axons, giving it a greyish color, and it processes and integrates information locally. In the brain, grey matter forms the cortex, while white matter lies deeper. In the spinal cord, grey matter is central, and white matter surrounds it. This arrangement supports efficient processing and communication.

MARKING SCHEME (3 Marks)
• Composition/appearance (1 mark), function (2 marks).

(o) Myelinated and Non-myelinated nerve fibers.

Ans. Myelinated nerve fibers are covered by a **myelin sheath**, which allows **faster saltatory conduction** of impulses. They appear **white** and are mainly found in **somatic nerves**. Non-myelinated fibers **lack myelin**, conduct impulses **slower** via continuous conduction, appear **grey**, and are common in **autonomic nerves**. Myelination **increases conduction speed, conserves energy, and ensures precise signaling**, while non-myelinated fibers are sufficient for slower, less urgent communication.

MARKING SCHEME (3 Marks)
• Conduction speed (1 mark), appearance (1 mark), example / location (1 mark).

(p) Cranial nerves and Spinal nerves.

Ans. Cranial nerves **originate from the brain and brainstem** and there are **12 pairs** in humans. They serve **sensory, motor, or mixed functions**, primarily controlling the **head, face, and neck**, including sensations, facial movements, and some autonomic functions. Spinal nerves **originate from the spinal cord with 31 pairs** and carry both sensory and motor fibers to the **limbs and trunk**. Cranial nerves often have specialized functions, while spinal nerves are segmentally organized for broader body innervation. Both are essential for transmitting information between the CNS and body.

MARKING SCHEME (3 Marks)
• Origin (1 mark), number (1 mark), function (1 mark).

(q) Drug addiction and Drug tolerance

Ans. Drug addiction is a **compulsive dependence** on a substance despite negative consequences, such as nicotine, heroin, or cocaine. It **alters brain function**, causes cravings, and requires **medical or psychological intervention to overcome**. Drug tolerance occurs when the body **adapts to repeated use** of a drug, like caffeine, and the same dose produces a

MARKING SCHEME (3 Marks)
• Definition (1 mark), effect (1 mark), reversibility/examples (1 mark).



reduced effect, necessitating higher doses for the same response. Tolerance is reversible after abstinence, whereas addiction involves long-term changes in behavior and physiology. Both lead to habitual use and may cause withdrawal symptoms.

(r) Somatic and Autonomic nervous system.

Ans. The somatic nervous system controls **voluntary movements** of skeletal muscles using a **single motor neuron** from CNS to effector. It has **no subdivisions** and enables conscious motor control. The autonomic nervous system regulates **involuntary actions** of **smooth muscles**, **cardiac muscles**, and **glands** through a **two-neuron pathway**: preganglionic and postganglionic neurons. It is divided into **sympathetic** and **parasympathetic** divisions to maintain homeostasis, modulate stress responses, and regulate organ function without conscious effort. Both systems coordinate body activities, but differ in control and neural pathways.

MARKING SCHEME (3 Marks)

- Control (1 mark), neuron pathway (1 mark), subdivisions/function (1 mark).

III

Long Answer Questions

Q.1 What is the basic organization of a nervous system?

Ans. Please See SLO [B-12-G-18]

Q.2 Describe the structure of a neuron.

Ans. Please See SLO [B-12-G-05]

Q.3 Describe the three types of neurons and write their functions.

Ans. Please See SLO [B-12-G-05]

Q.4 Describe the mechanism of synaptic transmission.

Ans. Please See SLO [B-12-G-16]

Q.5 Write the classification of neurotransmitters.

Ans. Please See SLO [B-12-G-17]

Q.6 Describe the human brain.

Ans. Please See SLO [B-12-G-20]

Q.7 Describe the structure of spinal cord with diagram.

Ans. Please See SLO [B-12-G-19]

Q.8 Describe the somatic and autonomic nervous system.

Ans. Please See SLO [B-12-G-22]

Q.9 Give an account of narcotic drugs.

Ans. Please See SLO [B-12-G-30]

Q.10 Describe the cause, symptoms and treatment of:

- a. Stroke, b. Headache, c. Meningitis, d. Tumour, e. Alzheimer disease

Ans. Please See SLO [B-12-G-37]

Q.11 Explain the principles of the following diagnostic tests of nervous disorders:

- a. EEG b. CT scan c. MRI

Ans. Please See SLO [B-12-G-38]

Q.12 Describe the way how pain medicines can reduce or numb pain in the human body.

Ans. Please See SLO [B-12-G-34]

